Lithium battery solid-state electrolyte catalog: topology and physics description

Release MIT

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CONTENTS:

1 B2Li2S5 1

2 As1Li1O4Rb2 5

3 Ge2Li1N3 9

4 In1Li1S2 13

5 Li4N2Na2 17

6 Be1Li1Sb1 21

7 K1Li1O1 25

8 Li1Mg1N1 29

9 As1Be1Li1 33

10 B2Ca1Li4O6 37

11 As1K1Li2 41

12 B1Li3N2 45

13 I1Li1 49

14 Al1Li5O4 53

15 Li6O4Zn1 57

16 Li3N1 61

17 Li7N4P1 65

18 F8Li4Zr1 69

19 Li1N1Sr1 73

20 C1Li1N1S1 77

21 Li1O5P1Ti1 81

22 B3Ca4Li1N6 85

23 B1Ca1Li1O3 89

24 Cl2Cs1Li1 93

25 Li4O4Ti1 97

26 B1Ge1Li1O4 101

27 F6Li3Ti1 105

28 Li2O4S1 109

29 C1Li4O4 113

30 Li2S1 117

31 Li14N6O3P2 121

32 Li1O2Ti1 125

33 C2B1F2Li1O4 129

34 Al1Li1O4Si1 133

35 Li5N2Na1 137

36 Li1N2Zr1 141

37 Cl1Li1O4 145

38 Li2N2Na4 149

39 Li3N2Na3 153

40 Li2N2Zr1 157

41 I6Li2O18Ti1 161

42 In1Li3O3 165

43 Li2O4Si1Sr1 169

44 Ca1Li1N1 173

45 Br2Li5N1 177

46 Al1La4Li1O8 181

47 Br3Cs1Li2 185

48 B1Li1Mg1N2 189

49 Ge1Li3O4Zn0 193

50 Ge1Li2Mg1O4 197

51 Ga1Li1Se2 201

52 Ba1Li2Mg1O8P2 205

53 Ga1Li3Na2O4 209

54 Li2Mg1O4Si1 213

55 Ge1Li4S4 217

56 Li3O4P1 221

57 K3Li1Si4 225

58 Li3P1S4 229

59 Li1O3Sb1 233

60 Li1Si1Y1 237

61 Li2O3Si1 241

62 Li1S2Ti1 245

63 Li1O1Rb1 249

64 Li4P2S6 253

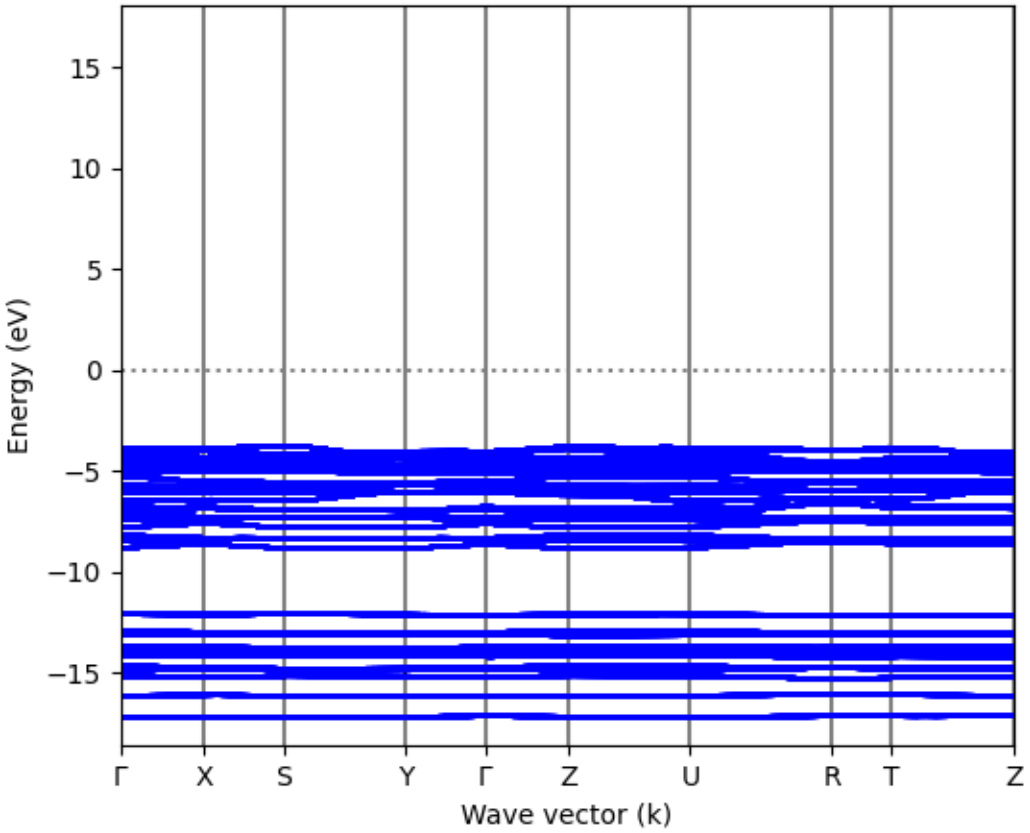
65 Al1Li1O2 257

CHAPTER

ONE

B2LI2S5

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
4.4325

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.988420357	0.000000000	-0.00000056
0.000000000	2.459407661	0.000000000
-0.00000059	0.000000000	1.065250444

ATOMIC_POSITIONS (crystal)

Li	0.0000000000	0.7691942067	0.0000000000
Li	0.5000000000	0.7308054143	0.5000000000
Li	0.5000000000	0.7308057224	0.0000000000
Li	0.0000000000	0.7691944037	0.5000000000
Li	0.5000000000	0.2691942776	0.0000000000
Li	0.0000000000	0.2308055963	0.5000000000
Li	0.0000000000	0.2308057933	0.0000000000
Li	0.5000000000	0.2691945857	0.5000000000
B	0.6472303916	0.9129660776	0.7500000354
B	0.1472300618	0.5870339009	0.7500000729
B	0.8527699382	0.5870339009	0.2499999271
B	0.3527696084	0.9129660776	0.2499999646
B	0.1472300618	0.4129660991	0.7500000729
B	0.6472303916	0.0870339224	0.7500000354
B	0.3527696084	0.0870339224	0.2499999646
B	0.8527699382	0.4129660991	0.2499999271
S	0.8688285561	0.5659151816	0.7499998964
S	0.3688289146	0.9340848171	0.7499998216
S	0.6311710854	0.9340848171	0.2500001784
S	0.1311714439	0.5659151816	0.2500001036
S	0.3688289146	0.0659151829	0.7499998216
S	0.8688285561	0.4340848184	0.7499998964
S	0.1311714439	0.4340848184	0.2500001036
S	0.6311710854	0.0659151829	0.2500001784
S	0.2299252396	0.6957448113	0.7500000095
S	0.7299253399	0.8042551482	0.7499999765
S	0.7700747604	0.6957448113	0.2499999905
S	0.2700746601	0.8042551482	0.2500000235

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Table 1 – continued from previous page

S	0.3319127684	0.5000000000	0.7500000809
S	0.6680872316	0.5000000000	0.2499999191
S	0.7299253399	0.1957448518	0.7499999765
S	0.2299252396	0.3042551887	0.7500000095
S	0.2700746601	0.1957448518	0.2500000235
S	0.7700747604	0.3042551887	0.2499999905
S	0.8319130941	0.0000000000	0.7500001369
S	0.1680869059	0.0000000000	0.2499998631

SYSTEM Parameters

System	
ntyp	3
nat	36
ibrav	0
celldm(1)	12.2044468
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

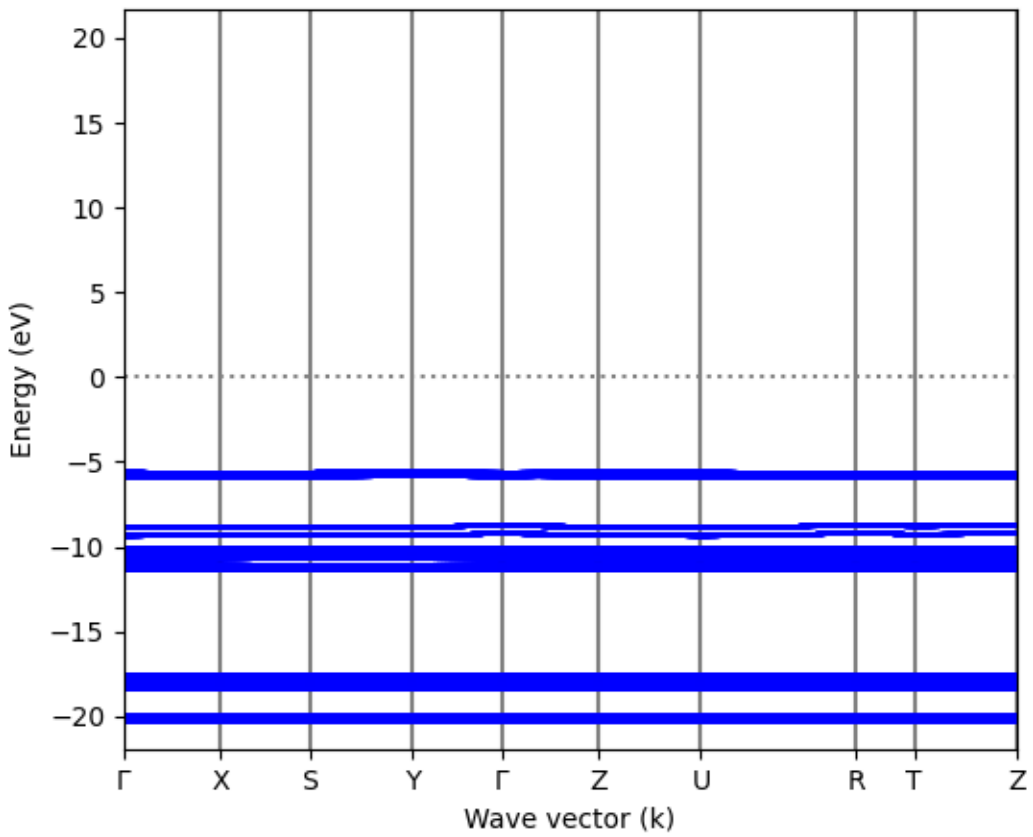
mp-29410: https://next-gen.materialsproject.org/materials/mp-29410?material_ids=mp-29410

CHAPTER

TWO

AS1LI1O4RB2

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
5.5936

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.985828700	0.000000000	0.000000000
0.000000000	1.986455961	-0.00000070
0.000000000	-0.00000047	1.337320991

ATOMIC_POSITIONS (crystal)

Rb	0.0000000000	0.0579301141	0.1505471165
Rb	0.5000000000	0.4420697840	0.6505473871
Rb	0.5000000000	0.1026654868	0.5052419232
Rb	0.0000000000	0.3973343953	0.0052416456
Rb	0.5000000000	0.5579301127	0.1505471465
Rb	0.0000000000	0.9420697702	0.6505473785
Rb	0.0000000000	0.6026655162	0.5052419818
Rb	0.5000000000	0.8973343617	0.0052415622
Li	0.5000000000	0.2951439594	0.2306032061
Li	0.0000000000	0.2048560832	0.7306030378
Li	0.0000000000	0.7951439809	0.2306031706
Li	0.5000000000	0.7048560616	0.7306030262
As	0.5000000000	0.1885832956	0.9436779321
As	0.0000000000	0.3114167599	0.4436779663
As	0.0000000000	0.6885832856	0.9436779120
As	0.5000000000	0.8114167561	0.4436780562
O	0.7427415705	0.1415047029	0.8380277287
O	0.7572583301	0.3584953421	0.3380279640
O	0.5000000000	0.1318939791	0.1477302518
O	0.0000000000	0.3681059944	0.6477304479
O	0.2572584295	0.1415047029	0.8380277287
O	0.2427416699	0.3584953421	0.3380279640
O	0.0000000000	0.1642028616	0.4752437712
O	0.5000000000	0.3357972207	0.9752438712
O	0.2427415853	0.6415046907	0.8380277033
O	0.2572583944	0.8584953403	0.3380280000
O	0.0000000000	0.6318939771	0.1477302269
O	0.5000000000	0.8681060487	0.6477305222

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Table 1 – continued from previous page

O	0.7572584147	0.6415046907	0.8380277033
O	0.7427416056	0.8584953403	0.3380280000
O	0.5000000000	0.6642028369	0.4752438032
O	0.0000000000	0.8357972063	0.9752438651

SYSTEM Parameters

System	
ntyp	4
nat	32
ibrav	0
celldm(1)	11.1701061
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

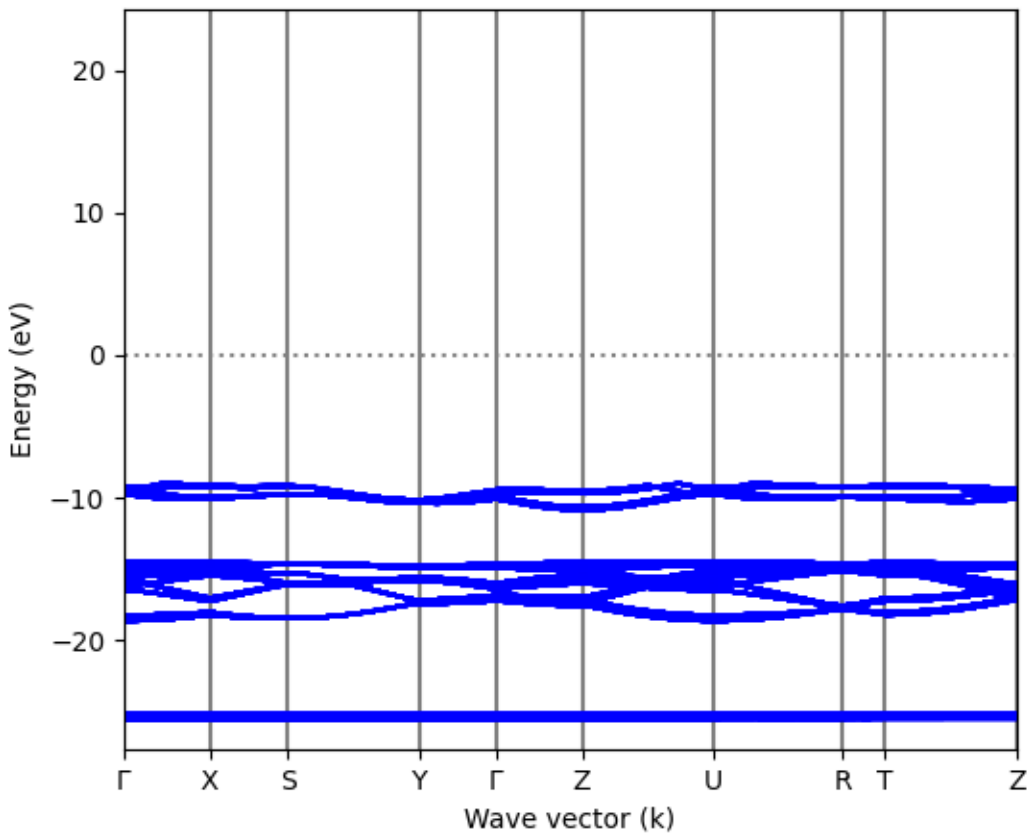
mp-14363: https://next-gen.materialsproject.org/materials/mp-14363?formula=As1Li1O4Rb2

CHAPTER

THREE

GE2LI1N3

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
9.8777

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

1.020938488	0.000000000	-0.00000023
0.000000000	1.765025137	0.000000000
-0.00000029	0.000000000	0.864877073

ATOMIC_POSITIONS (crystal)

Li	0.8383605963	0.5000000000	0.5157670895
Li	0.1616393304	0.5000000000	1.0157670815
Li	0.3383606534	0.0000000000	0.5157669911
Li	0.6616394393	0.0000000000	1.0157669468
Ge	0.3299941479	0.6617559171	0.5020812085
Ge	0.3299941479	0.3382440829	0.5020812085
Ge	0.6700058403	0.3382440844	1.0020812456
Ge	0.6700058403	0.6617559156	1.0020812456
Ge	0.8299939807	0.1617560564	0.5020812610
Ge	0.8299939807	0.8382439436	0.5020812610
Ge	0.1700060575	0.8382439672	1.0020812774
Ge	0.1700060575	0.1617560328	1.0020812774
N	0.8470093249	0.1808466631	0.1056641529
N	0.8470093249	0.8191533369	0.1056641529
N	0.1529906724	0.8191533344	0.6056641821
N	0.1529906724	0.1808466656	0.6056641821
N	0.8096130977	0.5000000000	0.0897401647
N	0.1903868419	0.5000000000	0.5897401471
N	0.3470096076	0.6808463025	0.1056643088
N	0.3470096076	0.3191536975	0.1056643088
N	0.6529903394	0.3191536975	0.6056643392
N	0.6529903394	0.6808463025	0.6056643392
N	0.3096128411	0.0000000000	0.0897397970
N	0.6903872584	0.0000000000	0.5897398311

SYSTEM Parameters

System	
ntyp	3
nat	24
ibrav	0
celldm(1)	10.4185629
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

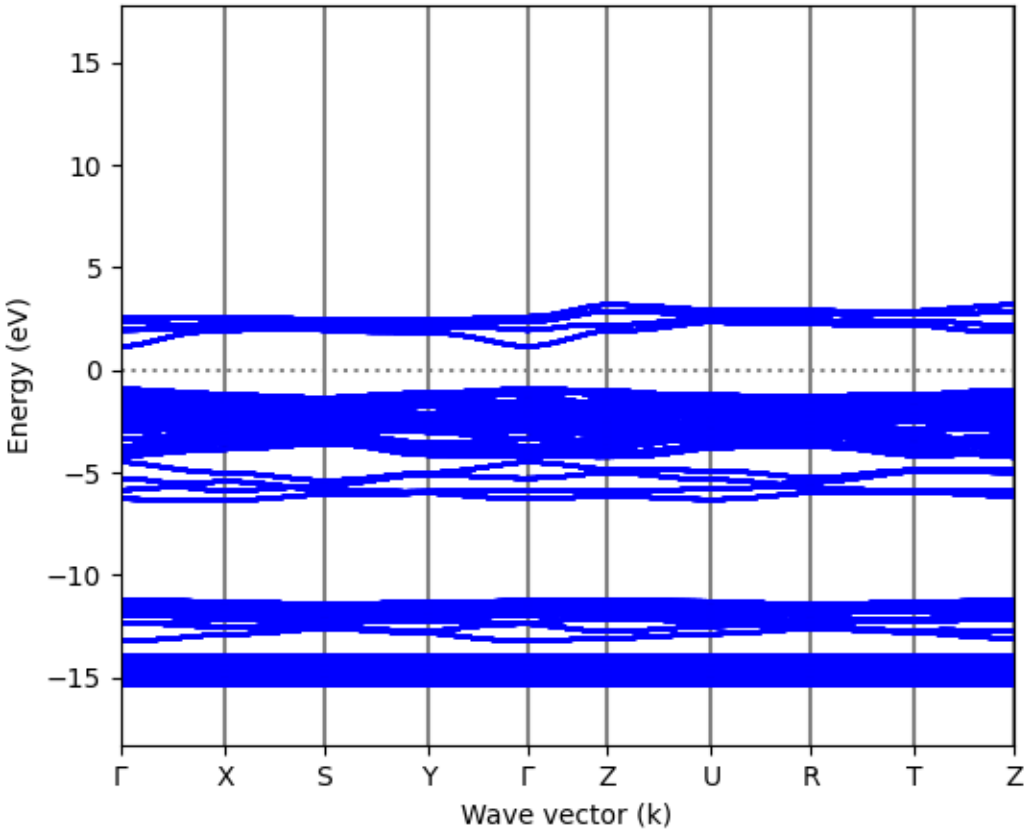
mp-1020059: https://next-gen.materialsproject.org/materials/mp-1020059?formula=Ge2Li1N3

CHAPTER

FOUR

IN1LI1S2

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
4.9700	2.0792	3.1196	3.2982	2.9422	2.0792	2.9281	3.5637	3.5938	3.4276	2.9281

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.982723615	0.000000000	0.000000000
0.000000000	1.034980419	0.000000000
0.000000000	0.000000000	1.292779148

ATOMIC_POSITIONS (crystal)

Li	0.5744092442	0.5763763013	0.1240848237
Li	0.0744092442	0.4236236987	0.8759151763
Li	0.0744092442	0.9236236987	0.6240848237
Li	0.5744092442	0.0763763013	0.3759151763
In	0.0637299588	0.9118412455	0.1246564576
In	0.5637299588	0.0881587545	0.8753435424
In	0.5637299588	0.5881587545	0.6246564576
In	0.0637299588	0.4118412455	0.3753435424
S	0.1924952076	0.0660052144	0.3730974488
S	0.6924952076	0.9339947856	0.6269025512
S	0.6924952076	0.4339947856	0.8730974488
S	0.1924952076	0.5660052144	0.1269025512
S	0.6765655894	0.9250617371	0.1240812279
S	0.1765655894	0.0749382629	0.8759187721
S	0.1765655894	0.5749382629	0.6240812279
S	0.6765655894	0.4250617371	0.3759187721

SYSTEM Parameters

System	
ntyp	3
nat	16
ibrav	0
celldm(1)	12.3218297
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

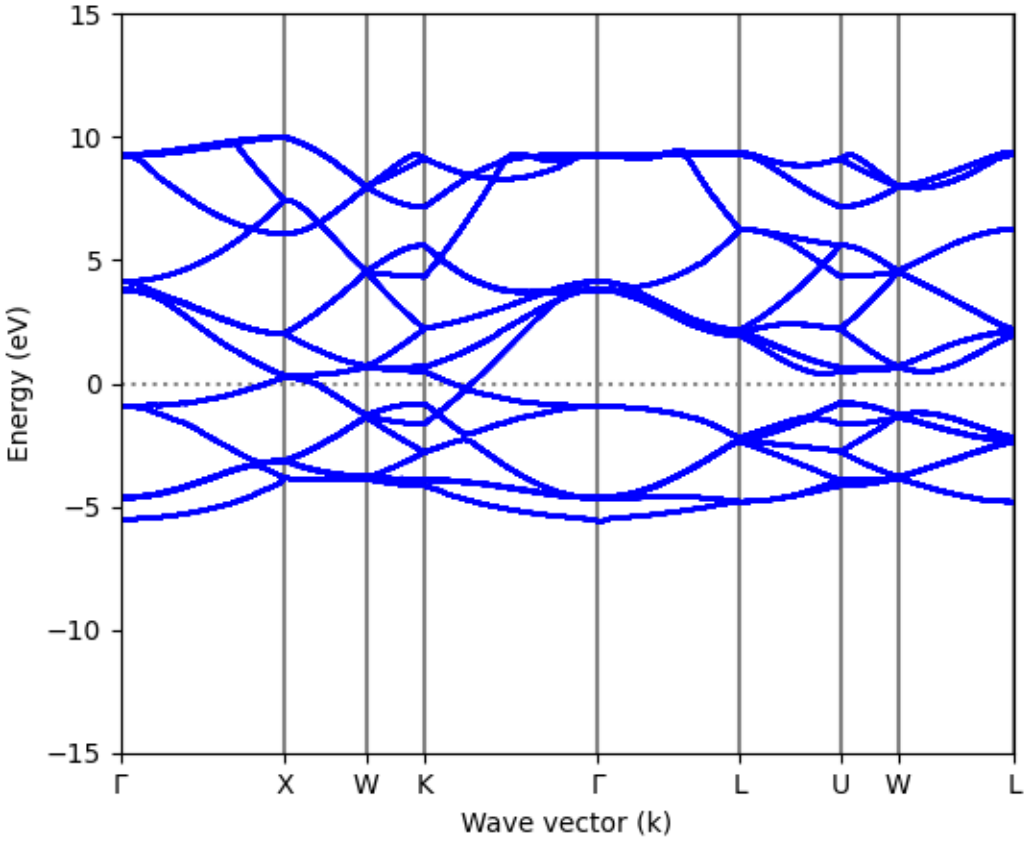
mp-1188392: https://next-gen.materialsproject.org/materials/mp-1188392?formula=In1Li1S2

CHAPTER

FIVE

LI4N2NA2

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	W	K	Γ	L	U	W2	L2
21.6453	4.7082	3.3604	1.8712	1.3149	4.7082	4.1279	1.2822	1.8712	4.1279

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.61514849	-0.00000052	0.615148496
0.000000520	0.615148496	0.615148496
-0.61514849	0.615148496	-0.00000052

ATOMIC_POSITIONS (crystal)

Na	-0.000000000	0.0000000000	-0.000000000
Na	0.0000000000	0.5000000000	0.5000000000
Na	0.5000000000	0.0000000000	0.5000000000
Na	0.5000000000	0.5000000000	-0.000000000
Li	0.2500000038	0.7499999997	0.7499999997
Li	0.2500000003	0.2500000003	0.7499999962
Li	0.2500000029	0.2500000029	0.2500000029
Li	0.2500000003	0.7499999962	0.2500000003
Li	0.7499999997	0.7499999997	0.2500000038
Li	0.7499999962	0.2500000003	0.2500000003
Li	0.7499999997	0.2500000038	0.7499999997
Li	0.7499999971	0.7499999971	0.7499999971
N	0.5000000000	0.0000000000	-0.000000000
N	0.5000000000	0.5000000000	0.5000000000
N	-0.000000000	-0.000000000	0.5000000000
N	0.0000000000	0.5000000000	-0.000000000

SYSTEM Parameters

System	
ntyp	3
nat	16
ibrav	0
celldm(1)	10.2355421
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

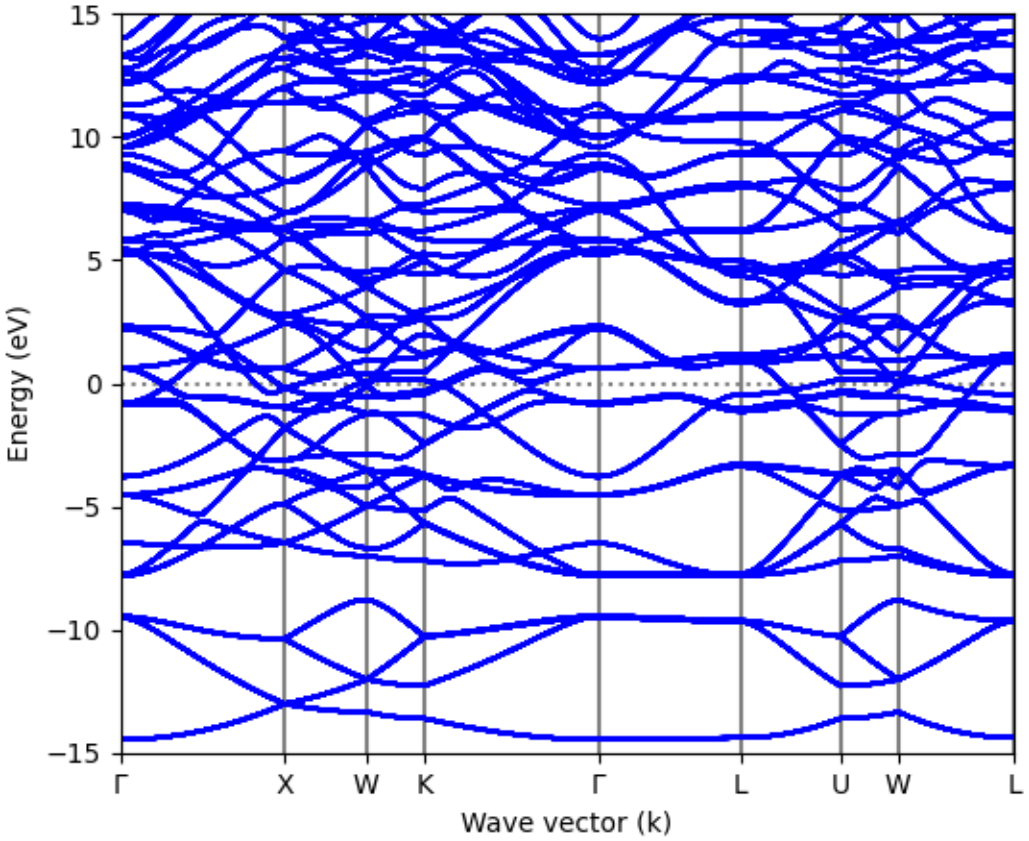
mp-10525: https://next-gen.materialsproject.org/materials/mp-10525?formula=Li4N2Na2

CHAPTER

SIX

BE1LI1SB1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	W	K	Γ	L	U	W2	L2
9.0759	1.3758	0.8080	0.1162	0.4464	1.3758	1.3445	0.4484	0.1162	1.3445

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.72243693	-0.00194602	0.722436930
0.001946023	0.722436930	0.722436930
-0.72243693	0.722436930	-0.00194602

ATOMIC_POSITIONS (crystal)

Li	0.0186220821	0.0186220821	0.0186220821
Li	0.0186220202	0.5186235148	0.5186235148
Li	0.5186235148	0.0186220202	0.5186235148
Li	0.5186235148	0.5186235148	0.0186220202
Be	0.2546181382	0.2546181382	0.2546181382
Be	0.2546228622	0.7546151514	0.7546151514
Be	0.7546151514	0.2546228622	0.7546151514
Be	0.7546151514	0.7546151514	0.2546228622
Sb	0.4767604136	-0.023240164	-0.023240164
Sb	0.4767574804	0.4767574804	0.4767574804
Sb	-0.023240164	-0.023240164	0.4767604136
Sb	-0.023240164	0.4767604136	-0.023240164

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	11.1551054
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

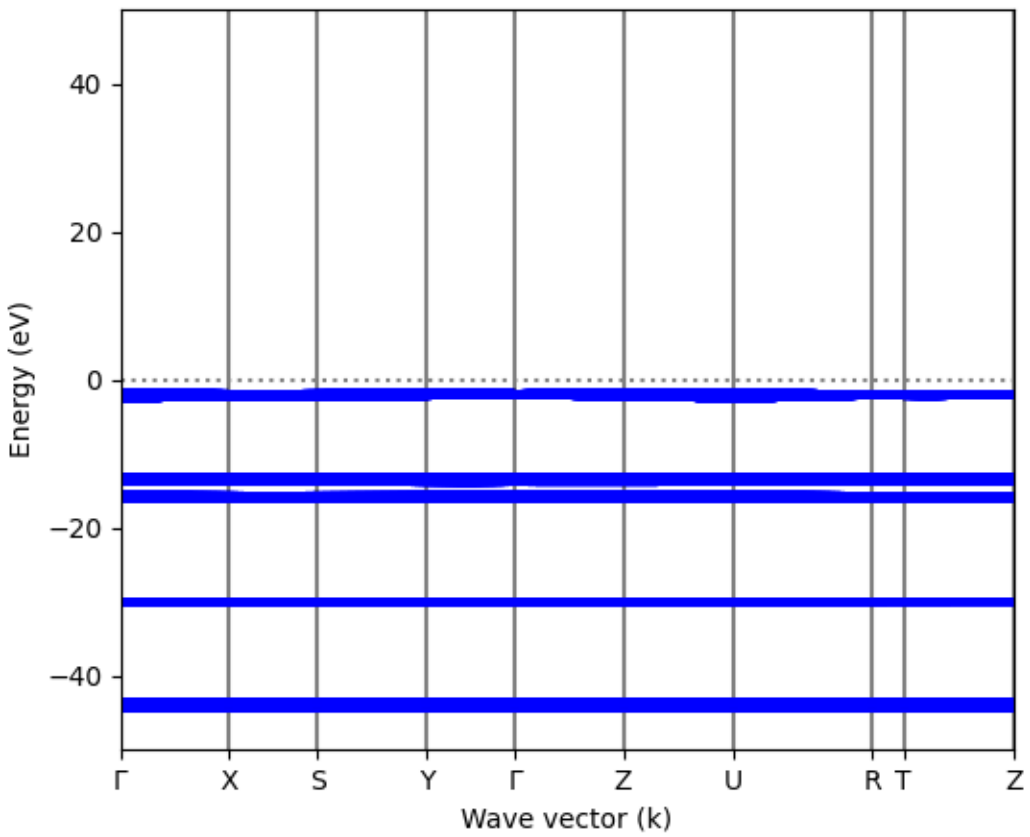
mp-1100398: https://next-gen.materialsproject.org/materials/mp-1100398?formula=Be1Li1Sb1

CHAPTER

SEVEN

K1LI1O1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
5.9212

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

ATOMIC_POSITIONS (crystal)

K	0.5	0.32239816	0.0
K	0.5	0.67760184	0.0
K	0.5	0.82239816	0.5
K	0.5	0.17760184	0.5
K	0.0	0.82239816	0.0
K	0.0	0.17760184	0.0
K	0.0	0.32239816	0.5
K	0.0	0.67760184	0.5
Li	0.12626311	0.0	0.37398296
Li	0.62626311	0.0	0.12601704
Li	0.37373689	0.0	0.8739829600
Li	0.87373689	0.0	0.62601704
Li	0.62626311	0.5	0.37398296
Li	0.12626311	0.5	0.12601704
Li	0.87373689	0.5	0.8739829600
Li	0.37373689	0.5	0.62601704
O	0.3311491	0.0	0.16734455
O	0.1688509	0.0	0.6673445500
O	0.6688509	0.0	0.83265545
O	0.8311491	0.0	0.33265545
O	0.8311491	0.5	0.16734455
O	0.6688509	0.5	0.6673445500
O	0.1688509	0.5	0.83265545
O	0.3311491	0.5	0.33265545

SYSTEM Parameters

System	
ntyp	3
nat	24
ibrav	8
celldm(1)	12.0324256
celldm(2)	1.34638996
celldm(3)	1.00462490
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0

Reference

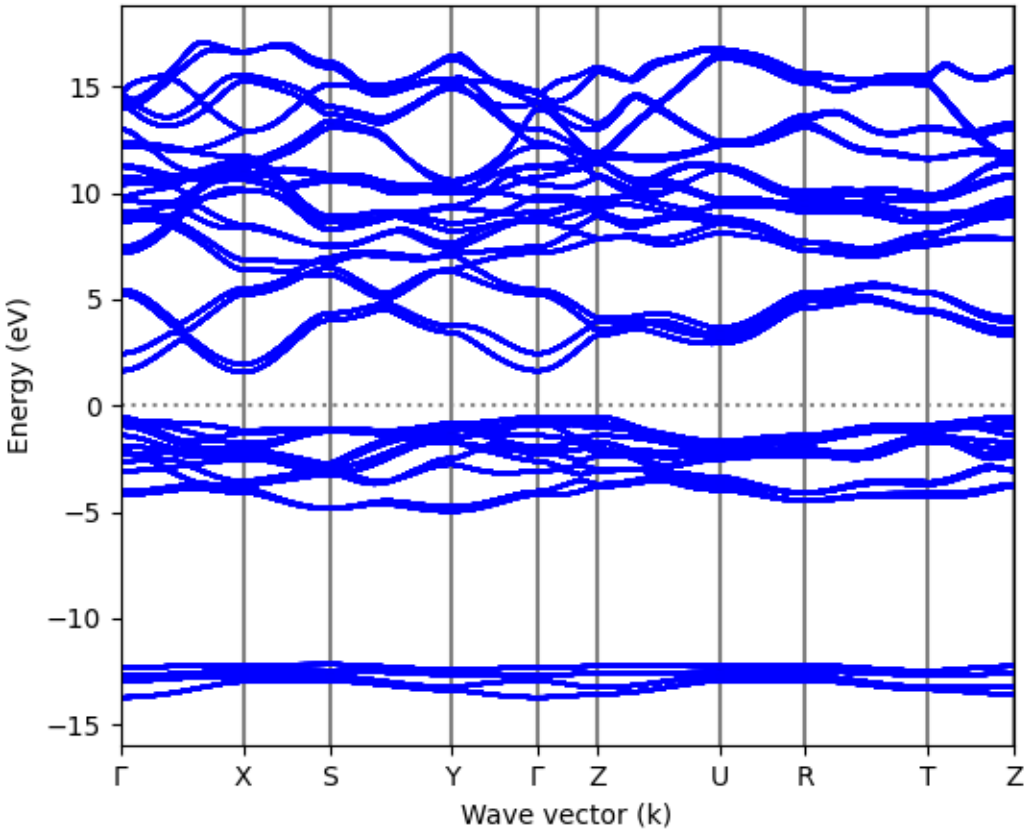
mp-559612: https://next-gen.materialsproject.org/materials/mp-559612?formula=K1Li1O1

CHAPTER

EIGHT

LI1MG1N1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
7.5820	2.1955	2.7387	5.0888	4.2804	2.1955	3.8275	4.6078	6.0173	5.2845	3.8275

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.965551792	0.000000000	0.000000000
0.000000000	1.373991533	0.000000000
0.000000000	0.000000000	1.999217869

ATOMIC_POSITIONS (crystal)

Li	0.2500000000	0.5191892272	0.0992841830
Li	0.7500000000	0.4808107728	0.9007158170
Li	0.7500000000	0.0191892272	0.4007158170
Li	0.2500000000	0.9808107728	0.5992841830
Mg	0.2500000000	0.4821437195	0.6434706983
Mg	0.7500000000	0.9821437195	0.8565293017
Mg	0.2500000000	0.0178562805	0.1434706983
Mg	0.7500000000	0.5178562805	0.3565293017
N	0.7500000000	0.7647382071	0.1196343399
N	0.7500000000	0.7352617929	0.6196343399
N	0.2500000000	0.2352617929	0.8803656601
N	0.2500000000	0.2647382071	0.3803656601

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	6.58062176
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

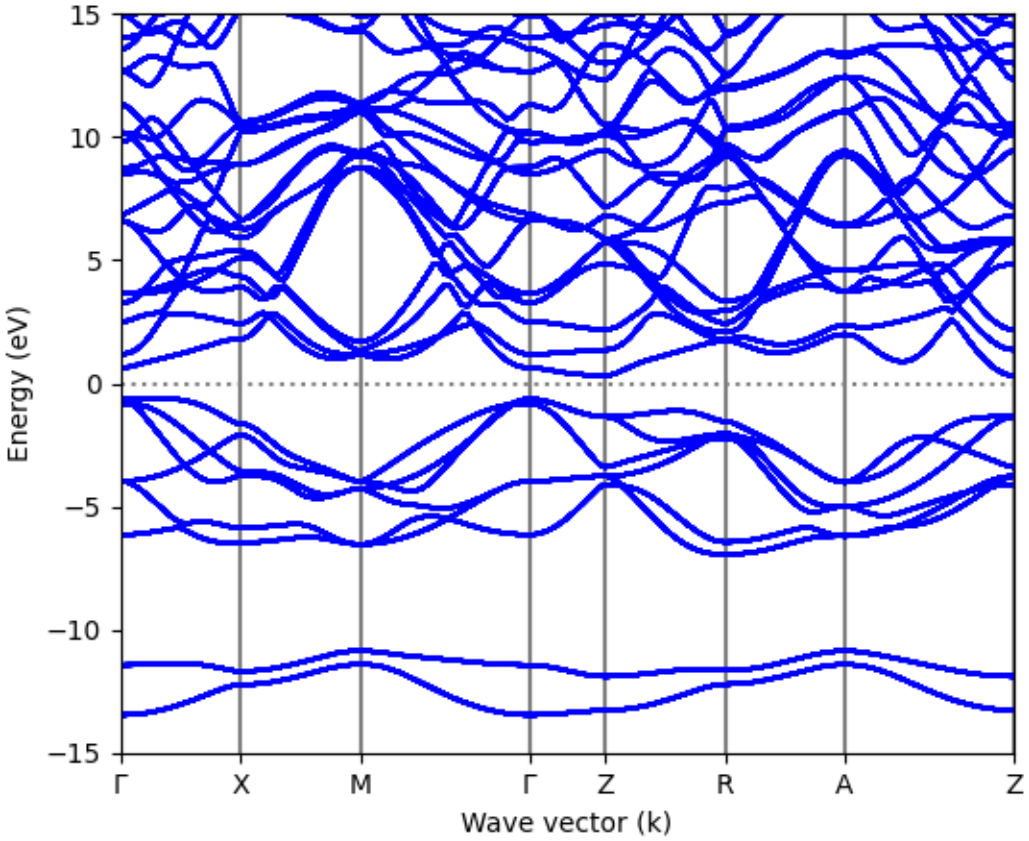
mp-30156: https://next-gen.materialsproject.org/materials/mp-30156?formula=Li1Mg1N1

CHAPTER

NINE

AS1BE1LI1

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
6.9878	1.2672	3.4474	5.1466	1.2672	1.6728	3.2738	5.9762	1.6728

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.996106514	0.000000000	0.000000000
0.000000000	0.996106514	0.000000000
0.000000000	0.000000000	1.566145375

ATOMIC_POSITIONS (crystal)

Li	0.5000000000	0.0000000000	0.3246577192
Li	0.0000000000	0.5000000000	0.6753422808
Be	0.5000000000	0.5000000000	-0.000000000
Be	0.0000000000	0.0000000000	0.0000000000
As	0.5000000000	0.0000000000	0.7661167482
As	0.0000000000	0.5000000000	0.2338832518

SYSTEM Parameters

System	
ntyp	3
nat	6
ibrav	0
celldm(1)	7.06109719
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

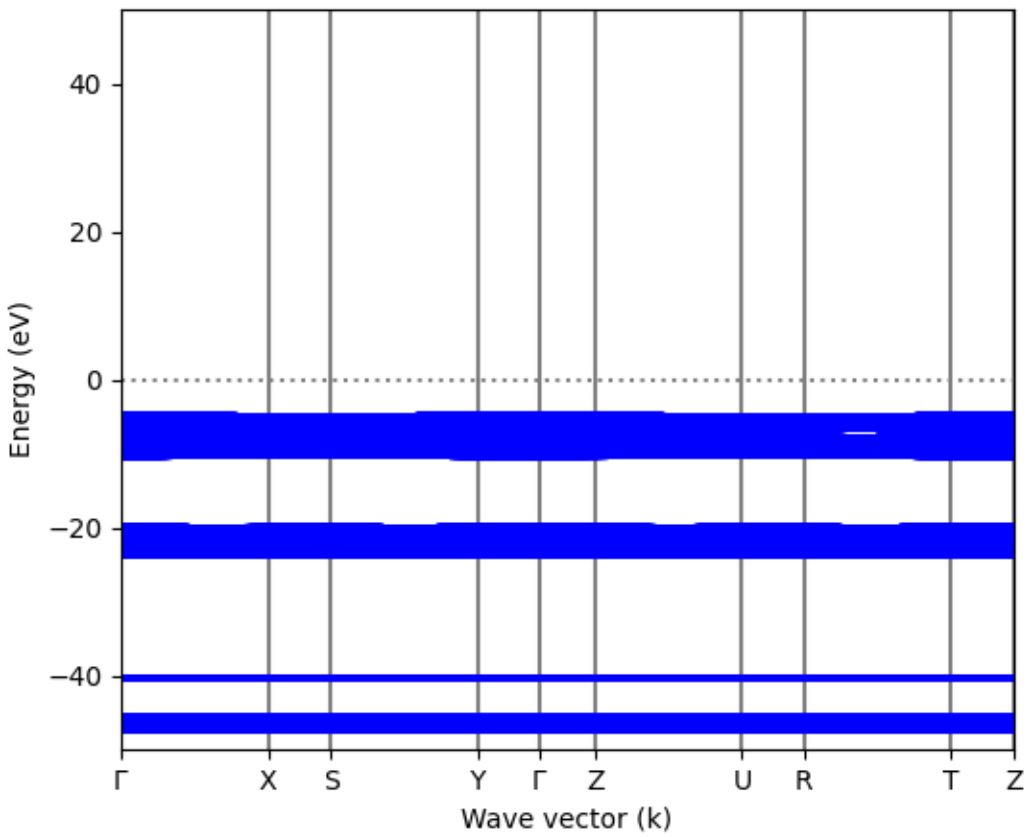
mp-9562: https://next-gen.materialsproject.org/materials/mp-9562?formula=As1Be1Li1

CHAPTER

TEN

B2CA1LI4O6

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
9.8923

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.919169094	0.000000000	0.000000000
0.000000000	2.161751086	0.000000000
0.000000000	0.000000000	2.446724489

ATOMIC_POSITIONS (crystal)

Li	0.0000000000	0.4839081233	0.1965258257
Li	-0.000000000	0.5160918767	0.8034741743
Li	0.5000000000	0.0160918767	0.6965258257
Li	0.5000000000	0.9839081233	0.3034741743
Li	0.5000000000	0.6675311428	0.9560354746
Li	0.5000000000	0.3324688572	0.0439645254
Li	-0.000000000	0.8324688572	0.4560354746
Li	-0.000000000	0.1675311428	0.5439645254
Ca	0.5000000000	0.5000000000	0.5000000000
Ca	0.0000000000	-0.000000000	-0.000000000
B	0.5000000000	0.7235727121	0.2150656058
B	0.5000000000	0.2764272879	0.7849343942
B	-0.000000000	0.7764272879	0.7150656058
B	0.0000000000	0.2235727121	0.2849343942
O	0.5000000000	0.7541550119	0.3719602323
O	0.5000000000	0.2458449881	0.6280397677
O	0.0000000000	0.7458449881	0.8719602323
O	0.0000000000	0.2541550119	0.1280397677
O	0.5000000000	0.5594863255	0.1507282342
O	0.5000000000	0.4405136745	0.8492717658
O	0.0000000000	0.9405136745	0.6507282342
O	-0.000000000	0.0594863255	0.3492717658
O	0.0000000000	0.3654885453	0.3817510373
O	0.0000000000	0.6345114547	0.6182489627
O	0.5000000000	0.1345114547	0.8817510373
O	0.5000000000	0.8654885453	0.1182489627

SYSTEM Parameters

System	
ntyp	4
nat	26
ibrav	0
celldm(1)	6.63075748
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

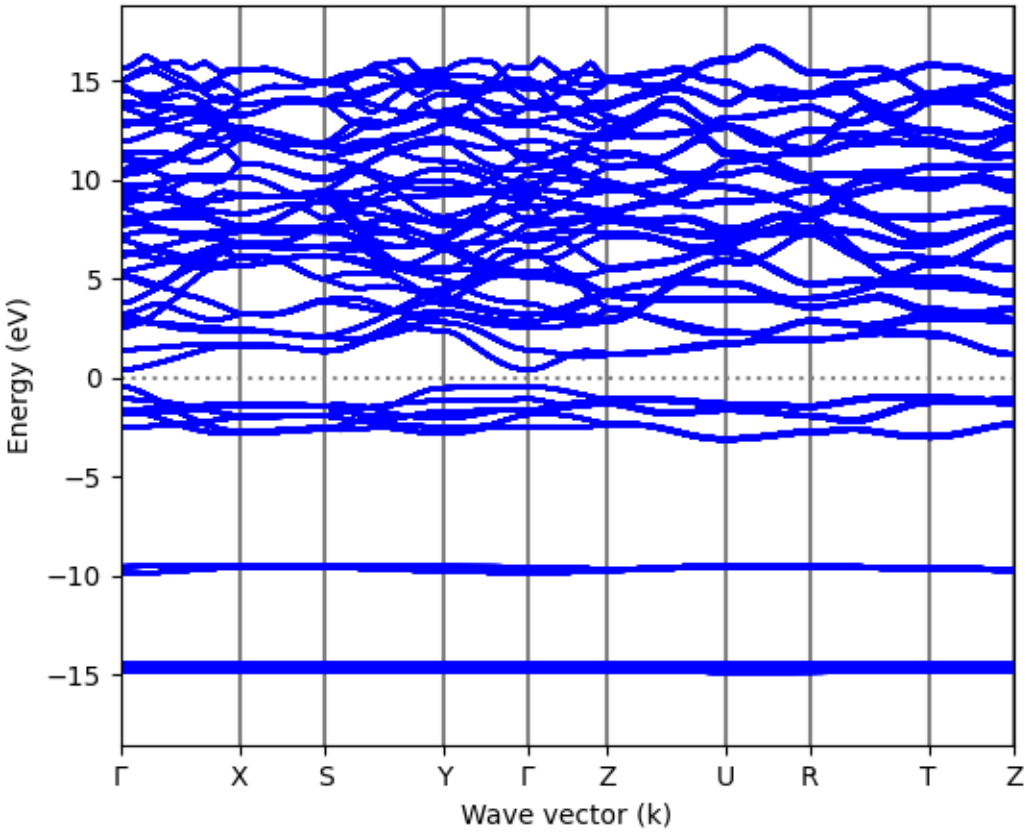
mp-557467: https://next-gen.materialsproject.org/materials/mp-557467?formula=B2Ca1Li4O6

CHAPTER

ELEVEN

AS1K1LI2

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
4.5135	0.8566	2.9539	2.7284	2.8764	0.8566	2.2276	2.9652	3.3030	3.0885	2.2276

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.977686260	0.000000000	0.000000000
0.000000000	1.360488749	0.000000000
0.000000000	0.000000000	1.486267815

ATOMIC_POSITIONS (crystal)

K	0.5000000000	0.0961160329	0.5000000000
K	-0.000000000	0.9038839671	-0.000000000
Li	0.0000000000	0.3981248983	0.1968448545
Li	0.5000000000	0.6018751017	0.3031551455
Li	0.5000000000	0.6018751017	0.6968448545
Li	-0.000000000	0.3981248983	0.8031551455
As	0.0000000000	0.6555333524	0.5000000000
As	0.5000000000	0.3444666476	0.0000000000

SYSTEM Parameters

System	
ntyp	3
nat	8
ibrav	0
celldm(1)	8.37553309
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

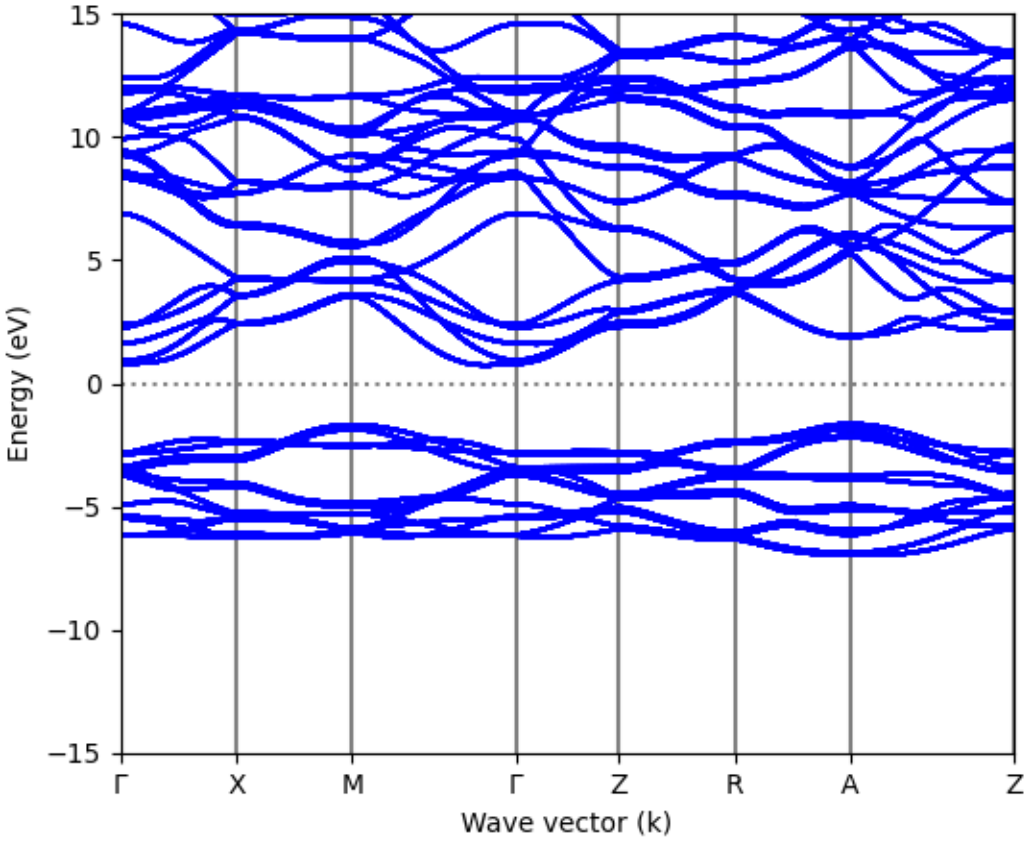
mp-28994: https://next-gen.materialsproject.org/materials/mp-28994?formula=As1K1Li2

CHAPTER

TWELVE

B1LI3N2

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
5.3556	3.6249	4.7371	5.1984	3.6249	5.0708	5.9837	3.5215	5.0708

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.981664419	0.000000000	0.000000000
0.000000000	0.981664419	0.000000000
0.000000000	0.000000000	1.100029010

ATOMIC_POSITIONS (crystal)

Li	0.5000000000	0.0000000000	0.2500000000
Li	0.5000000000	0.0000000000	0.7500000000
Li	0.5000000000	0.5000000000	0.0000000000
Li	0.0000000000	0.0000000000	0.5000000000
Li	0.0000000000	0.5000000000	0.2500000000
Li	0.0000000000	0.5000000000	0.7500000000
B	0.5000000000	0.5000000000	0.5000000000
B	0.0000000000	0.0000000000	0.0000000000
N	0.7910773663	0.7910773663	0.0000000000
N	0.7089226337	0.2910773663	0.5000000000
N	0.2910773663	0.7089226337	0.5000000000
N	0.2089226337	0.2089226337	0.0000000000

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	8.72223677
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	30.0
ecutrho	40.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

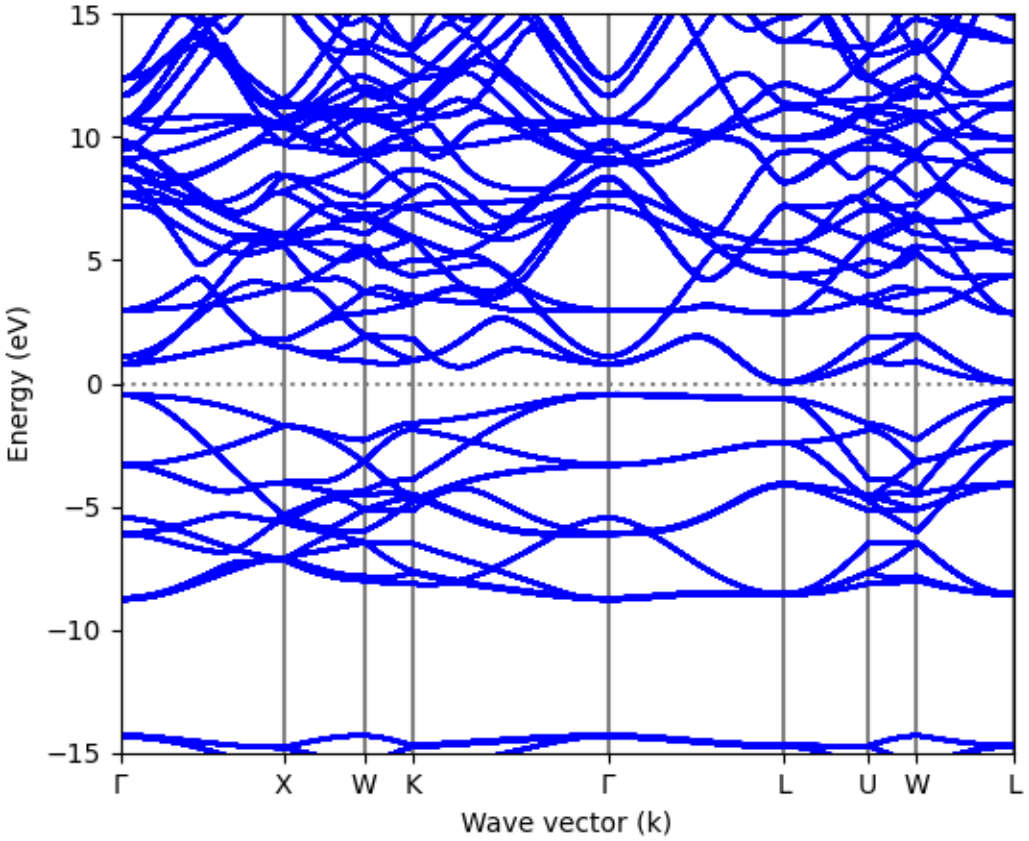
mp-8926: https://next-gen.materialsproject.org/materials/mp-8926?formula=B1Li3N2

CHAPTER

THIRTEEN

I1LI1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	W	K	Γ	L	U	W2	L2
10.9249	1.1924	3.2324	3.1989	2.5503	1.1924	0.6406	2.5676	3.1989	0.6406

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.60641984	-0.16607236	0.606419840
0.166072364	0.606419840	0.606419840
-0.60641984	0.606419840	-0.16607236

ATOMIC_POSITIONS (crystal)

Li	-0.000000000	0.0000000000	0.0000000000
Li	0.0000000000	0.5000000000	0.5000000000
Li	0.5000000000	0.0000000000	0.5000000000
Li	0.5000000000	0.5000000000	-0.000000000
I	0.0000000000	0.5000000000	0.0000000000
I	0.0000000000	-0.000000000	0.5000000000
I	0.5000000000	0.5000000000	0.5000000000
I	0.5000000000	0.0000000000	0.0000000000

SYSTEM Parameters

System	
ntyp	2
nat	8
ibrav	0
celldm(1)	11.2785552
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

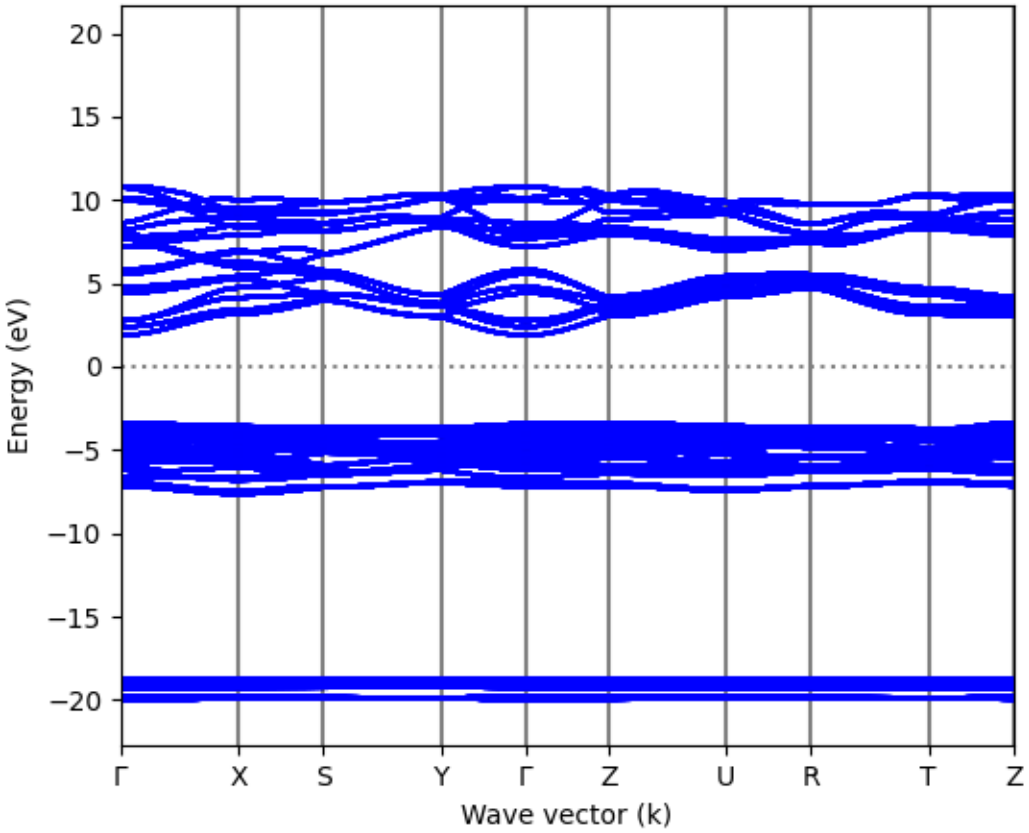
mp-22899: https://next-gen.materialsproject.org/materials/mp-22899?formula=I1Li1

CHAPTER

FOURTEEN

AL1LI5O4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
6.6881	5.2921	6.7268	7.5878	6.6792	5.2921	6.3987	7.7190	8.3101	6.8345	6.3987

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.978698732	0.000000000	0.000000000
0.000000000	1.342439932	0.000000000
0.000000000	0.000000000	1.388040617

ATOMIC_POSITIONS (crystal)

Li	0.7443489864	0.2392602874	0.1985240463
Li	0.7443489864	0.7607397126	0.8014759537
Li	0.2556510136	0.7392602874	0.3014759537
Li	0.2556510136	0.2607397126	0.6985240463
Li	0.2556510136	0.2607397126	0.3014759537
Li	0.2556510136	0.7392602874	0.6985240463
Li	0.7443489864	0.7607397126	0.1985240463
Li	0.7443489864	0.2392602874	0.8014759537
Li	0.8029993817	-0.000000000	0.5000000000
Li	0.1970006183	0.5000000000	0.0000000000
Al	0.2242376422	-0.000000000	-0.000000000
Al	0.7757623578	0.5000000000	0.5000000000
O	0.0126260875	-0.000000000	0.2350053851
O	0.0126260875	0.0000000000	0.7649946149
O	0.5655956383	0.7415928976	0.5000000000
O	0.5655956383	0.2584071024	0.5000000000
O	0.4344043617	0.2415928976	0.0000000000
O	0.4344043617	0.7584071024	0.0000000000
O	0.9873739125	0.5000000000	0.7350053851
O	0.9873739125	0.5000000000	0.2649946149

SYSTEM Parameters

System	
ntyp	3
nat	20
ibrav	0
celldm(1)	8.64285862
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

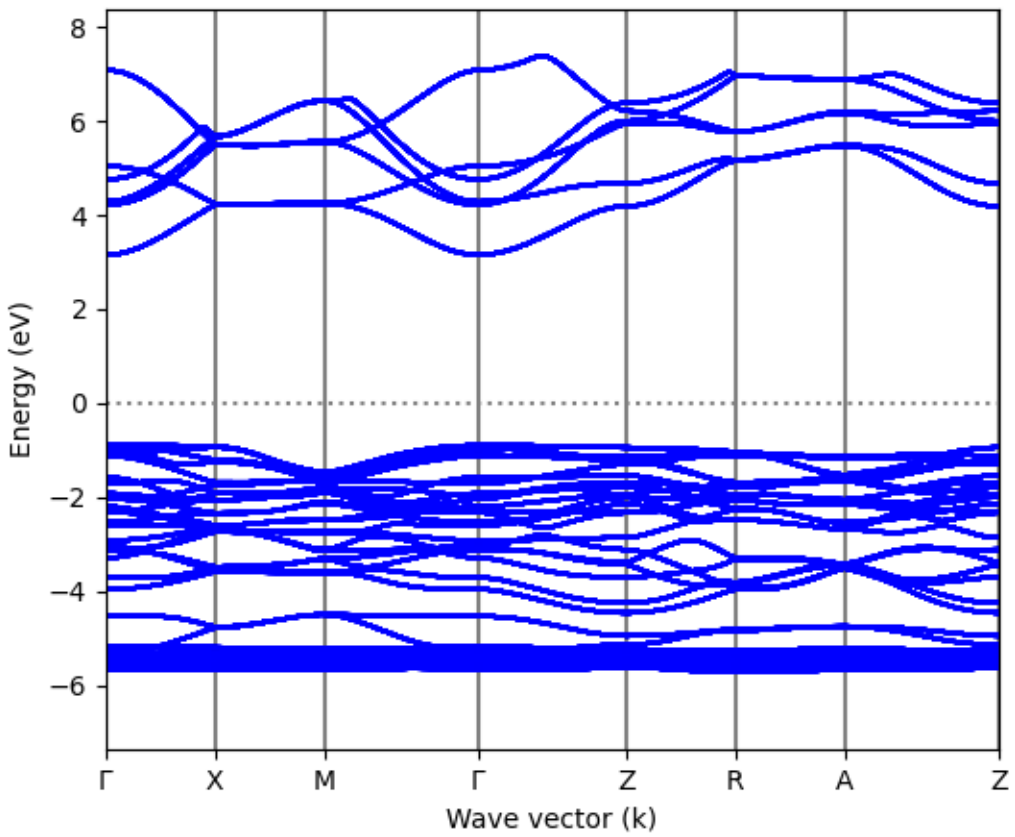
mp-7535: https://next-gen.materialsproject.org/materials/mp-7535?formula=Al1Li5O4

CHAPTER

FIFTEEN

LI6O4ZN1

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
5.3584	4.0487	5.1484	5.7070	4.0487	5.1287	6.1948	6.6179	5.1287

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.972009955	0.000000000	0.000000000
0.000000000	0.972009955	0.000000000
0.000000000	0.000000000	0.716460346

ATOMIC_POSITIONS (crystal)

Li	0.500000000	0.000000000	0.461416624
Li	0.000000000	0.500000000	0.038583376
Li	0.716452188	0.283547812	0.500000000
Li	0.216452188	0.216452188	-0.000000000
Li	0.783547812	0.783547812	0.000000000
Li	0.283547812	0.716452188	0.500000000
Li	0.783547812	0.216452188	0.000000000
Li	0.283547812	0.283547812	0.500000000
Li	0.716452188	0.716452188	0.500000000
Li	0.000000000	0.500000000	0.538583376
Li	0.500000000	0.000000000	0.961416624
Li	0.216452188	0.783547812	0.000000000
Zn	0.500000000	0.500000000	-0.000000000
Zn	0.000000000	0.000000000	0.500000000
O	0.000000000	0.734250120	0.278602775
O	0.234250120	0.500000000	0.778602775
O	0.500000000	0.234250120	0.221397225
O	0.265749880	-0.000000000	0.721397225
O	0.734250120	-0.000000000	0.721397225
O	0.765749880	0.500000000	0.778602775
O	-0.000000000	0.265749880	0.278602775
O	0.500000000	0.765749880	0.221397225

SYSTEM Parameters

System	
ntyp	3
nat	22
ibrav	0
celldm(1)	12.2436094
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

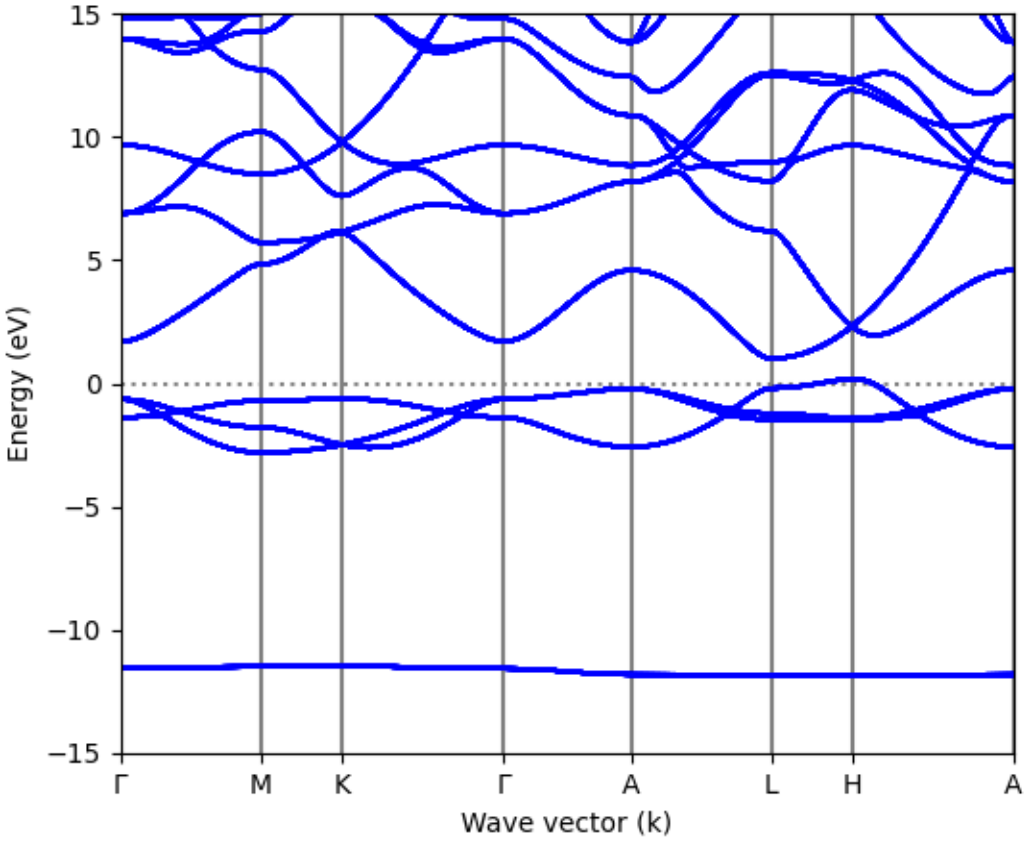
mp-14495: https://next-gen.materialsproject.org/materials/mp-14495?formula=Li6O4Zn1

CHAPTER

SIXTEEN

LI3N1

Band gap Diagram



Bravais lattice type

HexagonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	M	K	Γ	A	L	H	A2
3.3812	2.3421	5.5400	6.7534	2.3421	4.8060	1.2062	1.6279	4.8060

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.985487367	0.000000000	0.000000000
-0.49274368	0.853457095	-0.00000000
0.000000000	-0.00000000	0.940743887

ATOMIC_POSITIONS (crystal)

Li	0.000000000	-0.000000000	0.500000000
Li	0.333333330	0.666666670	-0.000000000
Li	0.666666670	0.333333330	0.000000000
N	0.000000000	0.000000000	0.000000000

SYSTEM Parameters

System	
ntyp	2
nat	4
ibrav	0
celldm(1)	6.82139567
celldm(3)	1.06784756
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

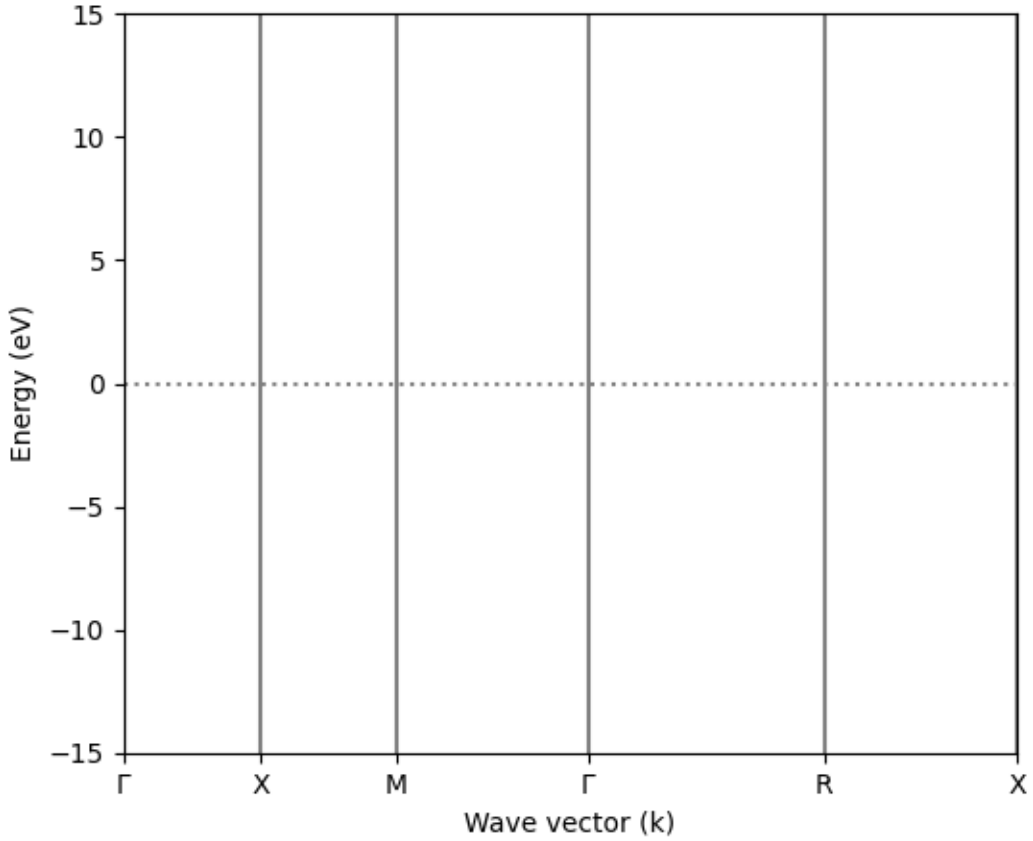
mp-2251: https://next-gen.materialsproject.org/materials/mp-2251?formula=Li3N1

CHAPTER

SEVENTEEN

LI7N4P1

Band gap Diagram



Bravais lattice type

CubicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
7.8542

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.980592596	0.000000000	0.000000000
0.000000000	0.980592596	0.000000000
0.000000000	0.000000000	0.980592596

ATOMIC_POSITIONS (crystal)

Li	0.000000000	0.500000000	0.500000000
Li	0.500000000	0.000000000	0.000000000
Li	0.000000000	0.000000000	0.500000000
Li	0.000000000	0.500000000	0.000000000
Li	0.500000000	0.000000000	0.500000000
Li	0.500000000	0.500000000	0.000000000
Li	0.250000000	-0.000000000	0.500000000
Li	0.000000000	0.500000000	0.750000000
Li	0.000000000	0.500000000	0.250000000
Li	0.500000000	0.250000000	0.000000000
Li	0.500000000	0.750000000	0.000000000
Li	0.750000000	-0.000000000	0.500000000
Li	0.228753188	0.228753188	0.228753188
Li	0.271246812	0.728753188	0.271246812
Li	0.271246812	0.271246812	0.728753188
Li	0.728753188	0.271246812	0.271246812
Li	0.771246812	0.771246812	0.228753188
Li	0.228753188	0.771246812	0.771246812
Li	0.771246812	0.228753188	0.771246812
Li	0.728753188	0.728753188	0.728753188
Li	0.252689117	0.234838909	0.978161796
Li	0.247310883	0.478161796	0.265161091
Li	0.247310883	0.521838204	0.734838909
Li	0.521838204	0.265161091	0.752689117
Li	0.478161796	0.265161091	0.247310883
Li	0.734838909	0.247310883	0.521838204
Li	0.265161091	0.752689117	0.521838204
Li	0.978161796	0.252689117	0.234838909

continues on next page

Table 1 – continued from previous page

Li	0.234838909	0.978161796	0.252689117
Li	0.765161091	0.021838204	0.252689117
Li	0.978161796	0.747310883	0.765161091
Li	0.765161091	0.978161796	0.747310883
Li	0.021838204	0.747310883	0.234838909
Li	0.021838204	0.252689117	0.765161091
Li	0.234838909	0.021838204	0.747310883
Li	0.252689117	0.765161091	0.021838204
Li	0.747310883	0.234838909	0.021838204
Li	0.747310883	0.765161091	0.978161796
Li	0.265161091	0.247310883	0.478161796
Li	0.734838909	0.752689117	0.478161796
Li	0.521838204	0.734838909	0.247310883
Li	0.478161796	0.734838909	0.752689117
Li	0.752689117	0.521838204	0.265161091
Li	0.752689117	0.478161796	0.734838909
Li	0.265372715	0.000000000	0.000000000
Li	0.234627285	0.500000000	0.500000000
Li	0.500000000	0.500000000	0.765372715
Li	0.500000000	0.500000000	0.234627285
Li	0.500000000	0.234627285	0.500000000
Li	0.500000000	0.765372715	0.500000000
Li	-0.000000000	0.265372715	-0.000000000
Li	-0.000000000	0.000000000	0.265372715
Li	-0.000000000	0.734627285	0.000000000
Li	0.000000000	0.000000000	0.734627285
Li	0.734627285	0.000000000	0.000000000
Li	0.765372715	0.500000000	0.500000000
P	0.500000000	0.500000000	0.500000000
P	0.000000000	0.000000000	0.000000000
P	0.500000000	0.000000000	0.750000000
P	0.750000000	0.500000000	0.000000000
P	0.250000000	0.500000000	0.000000000
P	0.000000000	0.250000000	0.500000000
P	0.000000000	0.750000000	0.500000000
P	0.500000000	0.000000000	0.250000000
N	0.605960983	0.605960983	0.605960983
N	0.894039017	0.105960983	0.894039017
N	0.105960983	0.894039017	0.894039017
N	0.894039017	0.894039017	0.105960983
N	0.605960983	0.394039017	0.394039017
N	0.394039017	0.394039017	0.605960983
N	0.394039017	0.605960983	0.394039017
N	0.105960983	0.105960983	0.105960983
N	0.352813678	0.385478589	0.102685787
N	0.147186322	0.602685787	0.114521411
N	0.147186322	0.397314213	0.885478589
N	0.397314213	0.114521411	0.852813678
N	0.602685787	0.114521411	0.147186322
N	0.885478589	0.147186322	0.397314213
N	0.114521411	0.852813678	0.397314213
N	0.102685787	0.352813678	0.385478589

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Table 1 – continued from previous page

N	0.385478589	0.102685787	0.352813678
N	0.614521411	0.897314213	0.352813678
N	0.102685787	0.647186322	0.614521411
N	0.614521411	0.102685787	0.647186322
N	0.897314213	0.647186322	0.385478589
N	0.897314213	0.352813678	0.614521411
N	0.385478589	0.897314213	0.647186322
N	0.352813678	0.614521411	0.897314213
N	0.647186322	0.385478589	0.897314213
N	0.647186322	0.614521411	0.102685787
N	0.114521411	0.147186322	0.602685787
N	0.885478589	0.852813678	0.602685787
N	0.397314213	0.885478589	0.147186322
N	0.602685787	0.885478589	0.852813678
N	0.852813678	0.397314213	0.114521411
N	0.852813678	0.602685787	0.885478589

SYSTEM Parameters

System	
ntyp	3
nat	96
ibrav	0
celldm(1)	17.5515971
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

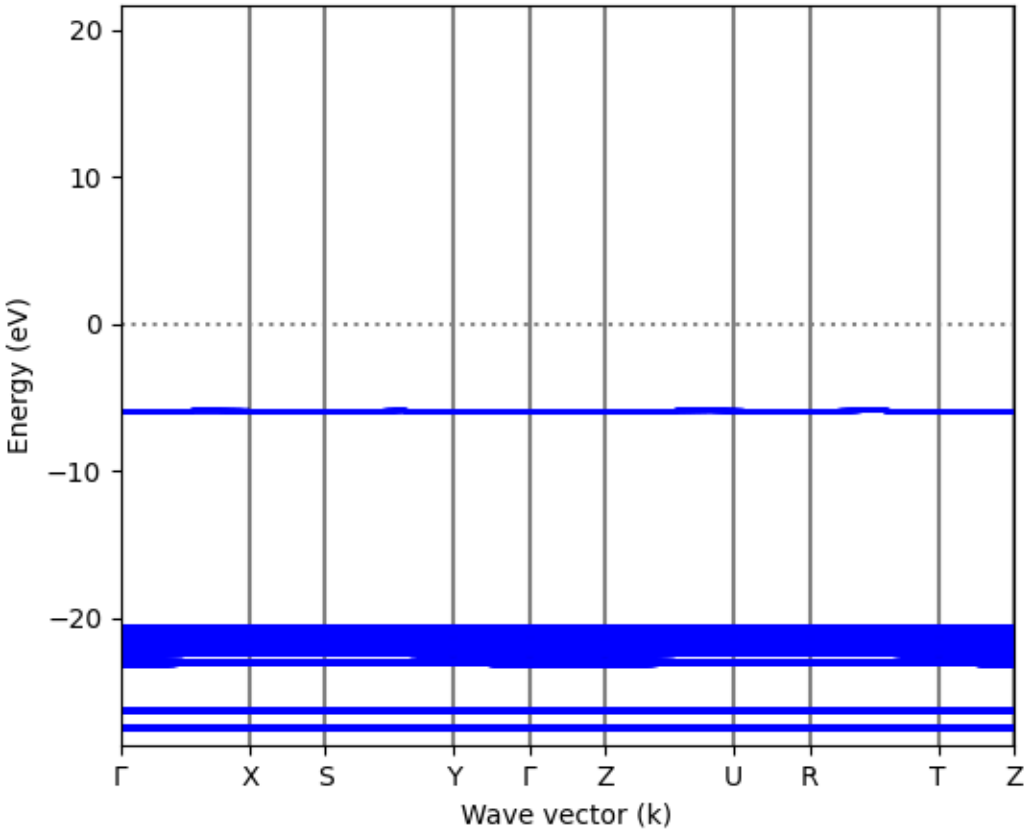
mp-14712: https://next-gen.materialsproject.org/materials/mp-14712?formula=Li7N4P1

CHAPTER

EIGHTEEN

F8LI4ZR1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
4.4440

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.989621684	0.000000000	0.000000000
0.000000000	1.681594491	0.000000000
0.000000000	0.000000000	1.695171377

ATOMIC_POSITIONS (crystal)

Li	0.118351514	0.370714585	0.058554045
Li	0.618351514	0.129285415	0.941445955
Li	0.381648486	0.870714585	0.441445955
Li	0.881648486	0.629285415	0.558554045
Li	0.881648486	0.629285415	0.941445955
Li	0.381648486	0.870714585	0.058554045
Li	0.618351514	0.129285415	0.558554045
Li	0.118351514	0.370714585	0.441445955
Li	0.629085621	0.395317956	0.079338649
Li	0.129085621	0.104682044	0.920661351
Li	0.870914379	0.895317956	0.420661351
Li	0.370914379	0.604682044	0.579338649
Li	0.370914379	0.604682044	0.920661351
Li	0.870914379	0.895317956	0.079338649
Li	0.129085621	0.104682044	0.579338649
Li	0.629085621	0.395317956	0.420661351
Zr	0.370944326	0.135867100	0.250000000
Zr	0.629055674	0.864132900	0.750000000
Zr	0.129055674	0.635867100	0.250000000
Zr	0.870944326	0.364132900	0.750000000
F	0.859155148	0.979055204	0.616126792
F	0.372706087	0.709454523	0.750000000
F	0.872706087	0.790545477	0.250000000
F	0.127293913	0.209454523	0.750000000
F	0.140844852	0.020944796	0.116126792
F	0.640844852	0.479055204	0.883873208
F	0.359155148	0.520944796	0.383873208
F	0.640844852	0.479055204	0.616126792

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Table 1 – continued from previous page

F	0.859155148	0.979055204	0.883873208
F	0.359155148	0.520944796	0.116126792
F	0.108281448	0.477005619	0.619972118
F	0.140844852	0.020944796	0.383873208
F	0.608281448	0.022994381	0.119972118
F	0.108281448	0.477005619	0.880027882
F	0.891718552	0.522994381	0.380027882
F	0.391718552	0.977005619	0.619972118
F	0.391718552	0.977005619	0.880027882
F	0.891718552	0.522994381	0.119972118
F	0.883649970	0.708674338	0.750000000
F	0.608281448	0.022994381	0.380027882
F	0.373198625	0.237527646	0.466196395
F	0.873198625	0.262472354	0.533803605
F	0.126801375	0.737527646	0.033803605
F	0.626801375	0.762472354	0.966196395
F	0.626801375	0.762472354	0.533803605
F	0.126801375	0.737527646	0.466196395
F	0.873198625	0.262472354	0.966196395
F	0.373198625	0.237527646	0.033803605
F	0.116350030	0.291325662	0.250000000
F	0.616350030	0.208674338	0.750000000
F	0.383649970	0.791325662	0.250000000
F	0.627293913	0.290545477	0.250000000

SYSTEM Parameters

System	
ntyp	3
nat	52
ibrav	0
celldm(1)	10.5996864
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

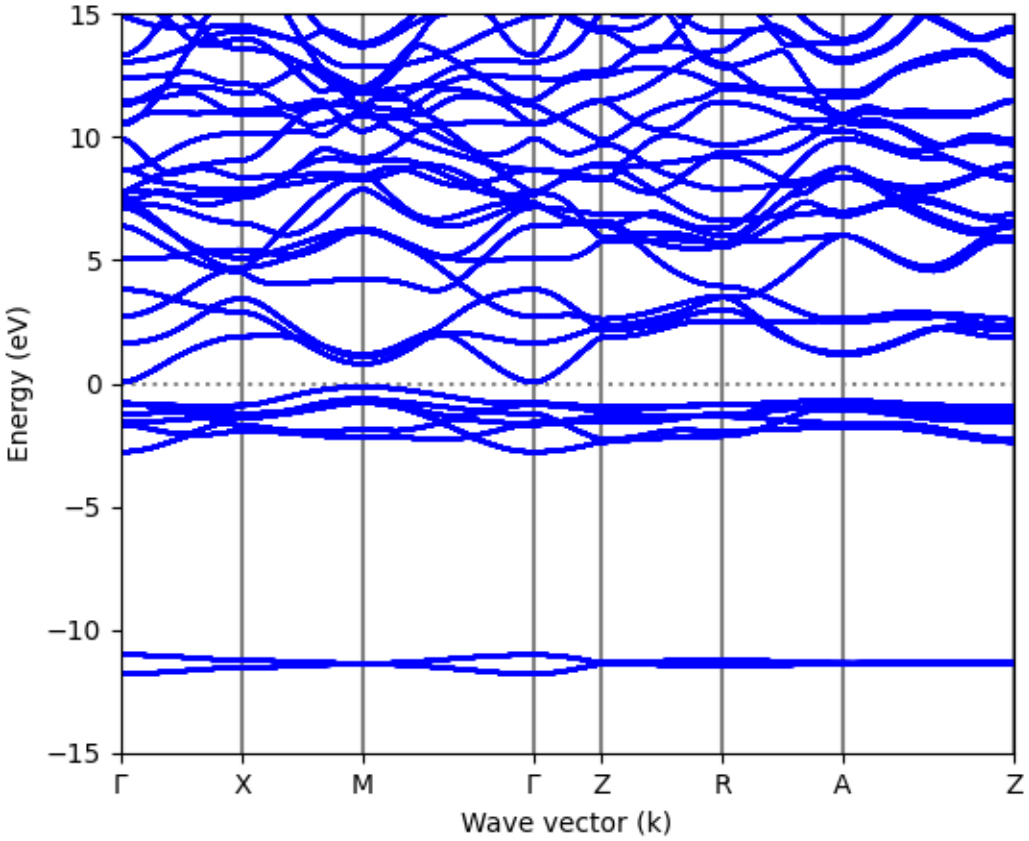
mp-9308: https://next-gen.materialsproject.org/materials/mp-9308?formula=F8Li4Zr1

CHAPTER

NINETEEN

LI1N1SR1

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
6.7392	0.8559	2.8002	0.9294	0.8559	2.7913	3.3598	1.8423	2.7913

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.993719653	0.000000000	0.000000000
0.000000000	0.993719653	0.000000000
0.000000000	0.000000000	1.760862932

ATOMIC_POSITIONS (crystal)

Sr	0.000000000	0.000000000	0.750000000
Sr	0.000000000	0.000000000	0.250000000
Li	0.500000000	0.500000000	0.500000000
Li	0.500000000	0.500000000	0.000000000
N	0.000000000	0.500000000	0.000000000
N	0.500000000	0.000000000	0.500000000

SYSTEM Parameters

System	
ntyp	3
nat	6
ibrav	0
celldm(1)	7.40996701
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

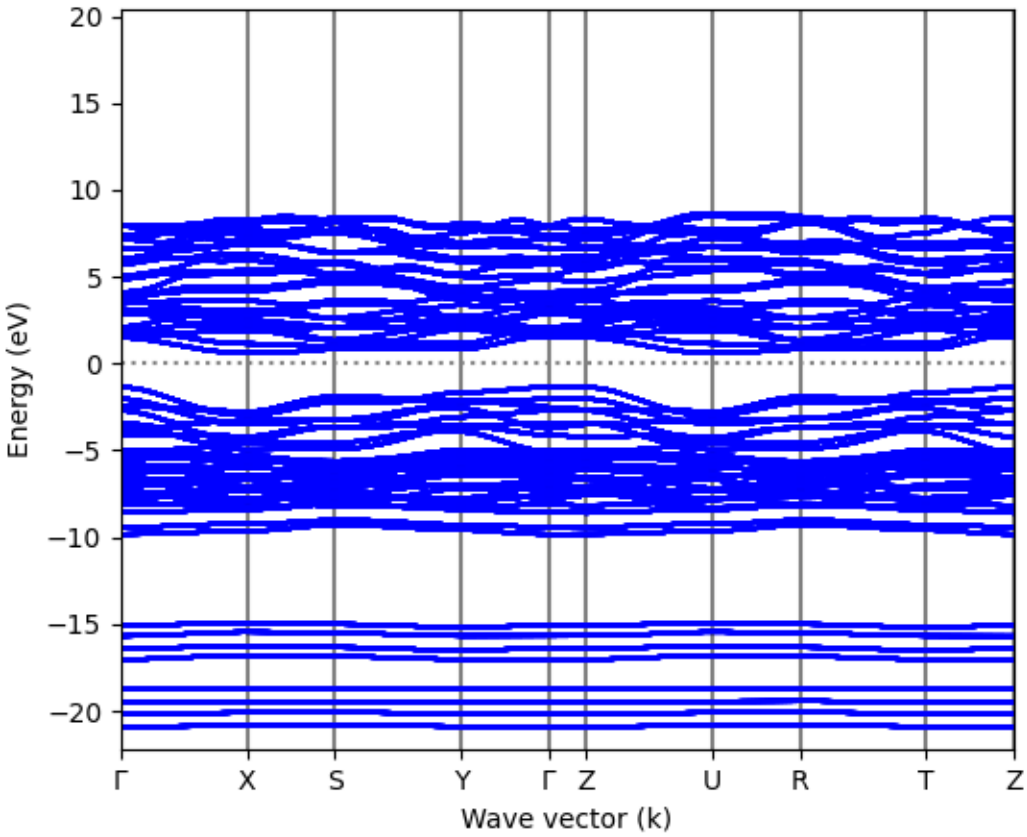
mp-29754: https://next-gen.materialsproject.org/materials/mp-29754?formula=Li1N1Sr1

CHAPTER

TWENTY

C1LI1N1S1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
5.0445	2.8571	3.3761	2.7358	2.5066	2.8571	2.8596	3.3711	2.7634	2.5060	2.8596

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.948477215	0.000000000	0.000000000
0.000000000	1.356522807	0.000000000
0.000000000	0.000000000	3.148826685

ATOMIC_POSITIONS (crystal)

Li	0.2500000000	0.4600667622	0.9049616078
Li	0.2500000000	0.0399332378	0.4049616078
Li	0.7500000000	0.5399332378	0.0950383922
Li	0.7500000000	0.9600667622	0.5950383922
C	0.2500000000	0.3920372707	0.6114623975
C	0.2500000000	0.1079627293	0.1114623975
C	0.7500000000	0.6079627293	0.3885376025
C	0.7500000000	0.8920372707	0.8885376025
S	0.2500000000	0.6469021520	0.6876792047
S	0.2500000000	0.8530978480	0.1876792047
S	0.7500000000	0.3530978480	0.3123207953
S	0.7500000000	0.1469021520	0.8123207953
N	0.2500000000	0.2016927229	0.5577667121
N	0.2500000000	0.2983072771	0.0577667121
N	0.7500000000	0.7983072771	0.4422332879
N	0.7500000000	0.7016927229	0.9422332879

SYSTEM Parameters

System	
ntyp	4
nat	16
ibrav	8
celldm(1)	7.29204252
celldm(2)	1.38525437
celldm(3)	3.19870047
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

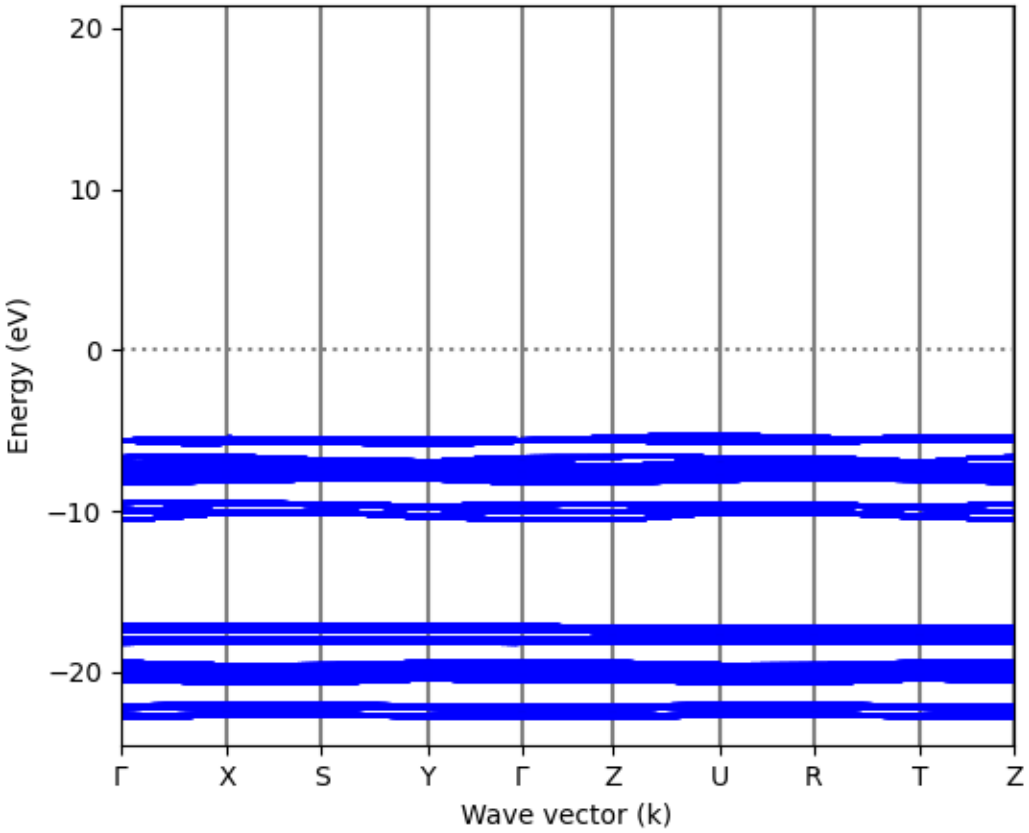
mp-1188861: https://next-gen.materialsproject.org/materials/mp-1188861?formula=C1Li1N1S1

CHAPTER

TWENTYONE

LI1O5P1TI1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
7.2597

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.996820401	0.000000000	0.000000000
0.000000000	1.125020331	0.000000000
0.000000000	0.000000000	1.158469101

ATOMIC_POSITIONS (crystal)

Li	-0.000000000	0.000000000	0.000000000
Li	0.500000000	0.500000000	0.500000000
Li	0.000000000	0.500000000	0.500000000
Li	0.500000000	0.000000000	-0.000000000
Ti	0.750000000	0.281195289	0.829940583
Ti	0.250000000	0.718804711	0.170059417
Ti	0.250000000	0.781195289	0.670059417
Ti	0.750000000	0.218804711	0.329940583
P	0.250000000	0.368876379	0.873192844
P	0.750000000	0.631123621	0.126807156
P	0.750000000	0.868876379	0.626807156
P	0.250000000	0.131123621	0.373192844
O	0.945243239	0.741426364	0.629270183
O	0.554756761	0.758573636	0.129270183
O	0.054756761	0.241426364	0.870729817
O	0.445243239	0.258573636	0.370729817
O	0.054756761	0.258573636	0.370729817
O	0.554756761	0.741426364	0.629270183
O	0.945243239	0.758573636	0.129270183
O	0.445243239	0.241426364	0.870729817
O	0.750000000	0.145413832	0.109681893
O	0.250000000	0.854586168	0.890318107
O	0.250000000	0.645413832	0.390318107
O	0.750000000	0.354586168	0.609681893
O	0.750000000	0.997360561	0.795825094
O	0.250000000	0.002639439	0.204174906
O	0.250000000	0.497360561	0.704174906
O	0.750000000	0.502639439	0.295825094

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Table 1 – continued from previous page

O	0.250000000	0.484958732	0.049404263
O	0.750000000	0.515041268	0.950595737
O	0.750000000	0.984958732	0.450595737
O	0.250000000	0.015041268	0.549404263

SYSTEM Parameters

System	
ntyp	4
nat	32
ibrav	0
celldm(1)	12.0439175
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	30
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

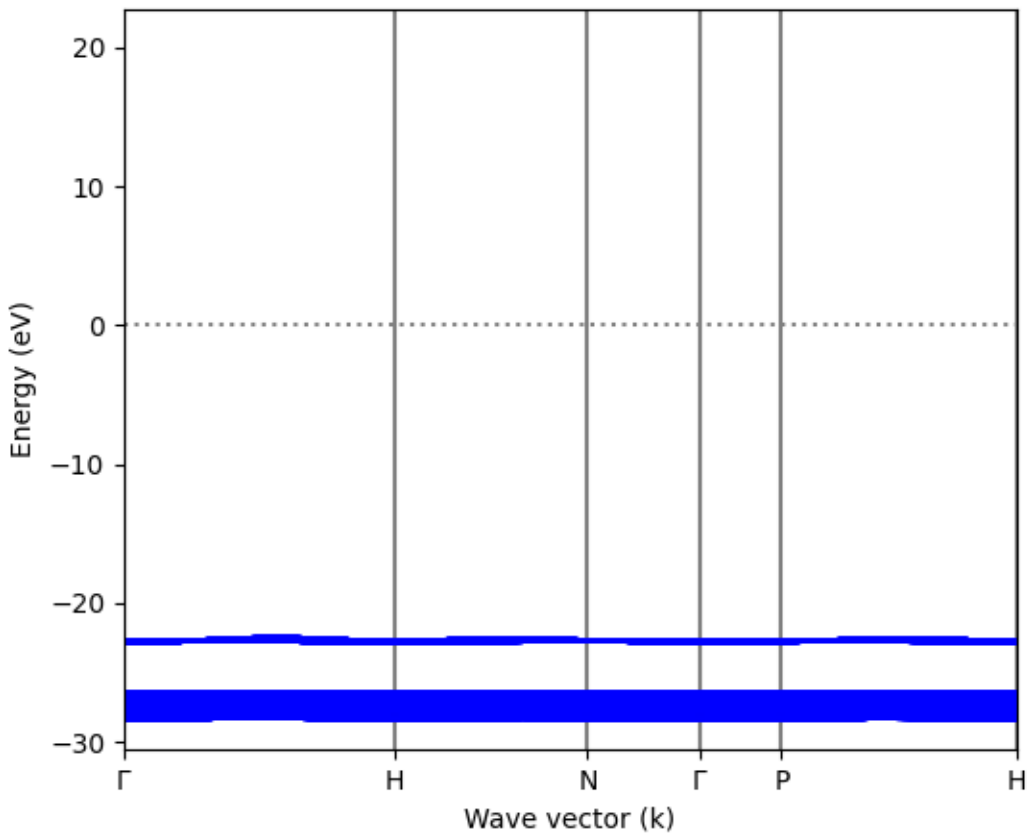
mp-6668: https://next-gen.materialsproject.org/materials/mp-6668?formula=Li1O5P1Ti1

CHAPTER

TWENTYTWO

B3CA4LI1N6

Band gap Diagram



Bravais lattice type

CubicI

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
13.0867

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.588077535	0.566526880	0.588077535
-0.58807753	0.588077535	0.566526880
-0.56652688	-0.58807753	0.588077535

ATOMIC_POSITIONS (crystal)

Li	0.000000000	-0.000000000	-0.000000000
Li	0.500000000	0.500000000	0.500000000
Ca	0.250012442	0.249991738	0.250012442
Ca	0.249984390	0.750015610	0.249984390
Ca	0.749987558	0.250012442	0.249991738
Ca	0.249991738	0.250012442	0.749987558
Ca	0.749987558	0.750008262	0.749987558
Ca	0.750015610	0.249984390	0.750015610
Ca	0.250012442	0.749987558	0.750008262
Ca	0.750008262	0.749987558	0.250012442
B	-0.000000000	0.500000000	0.000000000
B	0.500000000	-0.000000000	0.000000000
B	-0.000000000	0.000000000	0.500000000
B	0.500000000	-0.000000000	0.500000000
B	0.000000000	0.500000000	0.500000000
B	0.500000000	0.500000000	-0.000000000
N	0.019061309	-0.019061309	0.267330956
N	-0.019061309	0.019061309	0.732669044
N	-0.019061309	0.267330956	-0.019061309
N	0.732669044	0.019061309	-0.019061309
N	0.267330956	-0.019061309	0.019061309
N	0.019061309	0.732669044	0.019061309
N	0.519048840	0.480951160	0.767343606
N	0.480951160	0.519048840	0.232656394
N	0.480951160	0.767343606	0.480951160
N	0.232656394	0.519048840	0.480951160
N	0.767343606	0.480951160	0.519048840
N	0.519048840	0.232656394	0.519048840

SYSTEM Parameters

System	
ntyp	4
nat	28
ibrav	0
celldm(1)	13.4817096
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

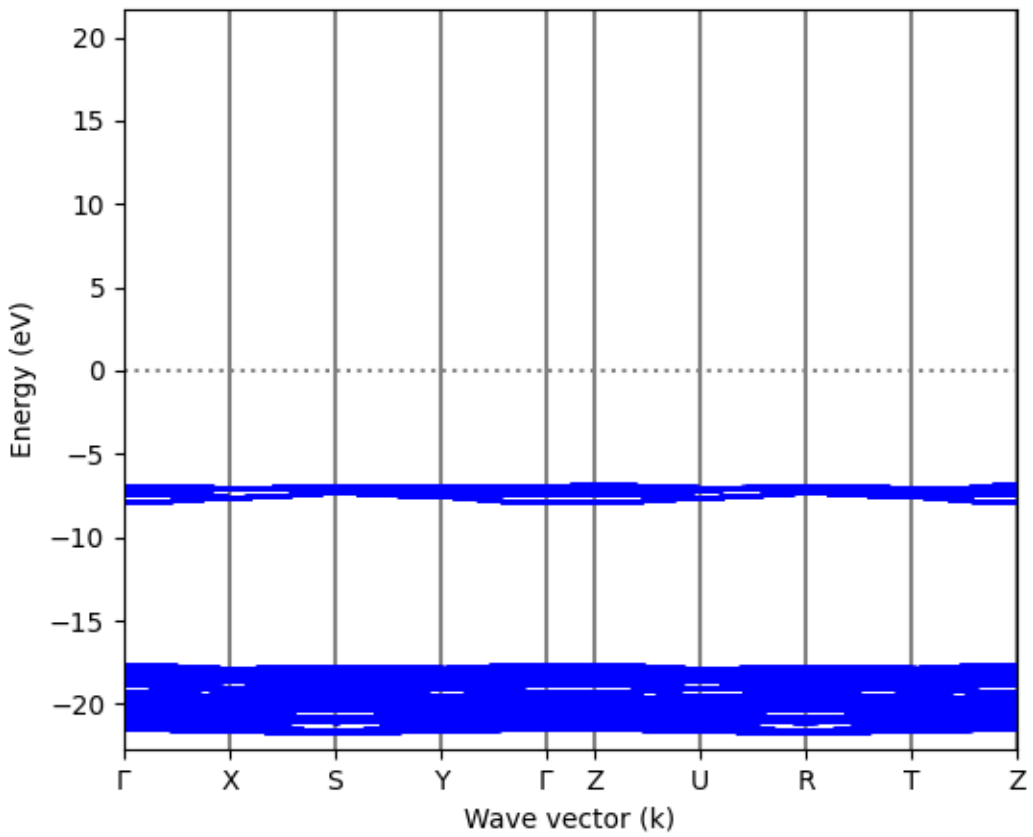
mp-6799: https://next-gen.materialsproject.org/materials/mp-6799?formula=B3Ca4Li1N6

CHAPTER

TWENTYTHREE

B1CA1LI1O3

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
7.6295

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.984857369	0.000000021	-0.00000001
0.000000021	0.993569119	-0.00000031
-0.00000001	-0.00000067	2.166883286

ATOMIC_POSITIONS (crystal)

Li	0.749984223	0.025688512	-0.000003098
Li	0.249991587	0.474325698	0.000002828
Li	0.250015777	0.974311488	0.000003098
Li	0.750008413	0.525674302	-0.000002828
Li	0.249991626	0.525674276	0.499997179
Li	0.749984246	0.974311501	0.500003107
Li	0.750008374	0.474325724	0.500002821
Li	0.250015754	0.025688499	0.499996893
Ca	-0.000003444	0.250001140	0.819091418
Ca	-0.000003425	0.749998835	0.680908597
Ca	0.500006619	0.249988871	0.680910894
Ca	0.500006644	0.750011122	0.819089100
Ca	0.499993381	0.750011129	0.319089106
Ca	0.499993356	0.249988878	0.180910900
Ca	0.000003444	0.749998860	0.180908582
Ca	0.000003425	0.250001165	0.319091403
B	0.499979325	0.749954728	0.601479634
B	0.499979296	0.250045272	0.898520354
B	0.000022793	0.749992516	0.898517658
B	0.000022763	0.250007481	0.601482354
B	-0.000022793	0.250007484	0.101482342
B	-0.000022763	0.749992519	0.398517646
B	0.500020675	0.250045272	0.398520366
B	0.500020704	0.749954728	0.101479646
O	0.857981736	0.611136348	0.845176827
O	0.857981708	0.388863653	0.654823176
O	0.142040887	0.888869815	0.845177327
O	0.142040847	0.111130169	0.654822686

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Table 1 – continued from previous page

O	0.642005471	0.888832707	0.654821703
O	0.642005446	0.111167263	0.845178298
O	0.357951110	0.611094636	0.654819651
O	0.357951073	0.388905390	0.845180346
O	-0.000019292	0.250007713	0.997399994
O	-0.000019236	0.749992282	0.502599982
O	0.500015870	0.250020296	0.502602421
O	0.500015930	0.749979687	0.997397563
O	0.357994529	0.111167293	0.345178297
O	0.357994554	0.888832737	0.154821702
O	0.642048890	0.388905364	0.345180349
O	0.642048927	0.611094610	0.154819654
O	0.142018264	0.388863652	0.154823173
O	0.142018292	0.611136347	0.345176824
O	0.857959113	0.111130185	0.154822673
O	0.857959153	0.888869831	0.345177314
O	0.499984130	0.749979704	0.497397579
O	0.499984070	0.250020313	0.002602437
O	0.000019292	0.749992287	0.002600006
O	0.000019236	0.250007718	0.497400018

SYSTEM Parameters

System	
ntyp	4
nat	48
ibrav	0
celldm(1)	11.5413616
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

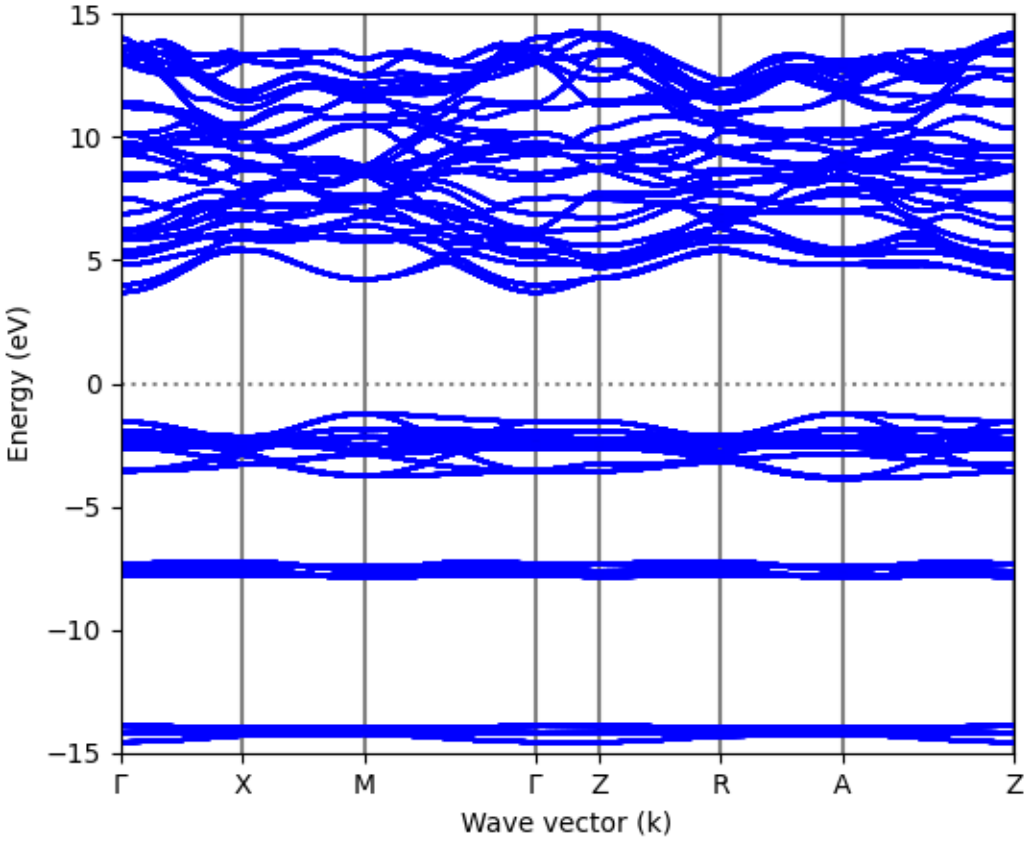
mp-561237: https://next-gen.materialsproject.org/materials/mp-561237?formula=B1Ca1Li1O3

CHAPTER

TWENTYFOUR

CL2CS1LI1

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
3.6525	5.2420	7.5718	5.4413	5.2420	5.8034	7.5032	6.0740	5.8034

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.993183991	0.000000000	0.000000000
0.000000000	0.993183991	0.000000000
0.000000000	0.000000000	1.908265518

ATOMIC_POSITIONS (crystal)

Cs	0.000000000	0.500000000	0.303443772
Cs	0.500000000	0.000000000	0.696556228
Li	0.500000000	0.000000000	0.080056886
Li	0.000000000	0.500000000	0.919943114
Cl	0.000000000	0.500000000	0.669345329
Cl	0.500000000	0.000000000	0.330654671
Cl	0.000000000	0.000000000	0.000000000
Cl	0.500000000	0.500000000	-0.000000000

SYSTEM Parameters

System	
ntyp	3
nat	8
ibrav	0
celldm(1)	9.26162636
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

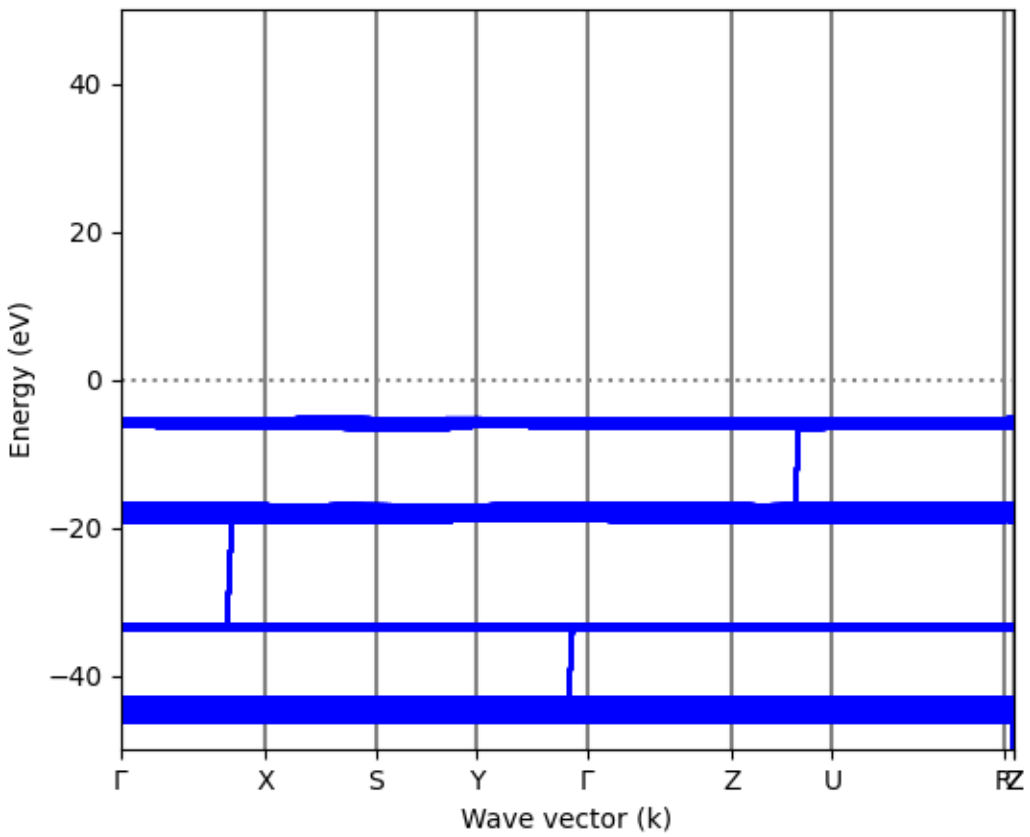
mp-23364: https://next-gen.materialsproject.org/materials/mp-23364?formula=Cl2Cs1Li1

CHAPTER

TWENTYFIVE

LI4O4TI1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
10.6441

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.526897400	0.586674589	-0.00000665
-0.54371061	0.603187805	0.000001677
-0.00000574	-0.00000361	0.915316731

ATOMIC_POSITIONS (crystal)

Li	0.489370745	0.712174311	-0.001256796
Li	0.510663061	0.287856139	0.498681499
Li	0.510629255	0.287825689	0.001256796
Li	0.489336939	0.712143861	0.501318501
Li	0.732270575	0.917398989	0.249959243
Li	0.267729425	0.082601011	0.750040757
Li	0.767186426	0.164749355	0.249960462
Li	0.232813574	0.835250645	0.750039538
Li	-0.010629279	0.212174377	-0.001256806
Li	0.010663056	0.787856229	0.498681501
Li	0.010629279	0.787825623	0.001256806
Li	-0.010663056	0.212143771	0.501318499
Li	0.232270467	0.417399015	0.249959259
Li	0.767729533	0.582600985	0.750040741
Li	0.267186505	0.664749324	0.249960482
Li	0.732813495	0.335250676	0.750039518
Ti	0.284389638	0.035096992	0.250028705
Ti	0.715610362	-0.035096992	0.749971295
Ti	0.784389629	0.535097001	0.250028080
Ti	0.215610371	0.464902999	0.749971920
O	0.525294397	0.758457600	0.250032322
O	0.474705603	0.241542400	0.749967678
O	0.240334535	-0.030291109	-0.007560820
O	0.759653092	0.030281531	0.492374833
O	0.240346908	-0.030281531	0.507625167
O	0.759665465	0.030291109	1.007560820
O	0.480840563	0.243859096	0.249974102
O	0.519159437	0.756140904	0.750025898

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Table 1 – continued from previous page

O	1.025294364	0.258457556	0.250032318
O	-0.025294364	0.741542444	0.749967682
O	0.740334565	0.469708937	-0.007560486
O	0.259653150	0.530281591	0.492374504
O	0.740346850	0.469718409	0.507625496
O	0.259665435	0.530291063	1.007560486
O	0.980840611	0.743859021	0.249974125
O	0.019159389	0.256140979	0.750025875

SYSTEM Parameters

System	
ntyp	3
nat	36
ibrav	0
celldm(1)	13.8864774
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

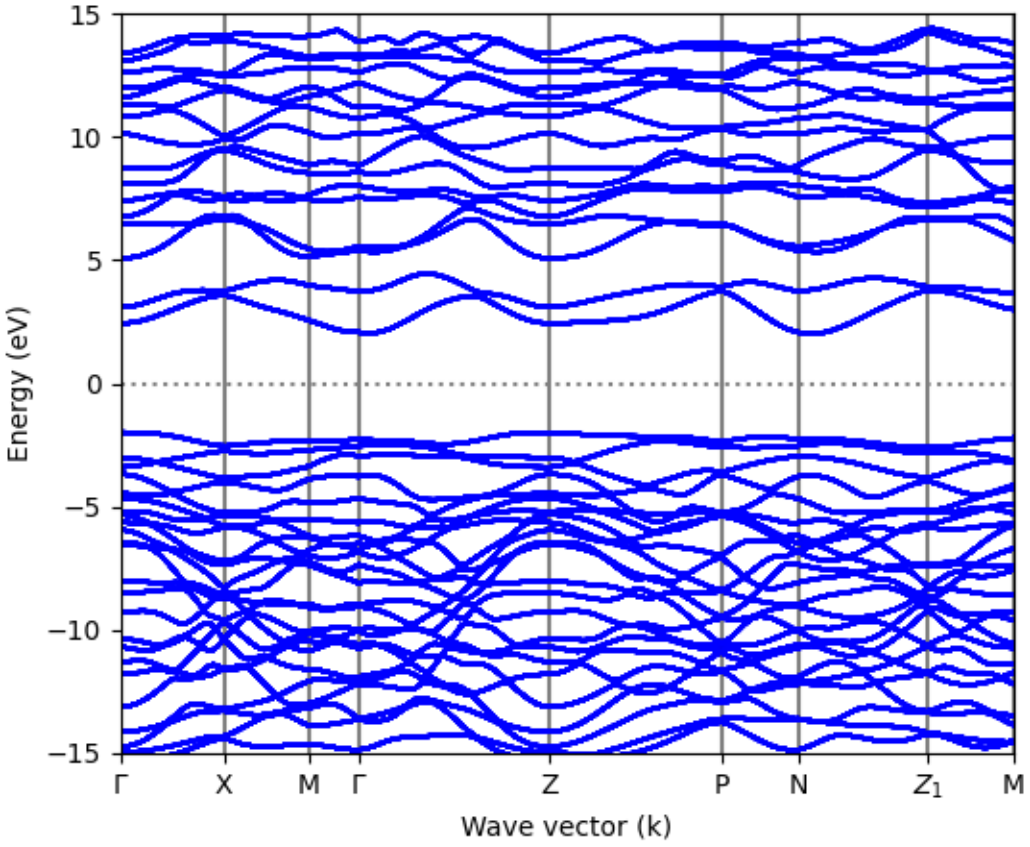
mp-9172: https://next-gen.materialsproject.org/materials/mp-9172?formula=Li4O4Ti1

CHAPTER

TWENTYSIX

B1GE1LI1O4

Band gap Diagram



Bravais lattice type

TetragonalI

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	P	N	Z1	M2
19.7531	4.4218	6.1072	4.8896	4.3364	4.4218	6.0734	4.3364	6.3598	5.2045

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.499105949	-0.50928154	0.755915298
0.495260866	0.490139869	0.755184689
-0.50050262	-0.50144957	0.758522851

ATOMIC_POSITIONS (crystal)

Li	0.219697506	0.587457707	0.231751422
Li	-0.280302291	0.087457484	0.731751087
Ge	0.206690623	-0.048016106	-0.140725604
Ge	0.706682745	0.451989245	0.359282199
B	0.390005141	0.053021860	0.277807519
B	-0.109994701	0.553022429	0.777806465
O	0.238027426	0.237513274	0.101832660
O	0.791632679	0.795059631	0.020617802
O	0.244778488	0.867194589	0.475085754
O	0.909171781	0.007765158	0.283629241
O	0.738025194	0.737513893	0.601833774
O	0.291633473	0.295059177	0.520617226
O	0.744777225	0.367195228	0.975085361
O	0.409174712	0.507766432	0.783625094

SYSTEM Parameters

System	
ntyp	4
nat	14
ibrav	0
celldm(1)	8.61997376
celldm(3)	1.53253517
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

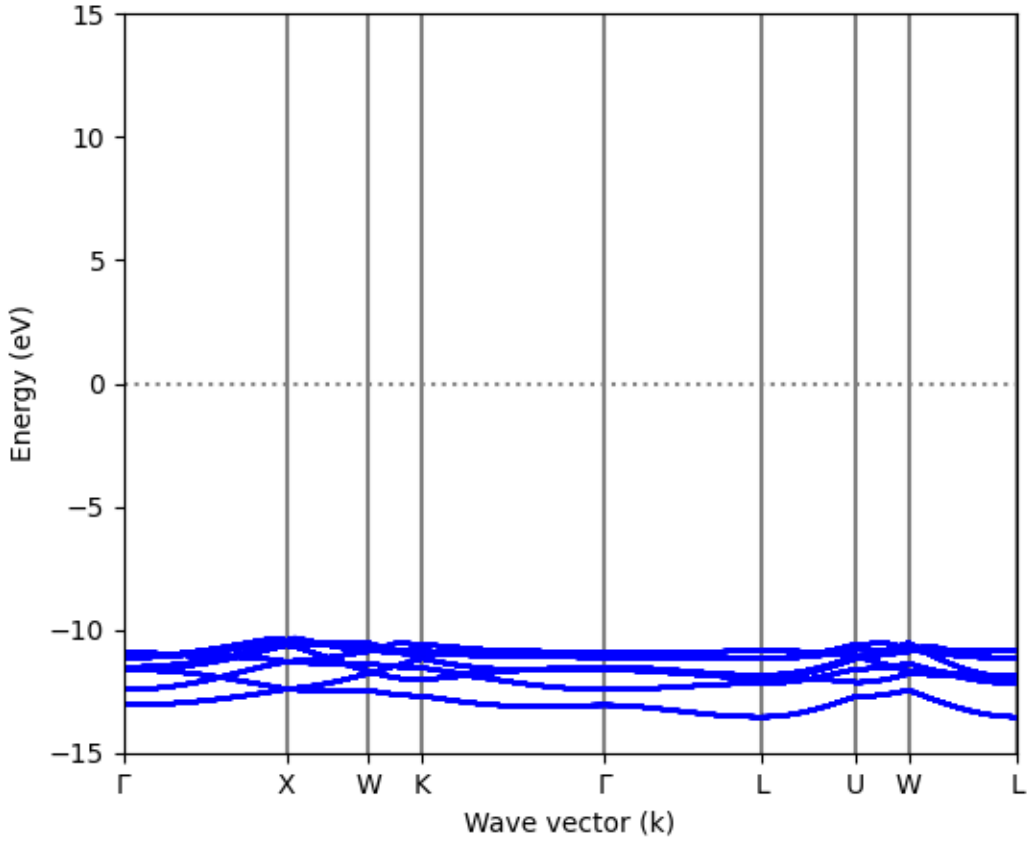
mp-8873: https://next-gen.materialsproject.org/materials/mp-8873?formula=B1Ge1Li1O4

CHAPTER

TWENTYSEVEN

F6LI3TI1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
14.7288

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.58308119	-0.06969178	0.583081189
0.069691779	0.583081189	0.583081189
-0.58308119	0.583081189	-0.06969178

ATOMIC_POSITIONS (crystal)

Li	0.294570530	0.294570530	0.794553002
Li	0.205446998	0.705429470	0.705429470
Li	0.500000000	0.000000000	0.000000000
Li	0.294570530	0.794553002	0.294570530
Li	0.205421277	0.205421277	0.205421277
Li	0.500000000	0.500000000	0.500000000
Li	0.794553002	0.294570530	0.294570530
Li	0.705429470	0.705429470	0.205446998
Li	-0.000000000	0.000000000	0.500000000
Li	0.794578723	0.794578723	0.794578723
Li	0.705429470	0.205446998	0.705429470
Li	0.000000000	0.500000000	0.000000000
Ti	-0.000000000	0.000000000	0.000000000
Ti	0.000000000	0.500000000	0.500000000
Ti	0.500000000	0.000000000	0.500000000
Ti	0.500000000	0.500000000	0.000000000
F	0.261695980	0.009993928	0.009993928
F	0.509980040	0.009980734	0.761699215
F	0.738304020	-0.009993928	-0.009993928
F	0.509980040	0.761699215	0.009980734
F	0.490019960	0.238300785	-0.009980734
F	-0.009993928	-0.009993928	0.738304020
F	0.261695352	0.510008835	0.510008835
F	0.510008835	0.510008835	0.261695352
F	0.738304648	0.489991165	0.489991165
F	0.510008835	0.261695352	0.510008835
F	0.489991165	0.738304648	0.489991165
F	-0.009980734	0.490019960	0.238300785

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Table 1 – continued from previous page

F	0.761699215	0.009980734	0.509980040
F	0.009993928	0.009993928	0.261695980
F	0.238300785	-0.009980734	0.490019960
F	0.009980734	0.761699215	0.509980040
F	-0.009980734	0.238300785	0.490019960
F	0.490019960	-0.009980734	0.238300785
F	0.761699215	0.509980040	0.009980734
F	0.009980734	0.509980040	0.761699215
F	0.238300785	0.490019960	-0.009980734
F	0.009993928	0.261695980	0.009993928
F	-0.009993928	0.738304020	-0.009993928
F	0.489991165	0.489991165	0.738304648

SYSTEM Parameters

System	
ntyp	3
nat	40
ibrav	0
celldm(1)	14.8231165
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

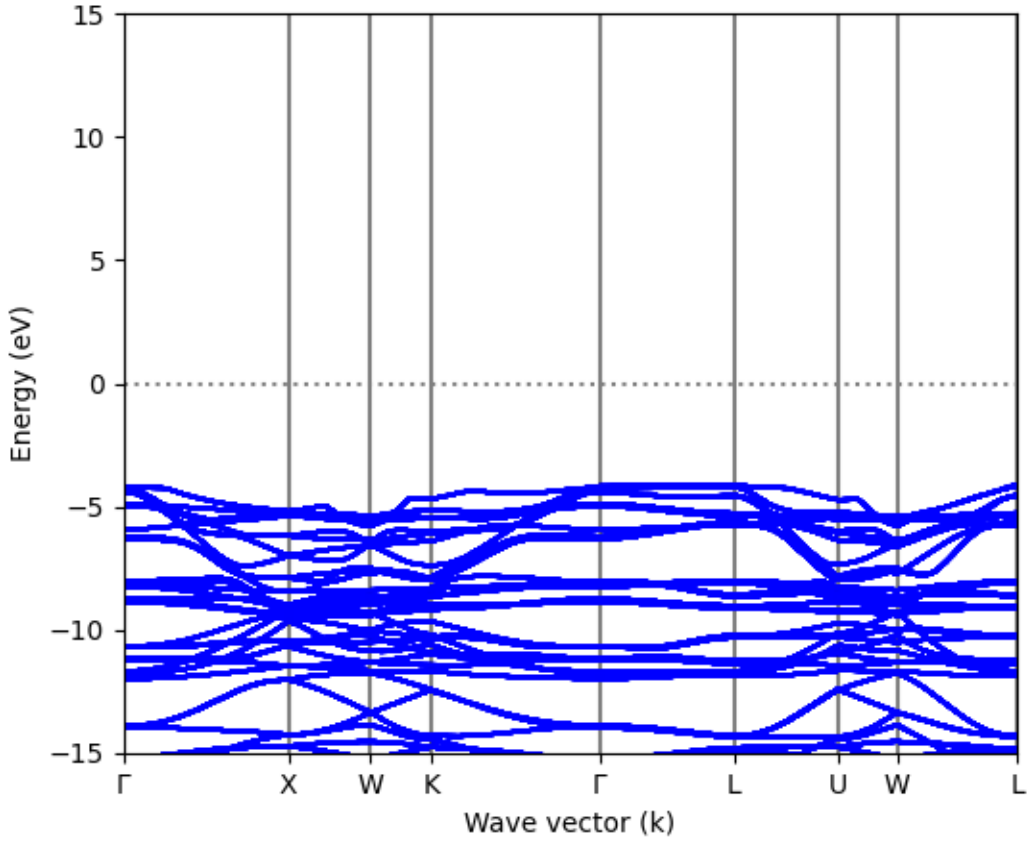
mp-1111280: https://next-gen.materialsproject.org/materials/mp-1111280?formula=F6Li3Ti1

CHAPTER

TWENTYEIGHT

LI2O4S1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
13.2465

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.601400842	0.035796332	0.60140084
-0.035796332	0.601400842	0.60140084
-0.601400842	0.601400842	0.03579633

ATOMIC_POSITIONS (crystal)

Li	0.209673811	0.209673811	0.209673811
Li	0.288609303	0.288609303	0.788603838
Li	0.209643549	0.709688827	0.709688827
Li	0.288609303	0.788603838	0.288609303
Li	0.709688827	0.209643549	0.709688827
Li	0.788603838	0.288609303	0.288609303
Li	0.709688827	0.709688827	0.209643549
Li	0.788614504	0.788614504	0.788614504
S	0.041879976	0.041879976	0.041879976
S	0.041834743	0.541895747	0.541895747
S	0.541895747	0.041834743	0.541895747
S	0.541895747	0.541895747	0.041834743
O	0.129698376	0.129698376	0.129698376
O	0.106634856	0.361776341	0.361776341
O	0.361776341	0.361776341	0.106634856
O	0.361776341	0.106634856	0.361776341
O	0.129690599	0.629685932	0.629685932
O	0.106640826	0.861765241	0.861765241
O	0.361767982	0.861755792	0.606597297
O	0.361767982	0.606597297	0.861755792
O	0.629685932	0.129690599	0.629685932
O	0.606597297	0.361767982	0.861755792
O	0.861755792	0.361767982	0.606597297
O	0.861765241	0.106640826	0.861765241
O	0.629685932	0.629685932	0.129690599
O	0.606597297	0.861755792	0.361767982
O	0.861765241	0.861765241	0.106640826
O	0.861755792	0.606597297	0.361767982

SYSTEM Parameters

System	
ntyp	3
nat	28
ibrav	0
celldm(1)	14.5713407
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

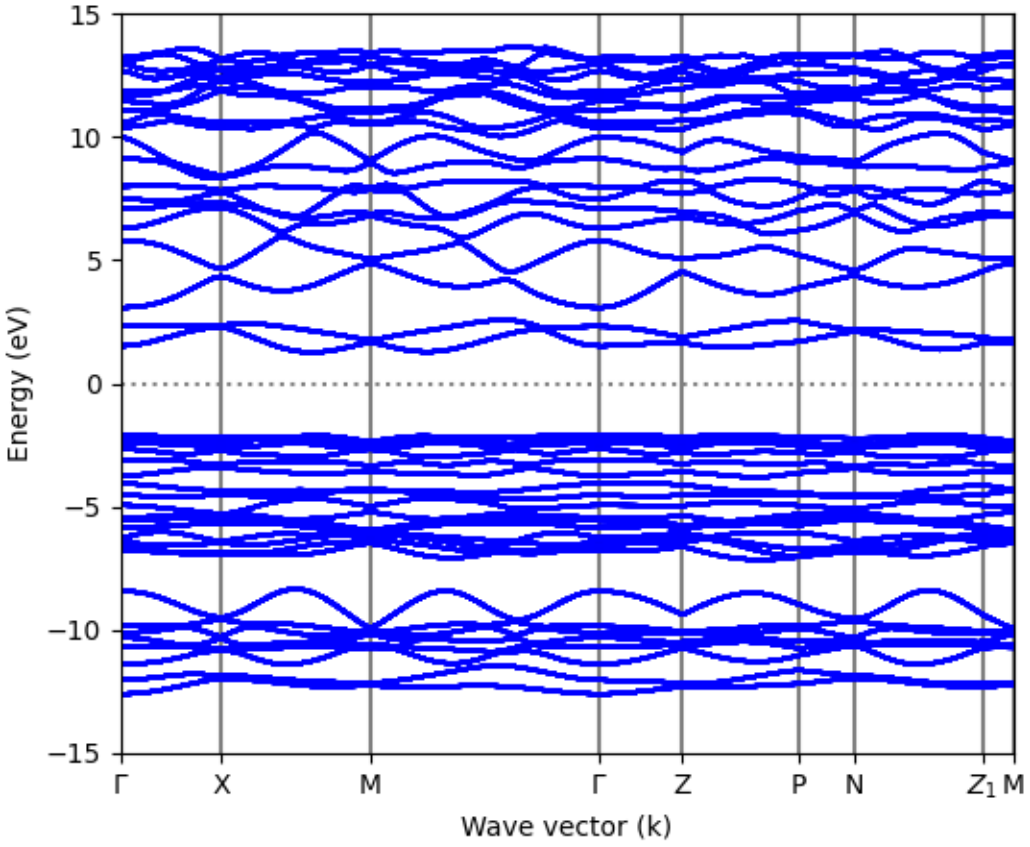
mp-1222624: https://next-gen.materialsproject.org/materials/mp-1222624?formula=Li2O4S1

CHAPTER

TWENTYNINE

C1LI4O4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	P	N	Z1	M2
9.1954	3.5865	4.4670	3.9801	3.5865	3.8590	3.8362	4.3108	3.8590	3.9801

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.538919228	-0.56540732	0.358602014
0.516227015	0.523358702	0.345170419
-0.52201811	-0.53537536	0.384544757

ATOMIC_POSITIONS (crystal)

Li	0.787960083	0.843264479	0.278989117
Li	0.955112456	0.140461709	0.575005508
Li	0.556835076	0.186798875	-0.007122481
Li	0.745469958	0.568010745	0.747777442
Li	0.288169739	0.343044409	0.778813322
Li	0.454995227	0.640606953	0.074969952
Li	0.056972181	0.686789954	0.492794437
Li	0.245218343	0.068046480	0.247598455
C	-0.046056320	0.134023738	0.052560261
C	0.453911387	0.633969875	0.552599005
O	0.951471450	0.335036525	0.345707181
O	0.577441788	0.636695429	0.331851494
O	0.839917874	0.918055062	0.973833466
O	0.631706104	0.238107579	0.701856756
O	0.451579986	0.834776238	0.845703130
O	0.077755521	0.136381559	0.831635298
O	0.339973951	0.417855717	0.473772380
O	0.131565193	0.738074675	0.201655277

SYSTEM Parameters

System	
ntyp	3
nat	18
ibrav	0
celldm(1)	12.2726365
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

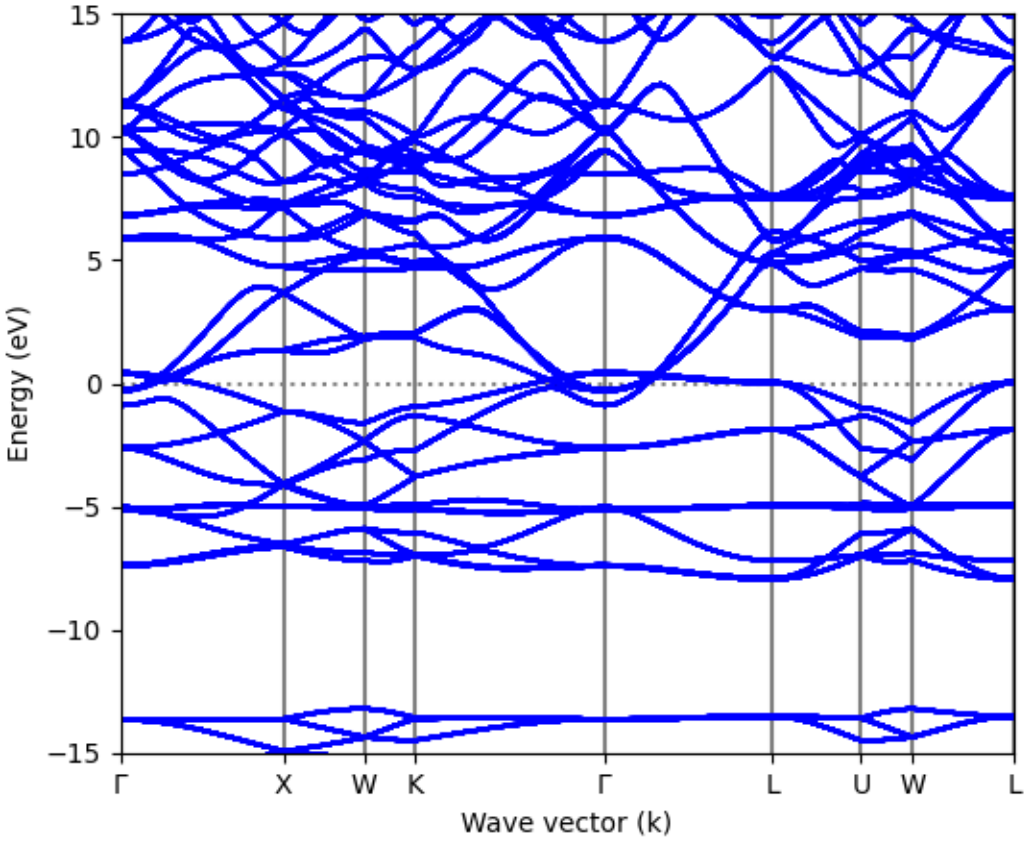
mp-555320: https://next-gen.materialsproject.org/materials/mp-555320?formula=C1Li4O4

CHAPTER

THIRTY

LI2S1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	W	K	Γ	L	U	W2	L2
10.5859	0.6611	2.5001	3.3951	2.8425	0.6611	1.8649	2.8787	3.3951	1.8649

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.60021415	-0.12332177	0.600214153
0.123321769	0.600214153	0.600214153
-0.60021415	0.600214153	-0.12332177

ATOMIC_POSITIONS (crystal)

Li	0.775337450	0.275337451	0.275337451
Li	0.224662550	0.724662549	0.724662549
Li	0.775337451	0.775337451	0.775337451
Li	0.224662549	0.224662549	0.224662549
Li	0.275337451	0.275337451	0.775337450
Li	0.724662549	0.724662549	0.224662550
Li	0.275337451	0.775337450	0.275337451
Li	0.724662549	0.224662550	0.724662549
S	-0.000000000	0.000000000	-0.000000000
S	0.000000000	0.500000000	0.500000000
S	0.500000000	-0.000000000	0.500000000
S	0.500000000	0.500000000	0.000000000

SYSTEM Parameters

System	
ntyp	2
nat	12
ibrav	0
celldm(1)	10.7153807
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

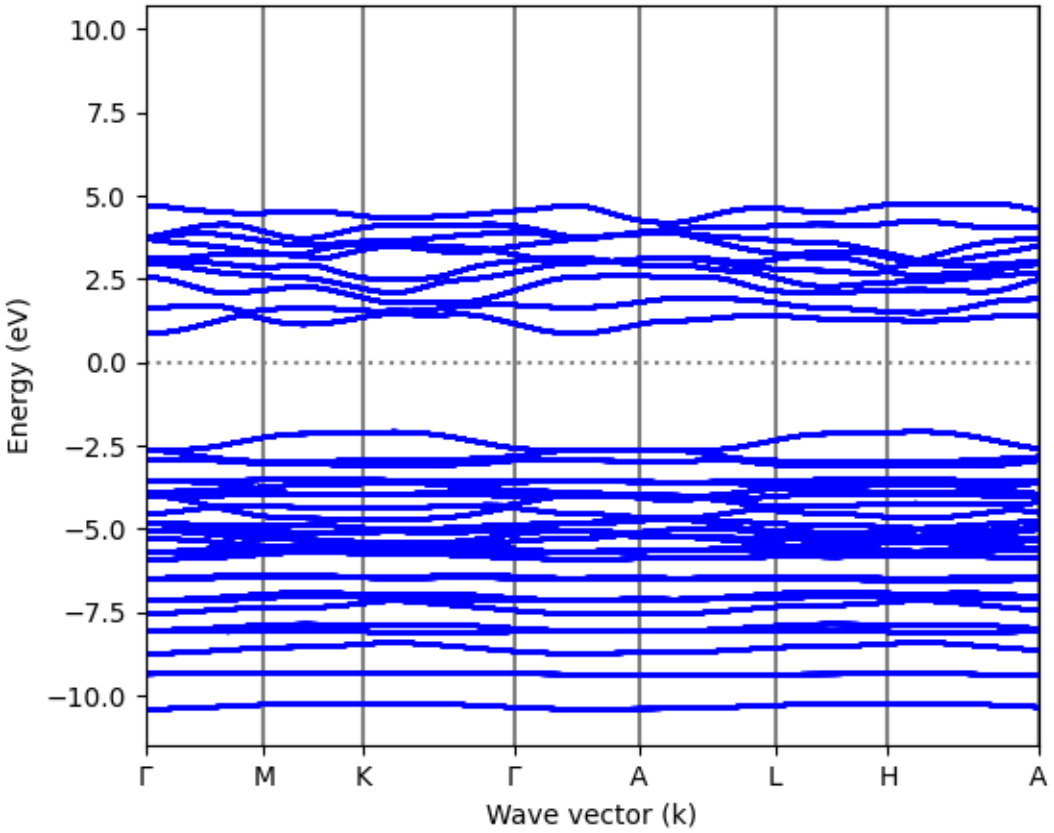
mp-1153: https://next-gen.materialsproject.org/materials/mp-1153?formula=Li2S1

CHAPTER

THIRTYONE

LI14N6O3P2

Band gap Diagram



Bravais lattice type

HexagonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	M	K	Γ	A	L	H	A2
7.5139	3.5014	3.5569	3.4655	3.7268	3.7799	3.7422	3.3715	3.9610

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.996512702	-0.00000000	0.000000000
-0.49825635	0.863005315	0.000000000
0.000000000	0.000000000	1.409363243

ATOMIC_POSITIONS (crystal)

Li	0.377803886	0.081339297	0.091328609
Li	0.703535411	0.622196114	0.091328609
Li	0.918660703	0.296464589	0.091328609
Li	0.622196114	0.918660703	0.908671391
Li	0.296464589	0.377803886	0.908671391
Li	0.081339297	0.703535411	0.908671391
Li	0.044271536	0.352079405	0.590067156
Li	0.307807870	0.955728464	0.590067156
Li	0.647920595	0.692192130	0.590067156
Li	0.955728464	0.647920595	0.409932844
Li	0.692192130	0.044271536	0.409932844
Li	0.352079405	0.307807870	0.409932844
Li	-0.000000000	-0.000000000	0.249537247
Li	0.000000000	0.000000000	0.750462753
P	0.666666670	0.333333330	0.773124497
P	0.333333330	0.666666670	0.226875503
N	0.375092000	0.070293314	0.825737428
N	0.695201314	0.624908000	0.825737428
N	0.929706686	0.304798686	0.825737428
N	0.624908000	0.929706686	0.174262572
N	0.304798686	0.375092000	0.174262572
N	0.070293314	0.695201314	0.174262572
O	0.666666670	0.333333330	0.551980356
O	0.333333330	0.666666670	0.448019644
O	0.000000000	0.000000000	0.500000000

SYSTEM Parameters

System	
ntyp	4
nat	25
ibrav	0
celldm(1)	10.7688639
celldm(3)	1.41648589
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	80
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

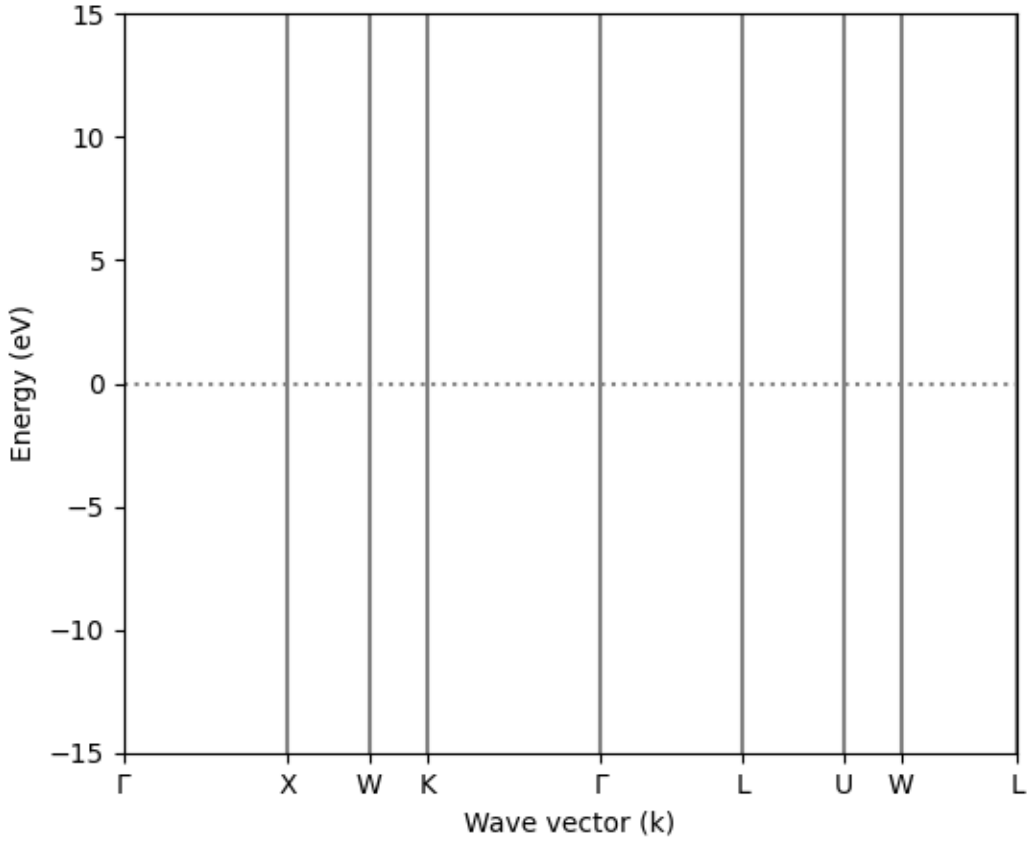
mp-1020011: https://next-gen.materialsproject.org/materials/mp-1020011?formula=Li14N6O3P2

CHAPTER

THIRTYTWO

LI1O2TI1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
52.6294

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.57291328	-0.06769881	0.572913285
0.067698808	0.572913285	0.572913285
-0.57291328	0.572913285	-0.06769881

ATOMIC_POSITIONS (crystal)

Li	0.124998862	0.875000635	0.875000635
Li	0.875000635	0.875000635	0.124998862
Li	0.375000038	0.125000278	0.375000038
Li	0.125000142	0.125000142	0.125000142
Li	0.125000278	0.375000038	0.375000038
Li	0.874999799	0.375001084	0.624999226
Li	0.375001084	0.624999226	0.874999799
Li	0.125000129	0.624999760	0.624999760
Li	0.624999226	0.874999799	0.375001084
Li	0.375001084	0.874999799	0.624999226
Li	0.875000635	0.124998862	0.875000635
Li	0.624999760	0.125000129	0.624999760
Li	0.624999226	0.375001084	0.874999799
Li	0.375000038	0.375000038	0.125000278
Li	0.874999799	0.624999226	0.375001084
Li	0.624999760	0.624999760	0.125000129
Ti	0.375002295	0.874996807	0.125001585
Ti	0.375002295	0.125001585	0.874996807
Ti	0.125001585	0.874996807	0.375002295
Ti	0.625000168	0.124998936	0.124998936
Ti	0.374988849	0.374988849	0.625020848
Ti	0.374988849	0.625020848	0.374988849
Ti	0.125001585	0.375002295	0.874996807
Ti	0.625000623	0.625000623	0.625000623
Ti	0.875009606	0.875009606	0.624983366
Ti	0.874996807	0.125001585	0.375002295
Ti	0.624983366	0.875009606	0.875009606
Ti	0.124998936	0.124998936	0.625000168

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Table 1 – continued from previous page

Ti	0.874996807	0.375002295	0.125001585
Ti	0.875009606	0.624983366	0.875009606
Ti	0.625020848	0.374988849	0.374988849
Ti	0.124998936	0.625000168	0.124998936
O	0.111567687	0.882810830	0.111567687
O	0.860665619	0.860665619	0.360668383
O	0.611567212	0.611567212	0.882810545
O	0.382807940	0.611568796	0.111570064
O	0.889334683	0.889334683	0.889334683
O	0.367194136	0.138430271	0.138430271
O	0.638428871	0.867194039	0.138429992
O	0.138430271	0.138430271	0.367194136
O	0.111570064	0.382807940	0.611568796
O	0.860665619	0.360668383	0.860665619
O	0.611568796	0.111570064	0.382807940
O	0.382807940	0.111570064	0.611568796
O	0.889331167	0.389332870	0.389332870
O	0.367193737	0.638431688	0.638431688
O	0.638431688	0.367193737	0.638431688
O	0.138429992	0.638428871	0.867194039
O	0.611567212	0.882810545	0.611567212
O	0.360668383	0.860665619	0.860665619
O	0.111570064	0.611568796	0.382807940
O	0.882810545	0.611567212	0.611567212
O	0.389332870	0.889331167	0.389332870
O	0.867194039	0.138429992	0.638428871
O	0.138429992	0.867194039	0.638428871
O	0.638428871	0.138429992	0.867194039
O	0.611568796	0.382807940	0.111570064
O	0.360664770	0.360664770	0.360664770
O	0.111567687	0.111567687	0.882810830
O	0.882810830	0.111567687	0.111567687
O	0.389332870	0.389332870	0.889331167
O	0.867194039	0.638428871	0.138429992
O	0.138430271	0.367194136	0.138430271
O	0.638431688	0.638431688	0.367193737

SYSTEM Parameters

System	
ntyp	3
nat	64
ibrav	0
celldm(1)	15.6810168
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

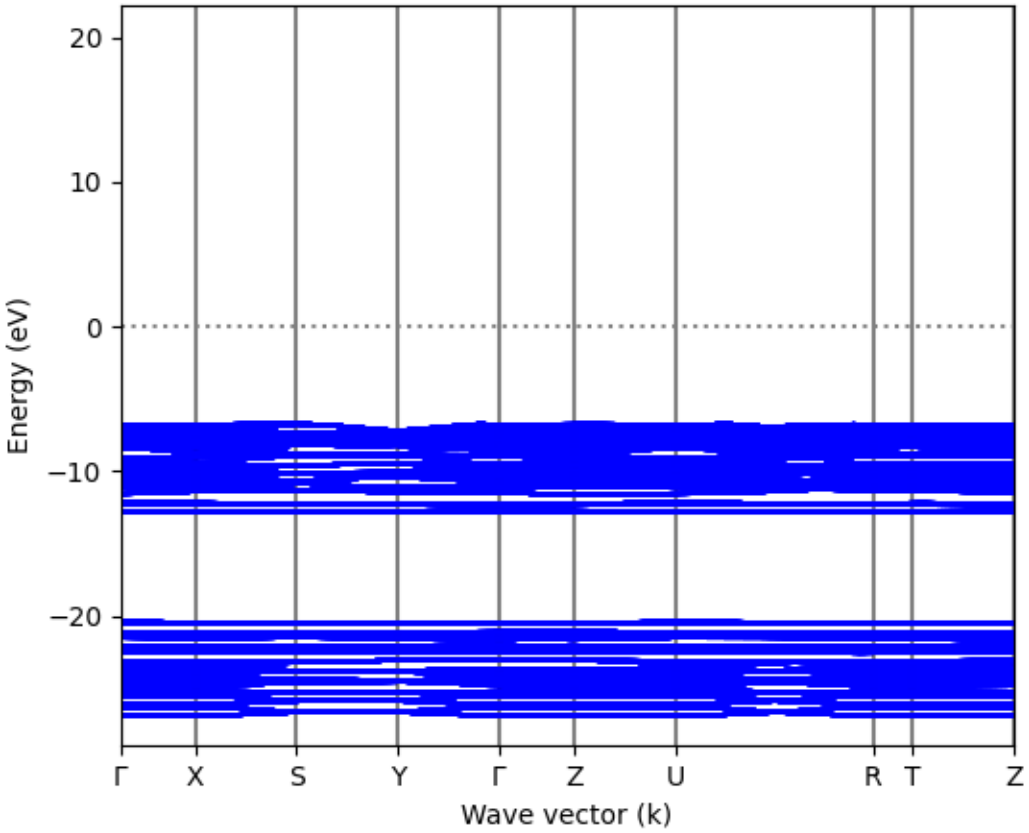
mp-6944: https://next-gen.materialsproject.org/materials/mp-6944?formula=Li1O2Ti1

CHAPTER

THIRTYTHREE

C2B1F2LI1O4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
6.8582

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.546258347	0.950521146	-0.00002835
-0.55031368	0.968606250	0.000033837
0.000192728	-0.00007858	1.063433586

ATOMIC_POSITIONS (crystal)

Li	0.582314061	0.335413718	0.751282955
Li	-0.082806906	0.164622782	0.251012546
Li	0.082806906	0.835377218	0.748987454
Li	0.417685939	0.664586282	0.248717045
B	0.121682044	0.476897360	0.749756007
B	0.378723791	0.022888975	0.249781808
B	0.621276209	0.977111025	0.750218192
B	-0.121682044	0.523102640	0.250243993
C	0.015245617	0.259092734	0.859080025
C	0.485074560	0.241210960	0.142136550
C	0.015332018	0.258593262	0.642021307
C	0.485138723	0.240664833	0.359184275
C	0.514925440	0.758789040	0.857863450
C	-0.015245617	0.740907266	0.140919975
C	0.514861277	0.759335167	0.640815725
C	-0.015332018	0.741406738	0.357978693
O	-0.062125719	0.180483509	0.964090111
O	0.560782033	0.321570139	0.037371208
O	-0.061354811	0.179043562	0.537407310
O	0.561638775	0.319897910	0.464421189
O	0.108518284	0.361625754	0.912996410
O	0.392464469	0.138491194	0.087303927
O	0.107904361	0.361402914	0.587170977
O	0.391873622	0.138129133	0.412916589
O	0.439217967	0.678429861	0.962628792
O	0.062125719	0.819516491	0.035909889
O	0.438361225	0.680102090	0.535578811
O	0.061354811	0.820956438	0.462592690

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Table 1 – continued from previous page

O	0.607535531	0.861508806	0.912696073
O	-0.108518284	0.638374246	0.087003590
O	0.608126378	0.861870867	0.587083411
O	-0.107904361	0.638597086	0.412829023
F	0.938457303	0.672422335	0.749880505
F	0.215672487	-0.025181531	0.248986509
F	0.561945095	-0.172600994	0.249641783
F	0.284625641	0.525188325	0.748876788
F	0.438054905	1.172600994	0.750358217
F	0.715374359	0.474811675	0.251123212
F	0.061542697	0.327577665	0.250119495
F	0.784327513	1.025181531	0.751013491

SYSTEM Parameters

System	
ntyp	5
nat	40
ibrav	0
celldm(1)	12.3319776
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

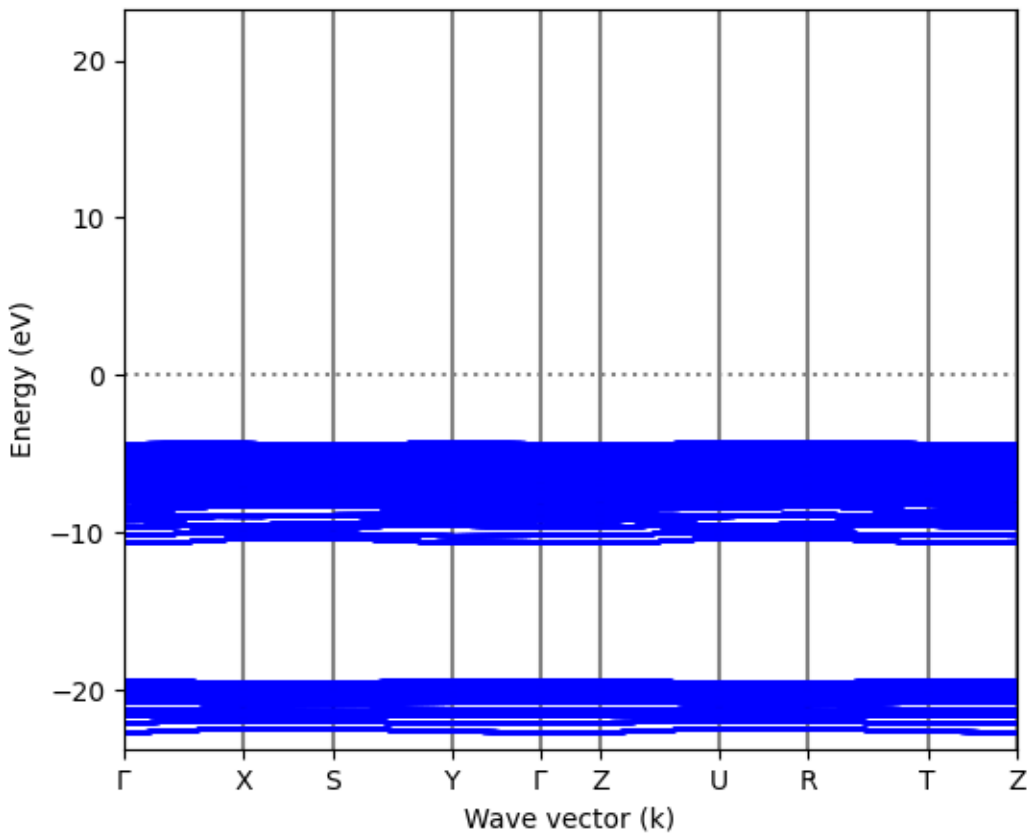
mp-1105683: https://next-gen.materialsproject.org/materials/mp-1105683?formula=C2B1F2Li1O4

CHAPTER

THIRTYFOUR

AL1LI1O4SI1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
6.5637

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.980874822	0.000000000	0.000000000
0.000000000	1.312653175	0.000000000
0.000000000	0.000000000	1.990949624

ATOMIC_POSITIONS (crystal)

Li	0.257650034	0.989525626	0.646259894
Li	0.757650034	0.489525626	0.853740106
Li	0.757650034	0.010474374	0.353740106
Li	0.257650034	0.510474374	0.146259894
Al	0.751953288	0.510538289	0.326855508
Al	0.251953288	0.010538289	0.173144492
Al	0.251953288	0.489461711	0.673144492
Al	0.751953288	0.989461711	0.826855508
Si	0.250875707	0.273149929	0.409273546
Si	0.250875707	0.226850071	0.909273546
Si	0.750875707	0.773149929	0.090726454
Si	0.750875707	0.726850071	0.590726454
O	0.397183833	0.036623483	0.830675906
O	0.897183833	0.963376517	0.169324094
O	0.921074677	0.288018833	0.381975037
O	0.421074677	0.711981167	0.618024963
O	0.897183833	0.536623483	0.669324094
O	0.817551062	0.727373236	0.429779857
O	0.421074677	0.788018833	0.118024963
O	0.397183833	0.463376517	0.330675906
O	0.858111399	0.559437392	0.158935504
O	0.317551062	0.272626764	0.570220143
O	0.358111399	0.440562608	0.841064496
O	0.858111399	0.940562608	0.658935504
O	0.358111399	0.059437392	0.341064496
O	0.317551062	0.227373236	0.070220143
O	0.921074677	0.211981167	0.881975037
O	0.817551062	0.772626764	0.929779857

SYSTEM Parameters

System	
ntyp	4
nat	28
ibrav	0
celldm(1)	9.43353103
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

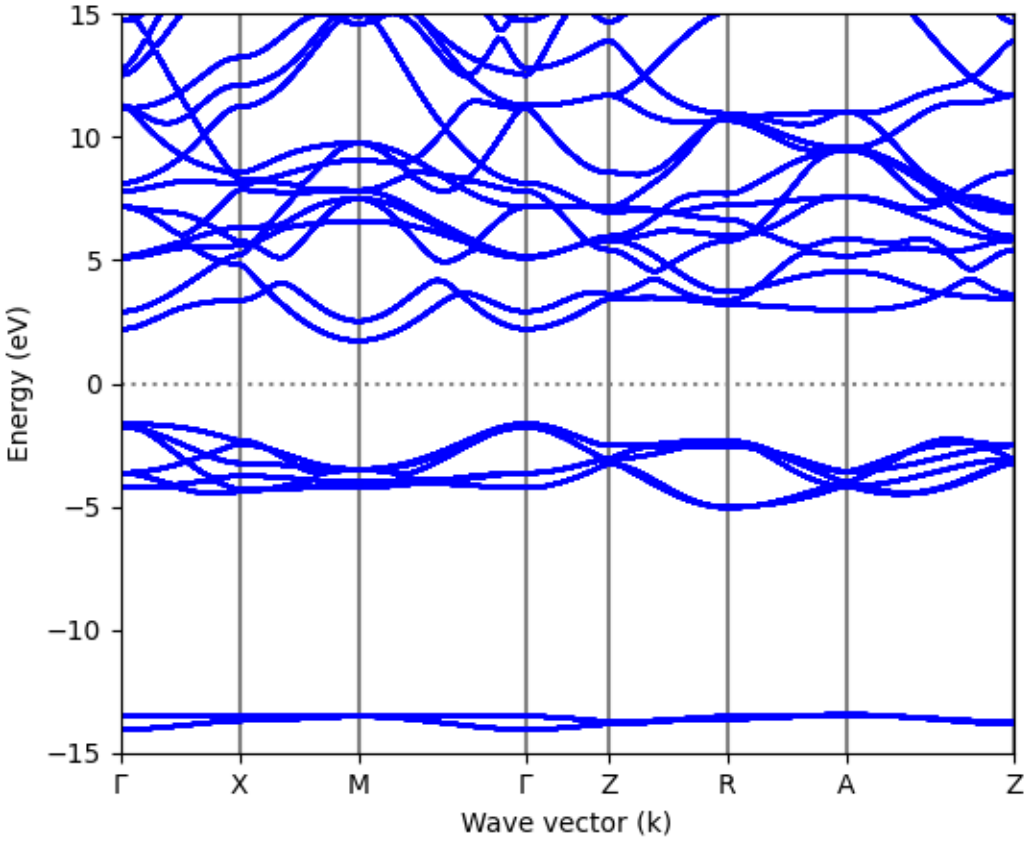
mp-559689: https://next-gen.materialsproject.org/materials/mp-559689?formula=Al1Li1O4Si1

CHAPTER

THIRTYFIVE

LI5N2NA1

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
7.9201	3.8423	5.6778	5.2121	3.8423	5.9434	5.5185	6.5287	5.9434

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.950088003	0.000000000	0.000000000
0.000000000	0.950088003	0.000000000
0.000000000	0.000000000	1.358613768

ATOMIC_POSITIONS (crystal)

Na	0.500000000	0.500000000	0.500000000
Li	0.500000000	0.000000000	0.241483733
Li	0.000000000	0.500000000	0.241483733
Li	0.500000000	0.000000000	0.758516267
Li	0.000000000	0.500000000	0.758516267
Li	0.000000000	0.000000000	0.000000000
N	0.500000000	0.500000000	0.000000000
N	0.000000000	0.000000000	0.500000000

SYSTEM Parameters

System	
ntyp	3
nat	8
ibrav	0
celldm(1)	6.99919357
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

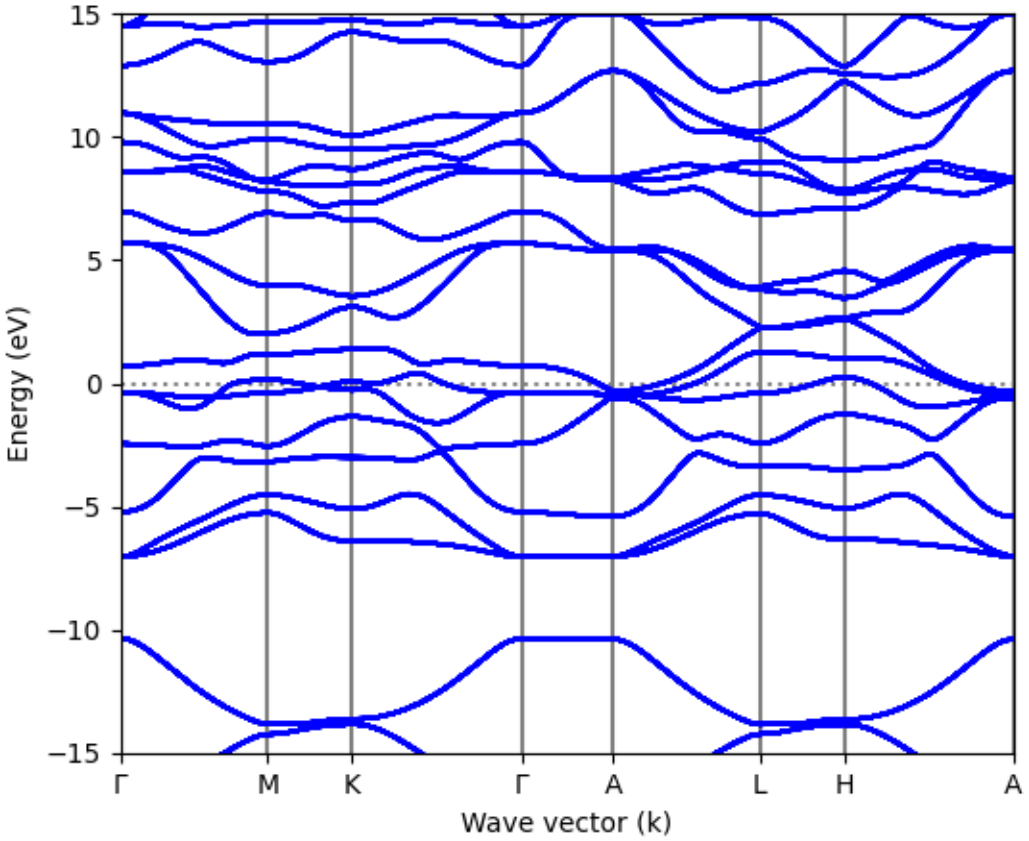
mp-29929: https://next-gen.materialsproject.org/materials/mp-29929?formula=Li5N2Na1

CHAPTER

THIRTYSIX

LI1N2ZR1

Band gap Diagram



Bravais lattice type

Trigonal

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	M	K	Γ	A	L	H	A2
10.0995	1.0832	0.5377	0.3523	1.0832	5.6433	1.6518	1.4959	5.6433

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

1.027103325	0.000000000	0.000000000
-0.51355166	0.889497572	0.000000000
-0.00000000	-0.00000000	1.419532338

ATOMIC_POSITIONS (crystal)

Li	0.000000000	0.000000000	0.432288432
Zr	0.333333330	0.666666670	-0.037274192
N	-0.000000000	-0.000000000	1.050283373
N	0.666666670	0.333333330	0.137702387

SYSTEM Parameters

System	
ntyp	3
nat	4
ibrav	0
celldm(1)	6.26638607
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

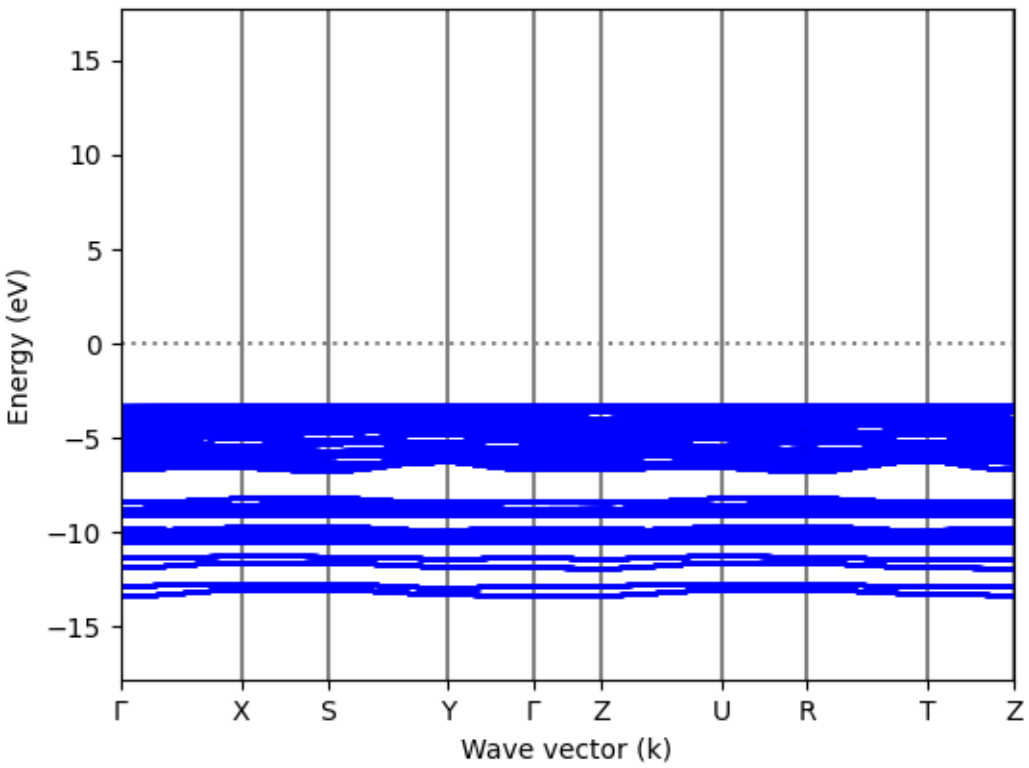
mp-1237264: https://next-gen.materialsproject.org/materials/mp-1237264?formula=Li1N2Zr1

CHAPTER

THIRTYSEVEN

CL1LI1O4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
4.8493

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.986714986	0.000000000	0.000000000
0.000000000	1.377569729	0.000000000
0.000000000	0.000000000	1.783581334

ATOMIC_POSITIONS (crystal)

Li	-0.000000000	0.000000000	0.500000000
Li	0.500000000	-0.000000000	0.000000000
Li	0.000000000	0.500000000	0.500000000
Li	0.500000000	0.500000000	-0.000000000
Cl	0.527858374	0.250000000	0.683484633
Cl	0.027858374	0.750000000	0.816515367
Cl	0.472141626	0.750000000	0.316515367
Cl	0.972141626	0.250000000	0.183484633
O	0.948500640	0.250000000	0.356147275
O	0.448500640	0.750000000	0.143852725
O	0.051499360	0.750000000	0.643852725
O	0.551499360	0.250000000	0.856147275
O	0.836368284	0.433332651	0.125515955
O	0.336368284	0.566667349	0.374484045
O	0.163631716	0.933332651	0.874484045
O	0.663631716	0.066667349	0.625515955
O	0.163631716	0.566667349	0.874484045
O	0.663631716	0.433332651	0.625515955
O	0.836368284	0.066667349	0.125515955
O	0.336368284	0.933332651	0.374484045
O	0.270562347	0.250000000	0.136118419
O	0.770562347	0.750000000	0.363881581
O	0.729437653	0.750000000	0.863881581
O	0.229437653	0.250000000	0.636118419

SYSTEM Parameters

System	
ntyp	3
nat	24
ibrav	0
celldm(1)	9.05644679
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40
ecutrho	400
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

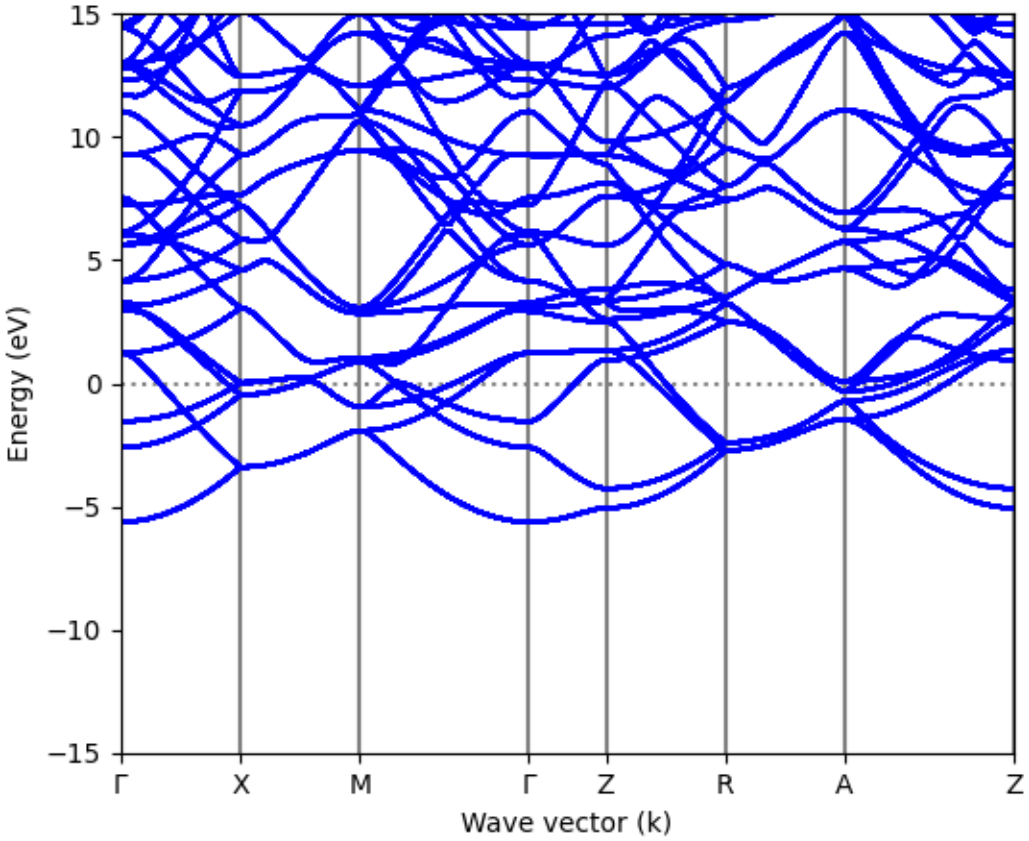
mp-30301: https://next-gen.materialsproject.org/materials/mp-30301?formula=Cl1Li1O4

CHAPTER

THIRTYEIGHT

LI2N2NA4

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
7.7029	2.7605	0.4896	1.8090	2.7605	5.2052	4.8834	0.4149	5.2052

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.998405122	0.000000000	0.000000000
0.000000000	0.998405122	0.000000000
0.000000000	0.000000000	1.551976840

ATOMIC_POSITIONS (crystal)

Na	0.0000000000	0.5000000000	0.2296332205
Na	0.5000000000	0.5000000000	0.5000000000
Na	0.5000000000	0.0000000000	0.7703667795
Na	0.0000000000	0.0000000000	0.5000000000
Li	0.0000000000	0.0000000000	0.0000000000
Li	0.5000000000	0.5000000000	-0.000000000
N	0.0000000000	0.5000000000	0.7669177240
N	0.5000000000	0.0000000000	0.2330822760

SYSTEM Parameters

System	
ntyp	3
nat	8
ibrav	0
celldm(1)	7.71178468
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

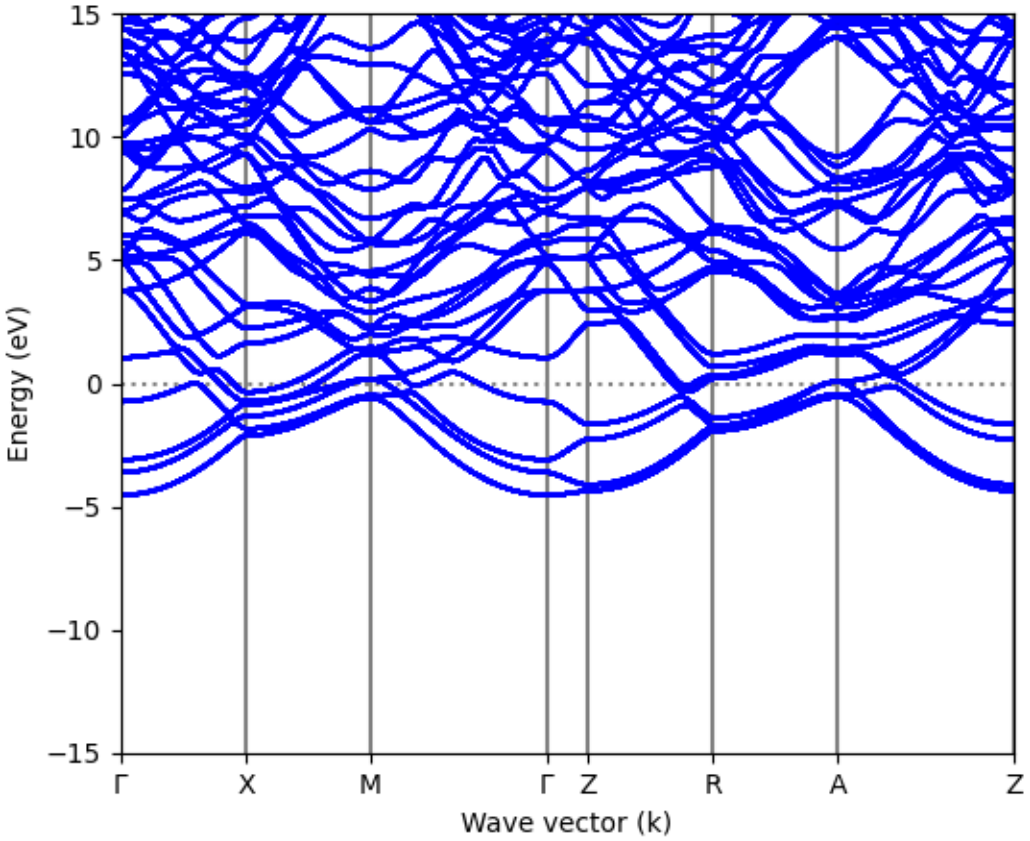
mp-571587: https://next-gen.materialsproject.org/materials/mp-571587?formula=Li2N2Na4

CHAPTER

THIRTYNINE

LI3N2NA3

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
4.3509	1.7795	1.9758	0.6714	1.7795	4.0880	1.5973	0.5127	4.0880

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.958724714	0.000000000	0.000000000
0.000000000	0.958724714	0.000000000
0.000000000	0.000000000	2.877703343

ATOMIC_POSITIONS (crystal)

Na	0.0000000000	0.0000000000	0.7053639264
Na	0.0000000000	0.0000000000	0.2946360736
Na	0.0000000000	0.0000000000	0.5000000000
Li	0.5000000000	0.5000000000	0.5000000000
Li	0.5000000000	0.5000000000	0.0000000000
Li	0.0000000000	0.0000000000	0.0000000000
N	-0.000000000	0.5000000000	0.1622152853
N	0.5000000000	0.0000000000	0.8377847147

SYSTEM Parameters

System	
ntyp	3
nat	8
ibrav	0
celldm(1)	7.17262506
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

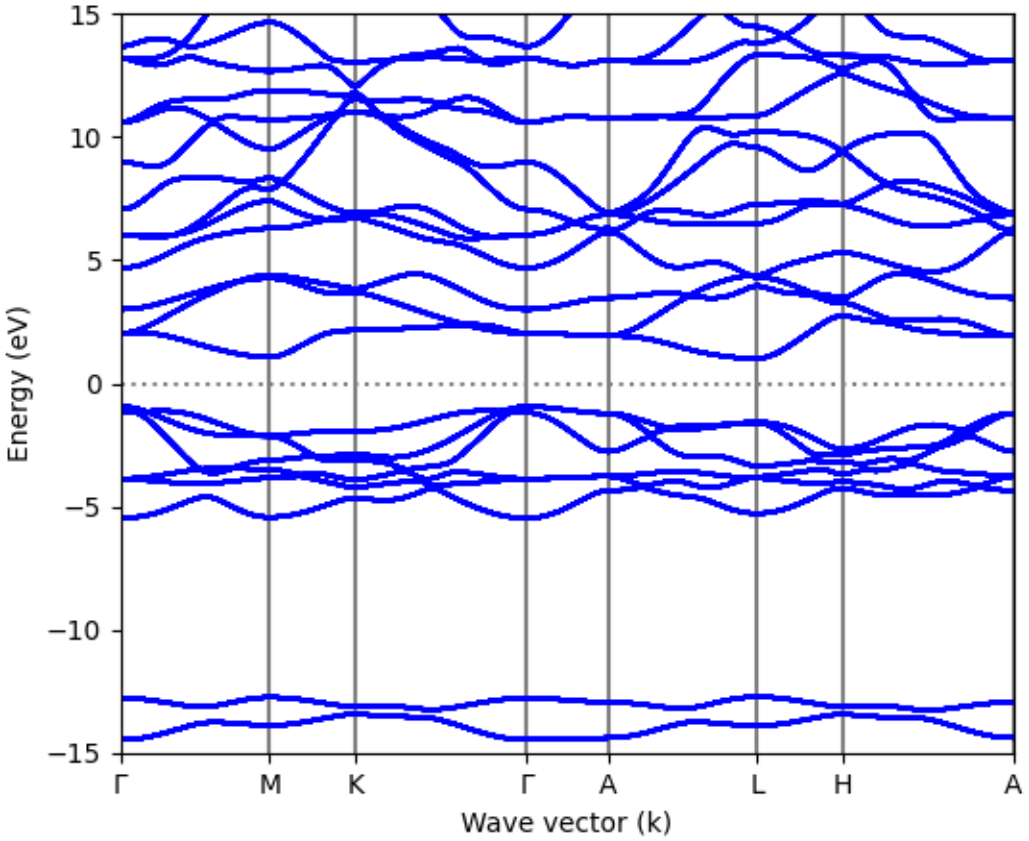
mp-628616: https://next-gen.materialsproject.org/materials/mp-628616?formula=Li3N2Na3

CHAPTER

FORTY

LI2N2ZR1

Band gap Diagram



Bravais lattice type

Trigonal

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	M	K	Γ	A	L	H	A2
10.9393	2.9723	3.1843	4.0852	2.9723	3.1454	2.5716	5.3985	3.1454

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

1.002479241	0.000000000	-0.00000000
-0.50123962	0.868172490	0.000000000
0.000000000	0.000000000	1.586757211

ATOMIC_POSITIONS (crystal)

Li	0.6666666700	0.3333333300	0.3849039555
Li	0.3333333300	0.6666666700	0.6150960445
Zr	0.0000000000	-0.000000000	0.0000000000
N	0.6666666700	0.3333333300	0.7739471641
N	0.3333333300	0.6666666700	0.2260528359

SYSTEM Parameters

System	
ntyp	3
nat	5
ibrav	0
celldm(1)	6.19510093
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

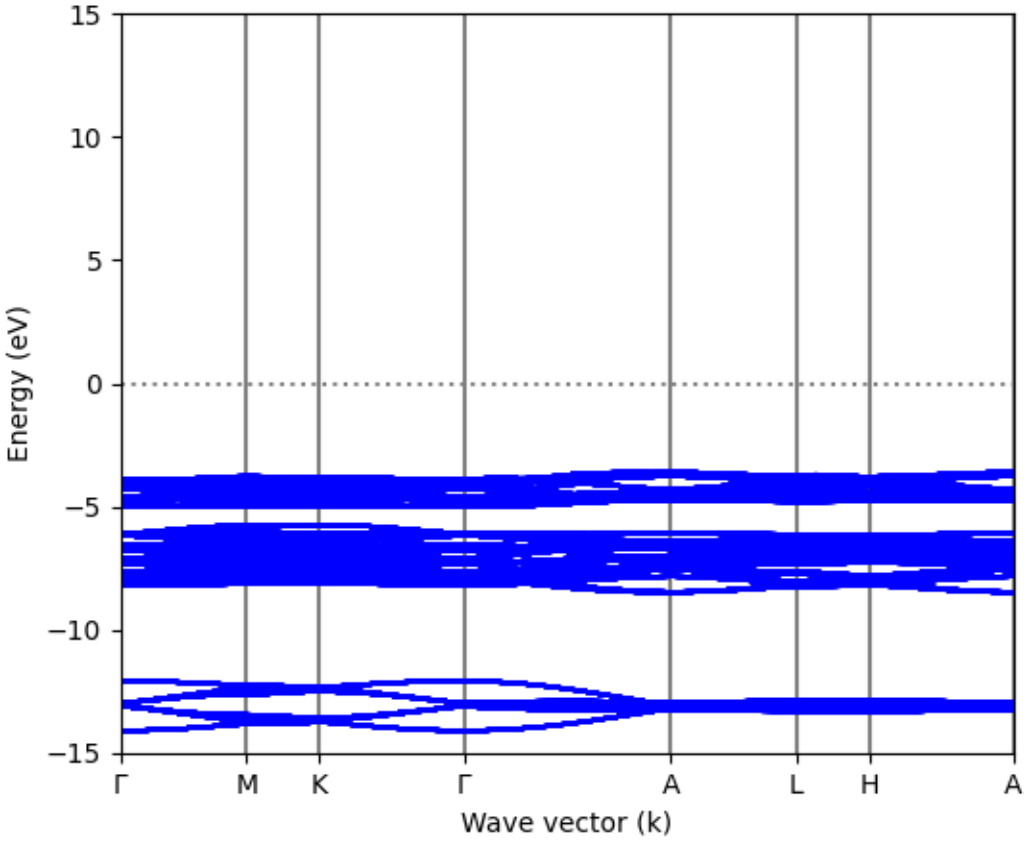
mp-3216: https://next-gen.materialsproject.org/materials/mp-3216?formula=Li2N2Zr1

CHAPTER

FORTYONE

I6LI2O18TI1

Band gap Diagram



Bravais lattice type

Trigonal

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
6.9882

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.982950352	0.000000000	0.000000000
-0.49147518	0.851259976	-0.00000000
-0.00000000	-0.00000000	0.515234508

ATOMIC_POSITIONS (crystal)

Li	0.6666666700	0.3333333300	0.2684050165
Li	-0.000000000	-0.000000000	0.7753029481
Ti	0.3333333300	0.6666666700	0.3130779548
I	0.6990403362	0.6985050759	0.2346941888
I	-0.000535270	0.3009596638	0.2346941888
I	0.3014949241	1.0005352702	0.2346941888
I	0.9980365246	0.6439477308	0.7247819224
I	0.6459111962	0.0019634754	0.7247819224
I	0.3560522692	0.3540888038	0.7247819224
O	0.5412165488	0.4302553190	0.5073681904
O	0.8890387602	0.4587834512	0.5073681904
O	0.5697446810	0.1109612398	0.5073681904
O	0.1281049675	0.9144545124	0.9899945770
O	0.7863495349	0.8718950325	0.9899945770
O	0.0855454876	0.2136504651	0.9899945770
O	0.4764389181	0.6156746679	0.0967144478
O	0.1392357399	0.5235610819	0.0967144478
O	0.3843253321	0.8607642601	0.0967144478
O	0.2092380608	0.7368975207	0.5500740642
O	0.5276594500	0.7907619392	0.5500740642
O	0.2631024793	0.4723405500	0.5500740642
O	0.7406004539	0.5544136712	0.0380474191
O	0.8138132073	0.2593995461	0.0380474191
O	0.4455863288	0.1861867927	0.0380474191
O	0.9442283017	0.7825706061	0.5364898838
O	0.8383422943	0.0557716983	0.5364898838
O	0.2174293939	0.1616577057	0.5364898838

SYSTEM Parameters

System	
ntyp	4
nat	27
ibrav	0
celldm(1)	18.0183747
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

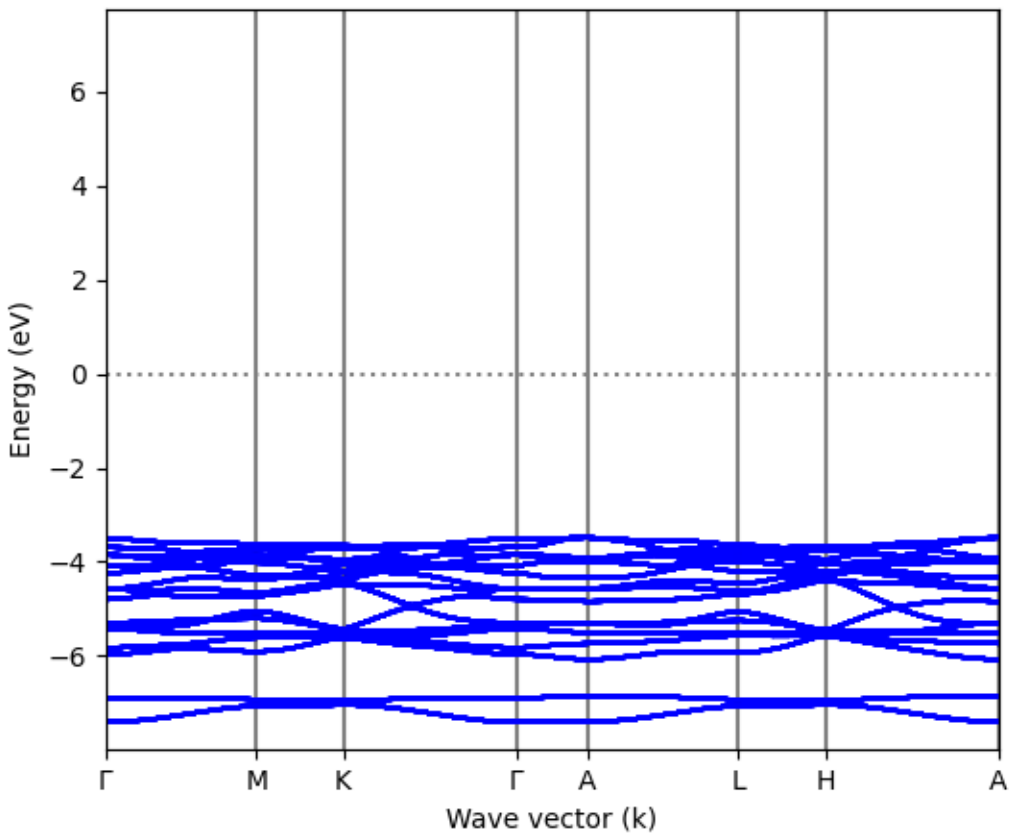
mp-1222778: https://next-gen.materialsproject.org/materials/mp-1222778?formula=I6Li2O18Ti1

CHAPTER

FORTYTWO

IN1LI3O3

Band gap Diagram



Bravais lattice type

HexagonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
7.4795

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.990553497	0.000000000	0.000000000
-0.49527675	0.857844492	0.000000000
0.000000000	-0.00000000	1.811823338

ATOMIC_POSITIONS (crystal)

Li	0.6268866725	0.9452960000	0.9301550379
Li	0.0547040000	0.6815906725	0.9301550379
Li	0.3184093275	0.3731133275	0.9301550379
Li	0.0547040000	0.3731133275	0.5698449621
Li	0.3184093275	0.9452960000	0.5698449621
Li	0.6268866725	0.6815906725	0.5698449621
Li	0.3731133275	0.0547040000	0.0698449621
Li	0.9452960000	0.3184093275	0.0698449621
Li	0.6815906725	0.6268866725	0.0698449621
Li	0.9452960000	0.6268866725	0.4301550379
Li	0.6815906725	0.0547040000	0.4301550379
Li	0.3731133275	0.3184093275	0.4301550379
In	0.6666666700	0.3333333300	0.7500000000
In	0.3333333300	0.6666666700	0.2500000000
In	-0.000000000	-0.000000000	0.7500000000
In	0.0000000000	0.0000000000	0.2500000000
O	0.6982403981	0.0362982123	0.6207341182
O	0.9637017877	0.6619421859	0.6207341182
O	0.3380578141	0.3017596019	0.6207341182
O	0.9637017877	0.3017596019	0.8792658818
O	0.3380578141	0.0362982123	0.8792658818
O	0.6982403981	0.6619421859	0.8792658818
O	0.3017596019	0.9637017877	0.3792658818
O	0.0362982123	0.3380578141	0.3792658818
O	0.6619421859	0.6982403981	0.3792658818
O	0.0362982123	0.6982403981	0.1207341182
O	0.6619421859	0.9637017877	0.1207341182
O	0.3017596019	0.3380578141	0.1207341182

SYSTEM Parameters

System	
ntyp	3
nat	28
ibrav	0
celldm(1)	10.6080773
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

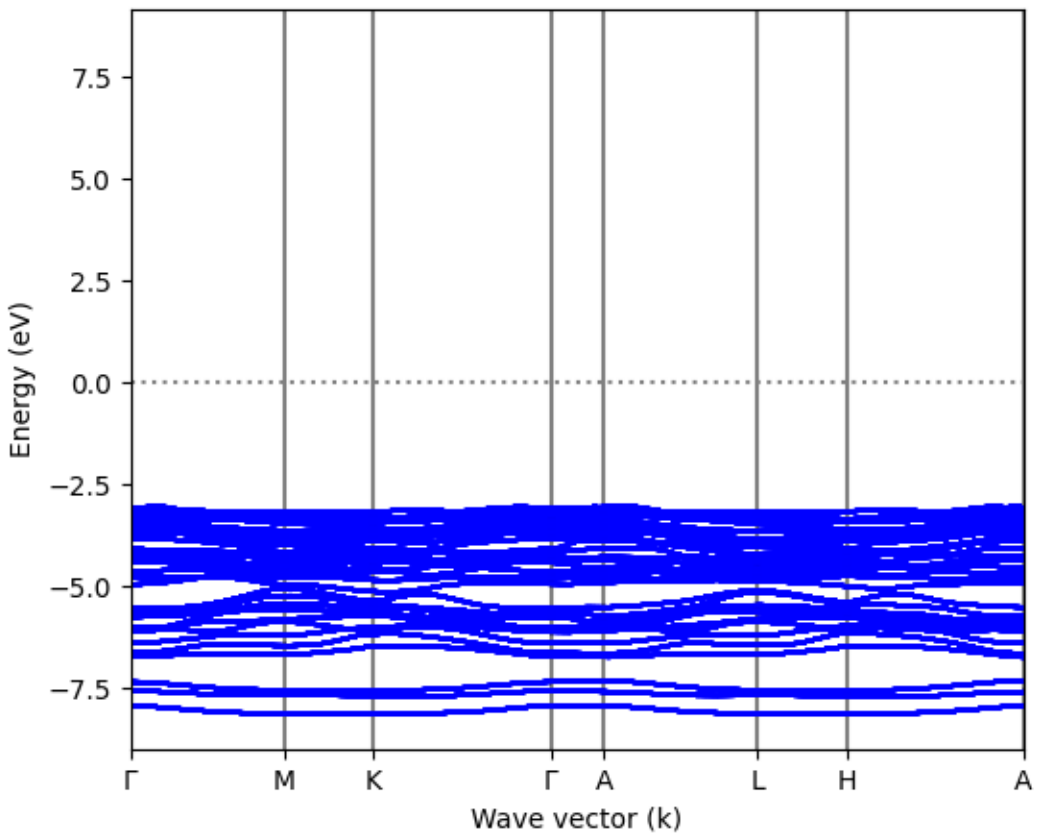
mp-27417: https://next-gen.materialsproject.org/materials/mp-27417?formula=In1Li3O3

CHAPTER

FORTYTHREE

LI2O4SI1SR1

Band gap Diagram



Bravais lattice type

Trigonal

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
8.0818

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.981692492	0.000000000	-0.00000000
-0.49084625	0.850170637	0.000000000
-0.00000000	-0.00000000	2.436044886

ATOMIC_POSITIONS (crystal)

Sr	0.4166569129	0.0000000000	0.8333333300
Sr	0.5833430871	0.5833430871	0.5000000000
Sr	0.0000000000	0.4166569129	0.1666666700
Li	0.6648072638	0.5914640868	0.7456947745
Li	0.9266568231	0.3351927362	0.4123614445
Li	0.4085359132	0.0733431769	0.0790281045
Li	0.3351927362	0.9266568231	0.5876385555
Li	0.5914640868	0.6648072638	0.2543052255
Li	0.0733431769	0.4085359132	0.9209718955
Si	0.2780560497	-0.000000000	0.3333333300
Si	0.7219439503	0.7219439503	0.0000000000
Si	0.0000000000	0.2780560497	0.6666666700
O	0.5466589421	0.5118582910	0.1073677885
O	0.9651993488	0.4533410579	0.7740344585
O	0.4881417090	0.0348006512	0.4407011185
O	0.4533410579	0.9651993488	0.2259655415
O	0.5118582910	0.5466589421	0.8926322115
O	0.0348006512	0.4881417090	0.5592988815
O	0.0716980453	0.7623797334	0.9966121285
O	0.6906816881	0.9283019547	0.6632787985
O	0.2376202666	0.3093183119	0.3299454585
O	0.9283019547	0.6906816881	0.3367212015
O	0.7623797334	0.0716980453	0.0033878715
O	0.3093183119	0.2376202666	0.6700545415

SYSTEM Parameters

System	
ntyp	4
nat	24
ibrav	0
celldm(1)	9.60210149
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

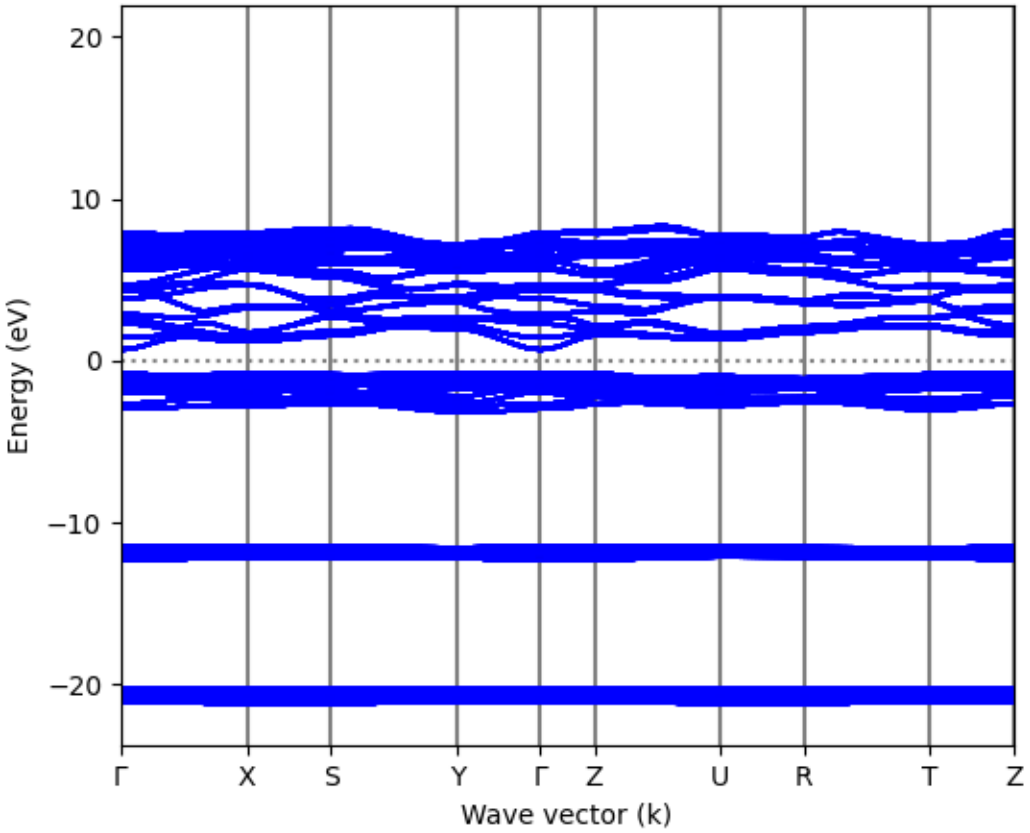
mp-1191141: https://next-gen.materialsproject.org/materials/mp-1191141?formula=Li2O4Si1Sr1

CHAPTER

FORTYFOUR

CA1LI1N1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
7.2868	1.4339	1.9941	2.3933	2.7997	1.4339	2.3243	2.2771	2.8415	2.9145	2.3243

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.987771386	0.000000000	0.000000000
0.000000000	1.484031089	0.000000000
0.000000000	0.000000000	2.310483741

ATOMIC_POSITIONS (crystal)

Li	0.2500000000	0.5732599937	0.0491346700
Li	0.7500000000	0.4267400063	0.9508653300
Li	0.7500000000	0.0732599937	0.4508653300
Li	0.2500000000	0.9267400063	0.5491346700
Ca	0.2500000000	0.4704305036	0.6557810449
Ca	0.7500000000	0.9704305036	0.8442189551
Ca	0.2500000000	0.0295694964	0.1557810449
Ca	0.7500000000	0.5295694964	0.3442189551
N	0.7500000000	0.7393493382	0.0939763262
N	0.7500000000	0.7606506618	0.5939763262
N	0.2500000000	0.2606506618	0.9060236738
N	0.2500000000	0.2393493382	0.4060236738

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	6.89366182
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

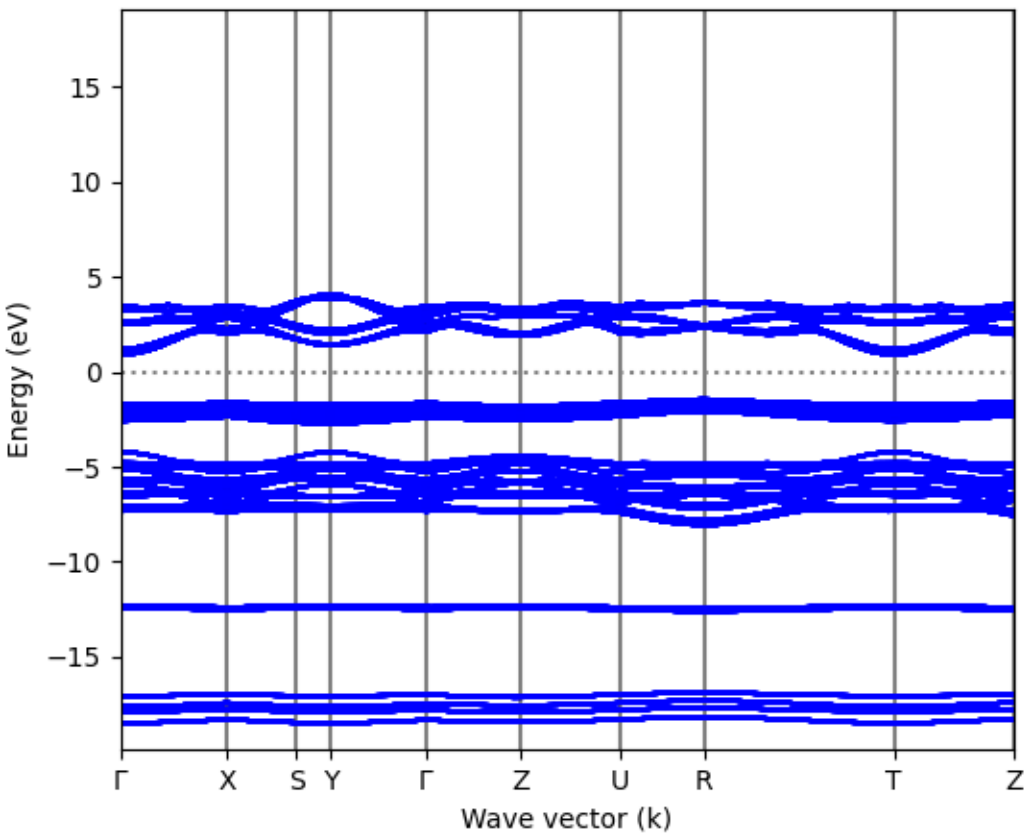
mp-31468: https://next-gen.materialsproject.org/materials/mp-31468?formula=Ca1Li1N1

CHAPTER

FORTYFIVE

BR2LI5N1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
6.8905	2.5980	3.6498	3.4201	3.0878	3.7280	3.7010	3.6498	3.7716	2.5980	3.5715

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.945105417	0.000000000	0.000000000
0.000000000	1.431875721	0.000000000
0.000000000	0.000000000	2.779844164

ATOMIC_POSITIONS (crystal)

Li	0.0000000000	0.7771530054	0.1253620449
Li	0.5000000000	0.7228438471	0.3746371136
Li	0.5000000000	0.2771561529	0.3746371136
Li	0.5000000000	0.0000000000	0.0000000000
Li	0.0000000000	0.2228469946	0.1253620449
Li	0.5000000000	0.2771561529	0.6253628864
Li	0.0000000000	0.2228469946	0.8746379551
Li	0.0000000000	0.7771530054	0.8746379551
Li	0.0000000000	0.5000000000	0.5000000000
Li	0.5000000000	0.7228438471	0.6253628864
Br	0.0000000000	0.0000000000	0.3310500685
Br	0.5000000000	0.5000000000	0.1689498111
Br	0.5000000000	0.5000000000	0.8310501889
Br	0.0000000000	0.0000000000	0.6689499315
N	0.0000000000	0.0000000000	0.0000000000
N	0.5000000000	0.5000000000	0.5000000000

SYSTEM Parameters

System	
ntyp	3
nat	16
ibrav	0
celldm(1)	7.36016780
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

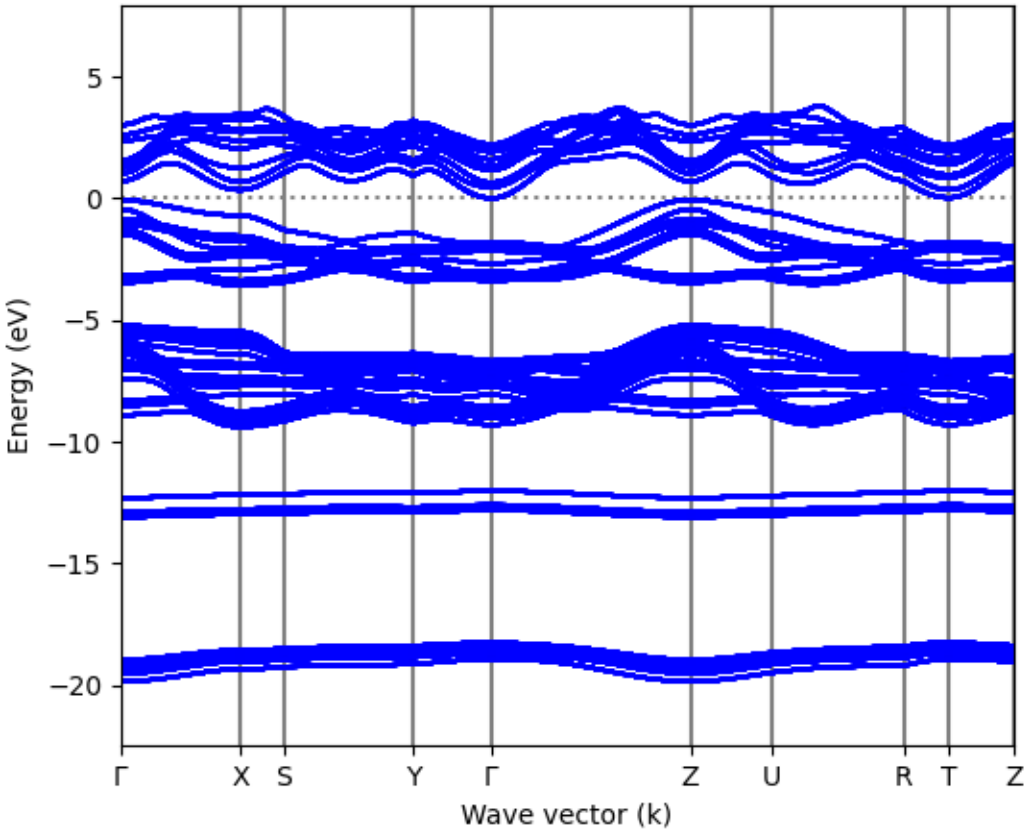
mp-29025: https://next-gen.materialsproject.org/materials/mp-29025?formula=Br2Li5N1

CHAPTER

FORTYSIX

AL1LA4LI1O8

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
7.2036	0.7588	1.0475	2.3468	2.3521	0.4537	0.7588	1.4839	2.5055	1.8193	3.4437

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.921515112	-0.16846581	-0.00261832
-0.14620926	0.884572496	0.003598519
0.004644047	0.022899649	7.631721375

ATOMIC_POSITIONS (crystal)

Li	0.6666887706	0.3332442414	0.8333327543
Li	-0.044877895	0.0335979229	0.9326688763
Li	0.3782282641	0.6329732737	0.7339975461
Li	0.0071970963	0.0064833555	0.6806973149
Li	0.3261307260	0.6601641935	0.9859702052
Li	0.3333112294	0.6667557586	0.1666672457
Li	0.6217717359	0.3670267263	0.2660024539
Li	0.0448778956	-0.033597923	0.0673311237
Li	0.6738692740	0.3398358065	0.0140297948
Li	-0.007197096	-0.006483355	0.3193026851
Li	1.0000000000	1.0000000000	0.5000000000
Li	0.2884617076	0.7003104848	0.5993349679
Li	0.7115382924	0.2996895152	0.4006650321
Li	0.3405316501	0.6731698081	0.3473639098
Li	0.6594683499	0.3268301919	0.6526360902
N	0.0000000000	0.0000000000	0.0000000000
N	0.6666636612	0.3333367240	0.3333336202
N	0.3333363388	0.6666632760	0.6666663798
Cl	0.3341491180	0.6751584224	0.8855925653
Cl	-0.000821901	-0.008761471	0.7810777234
Cl	0.0008219007	0.0087614710	0.2189222766
Cl	0.6658508820	0.3248415776	0.1144074347
Cl	0.6674656340	0.3417471080	0.5522590989
Cl	0.3325343660	0.6582528920	0.4477409011

SYSTEM Parameters

System	
ntyp	3
nat	24
ibrav	0
celldm(1)	6.83203463
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

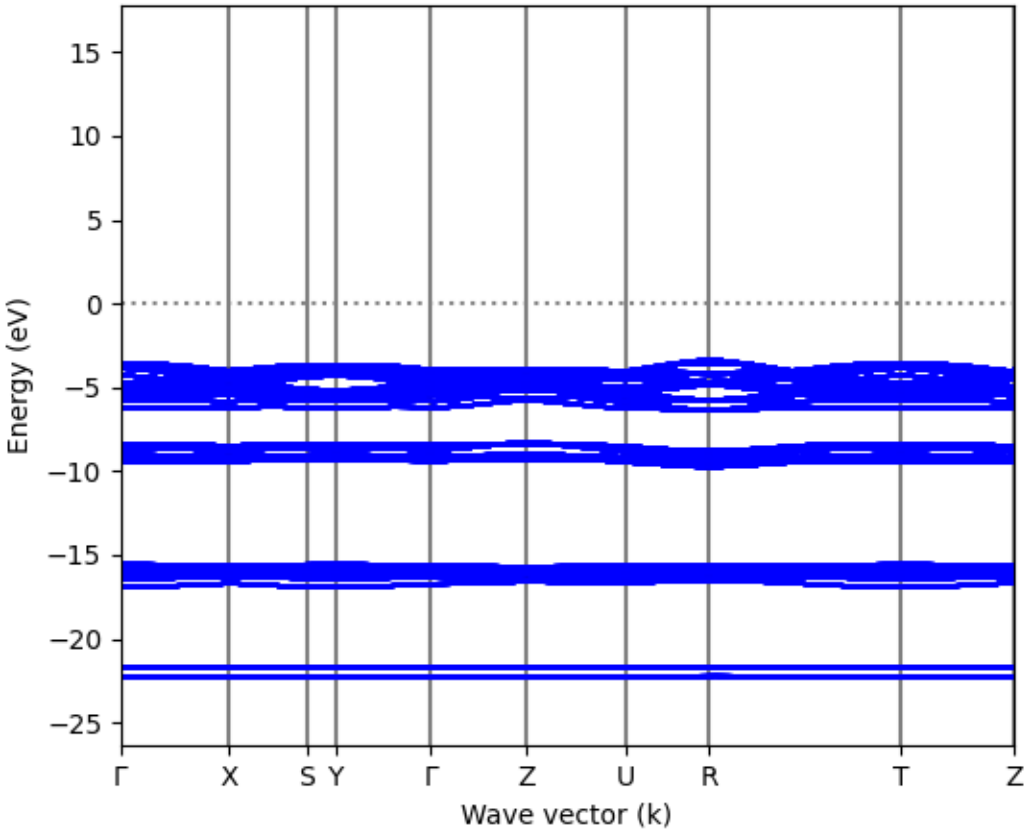
mp-1222363: https://next-gen.materialsproject.org/materials/mp-1222363?formula=Al1La4Li1O8

CHAPTER

FORTYSEVEN

BR3CS1LI2

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
7.6659

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.936496795	0.000000000	0.000000000
0.000000000	1.121410423	0.000000000
0.000000000	0.000000000	2.627731049

ATOMIC_POSITIONS (crystal)

Cs	0.5000000000	0.0000000000	0.0000000000
Cs	0.0000000000	0.5000000000	0.5000000000
Li	0.0000000000	0.5000000000	0.8060774670
Li	0.5000000000	0.0000000000	0.6939226189
Li	0.5000000000	0.0000000000	0.3060773811
Li	0.0000000000	0.5000000000	0.1939225330
Br	0.0000000000	0.0000000000	0.7805097566
Br	0.5000000000	0.5000000000	0.7194906288
Br	0.0000000000	0.5000000000	0.0000000000
Br	0.5000000000	0.5000000000	0.2805093712
Br	0.0000000000	0.0000000000	0.2194902434
Br	0.5000000000	0.0000000000	0.5000000000

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	8.81301747
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

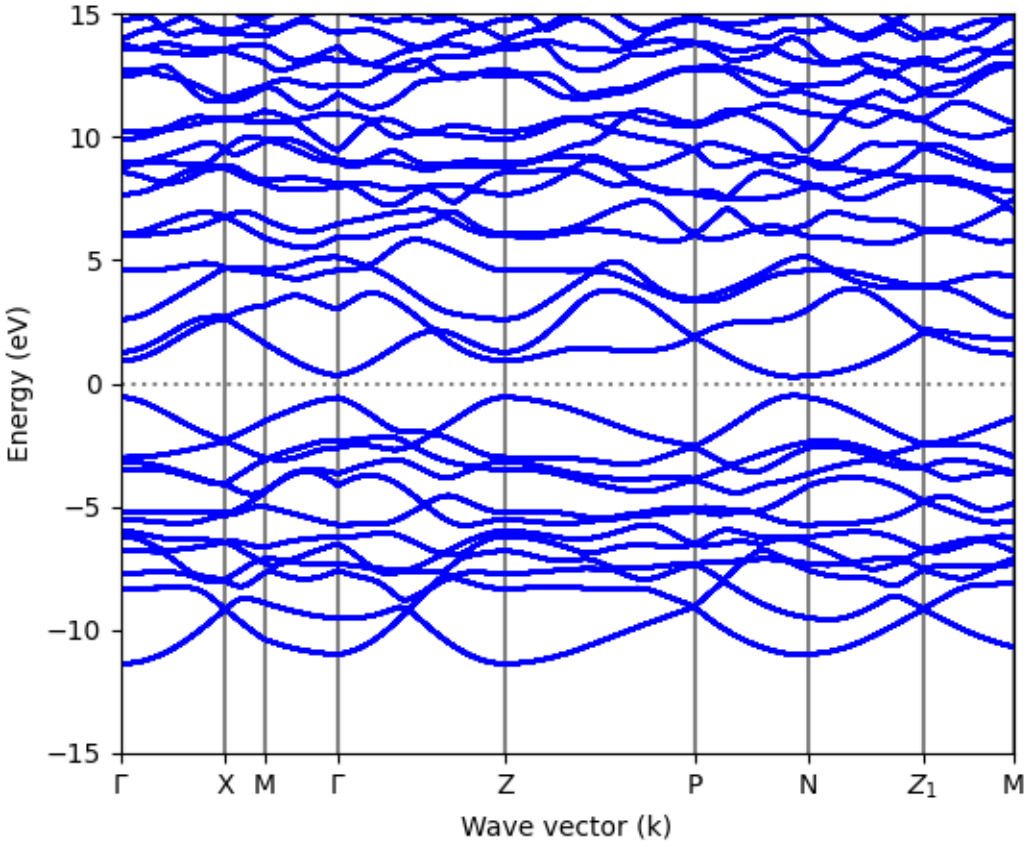
mp-606680: https://next-gen.materialsproject.org/materials/mp-606680?formula=Br3Cs1Li2

CHAPTER

FORTYEIGHT

B1LI1MG1N2

Band gap Diagram



Bravais lattice type

TetragonalI

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	P	N	Z1	M2
11.5110	1.4882	4.9050	3.1637	0.9008	1.4882	4.4045	0.9008	4.4689	2.5727

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.523952439	-0.511452063 | 1.20717270
0.527719019	0.5658833440 | 1.17815633
-0.50352126	-0.544660892 | 1.20145678

ATOMIC_POSITIONS (crystal)

Li	0.6590280096	0.0222370329	0.2067176574
Li	0.1588331472	0.5225277147	0.7069260202
Mg	-0.203680965	0.4764581964	0.3656219865
Mg	0.2967697882	-0.023492101	0.8652629651
B	0.6525589209	0.3905592116	-0.109461079
B	0.1526083067	-0.109333172	0.3903052888
N	0.6317442785	0.5551594276	0.0793237257
N	-0.239687706	0.0558975264	0.4573865097
N	0.1316928680	0.0547415909	0.5797381454
N	0.2601333527	0.5552445721	0.9581787806

SYSTEM Parameters

System	
ntyp	4
nat	10
ibrav	0
celldm(1)	7.21745314
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

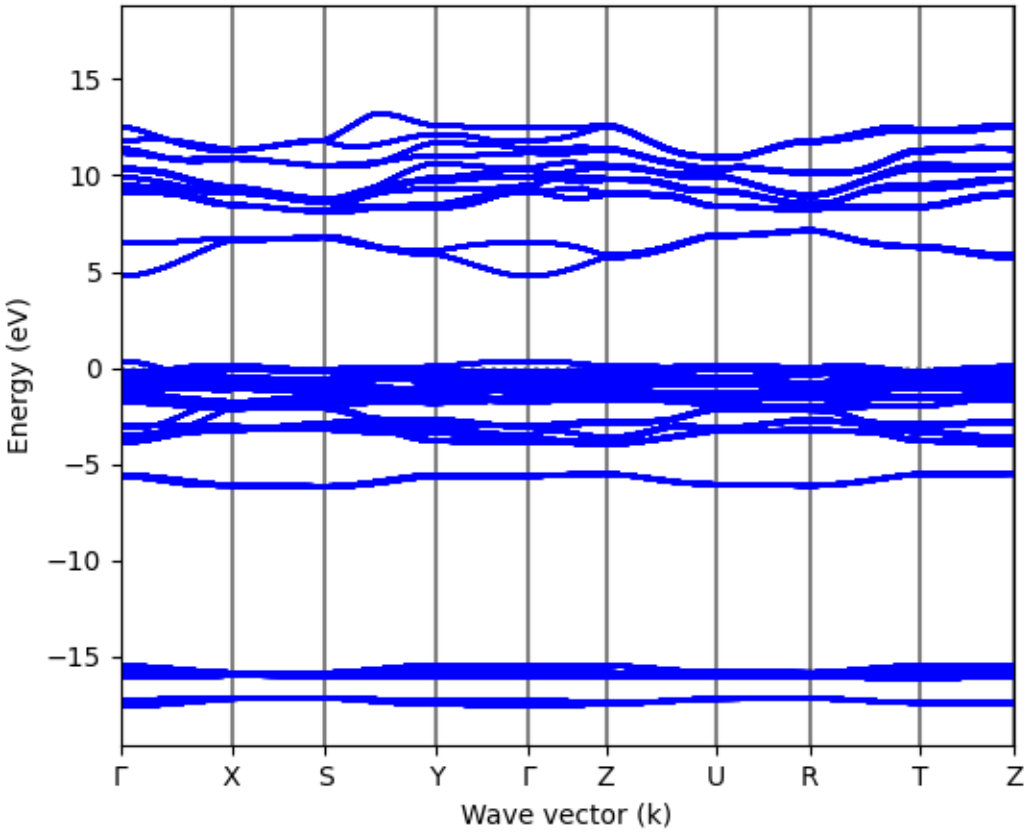
mp-1222299: https://next-gen.materialsproject.org/materials/mp-1222299?formula=B1Li1Mg1N2

CHAPTER

FORTYNINE

GE1LI3O4ZN0

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
3.4430	0.5036	0.1297	6.8443	0.0833	0.5036	0.2548	0.0773	0.0819	6.4240	0.2548

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.921190582	0.000000000	0.000000000
0.000000000	1.096398737	0.000000000
0.000000000	0.000000000	1.326194847

ATOMIC_POSITIONS (crystal)

Li	0.4669290388	0.8327887755	0.2553391013
Li	0.9669290388	0.1672112245	0.2446608987
Li	0.9669290388	0.1672112245	0.7553391013
Li	0.4669290388	0.8327887755	0.7446608987
Li	0.4696972349	0.3079325183	0.0000000000
Li	0.9696972349	0.6920674817	0.5000000000
Ge	-0.001442555	0.6598988129	-0.000000000
Ge	0.4985574453	0.3401011871	0.5000000000
O	0.9085572501	0.8157760787	0.2297124914
O	0.4085572501	0.1842239213	0.2702875086
O	0.4085572501	0.1842239213	0.7297124914
O	0.9085572501	0.8157760787	0.7702875086
O	0.9090794133	0.3420236412	-0.000000000
O	0.4090794133	0.6579763588	0.5000000000
O	0.3956943286	0.6741664645	-0.000000000
O	0.8956943286	0.3258335355	0.5000000000

SYSTEM Parameters

System	
ntyp	3
nat	16
ibrav	0
celldm(1)	9.30147509
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

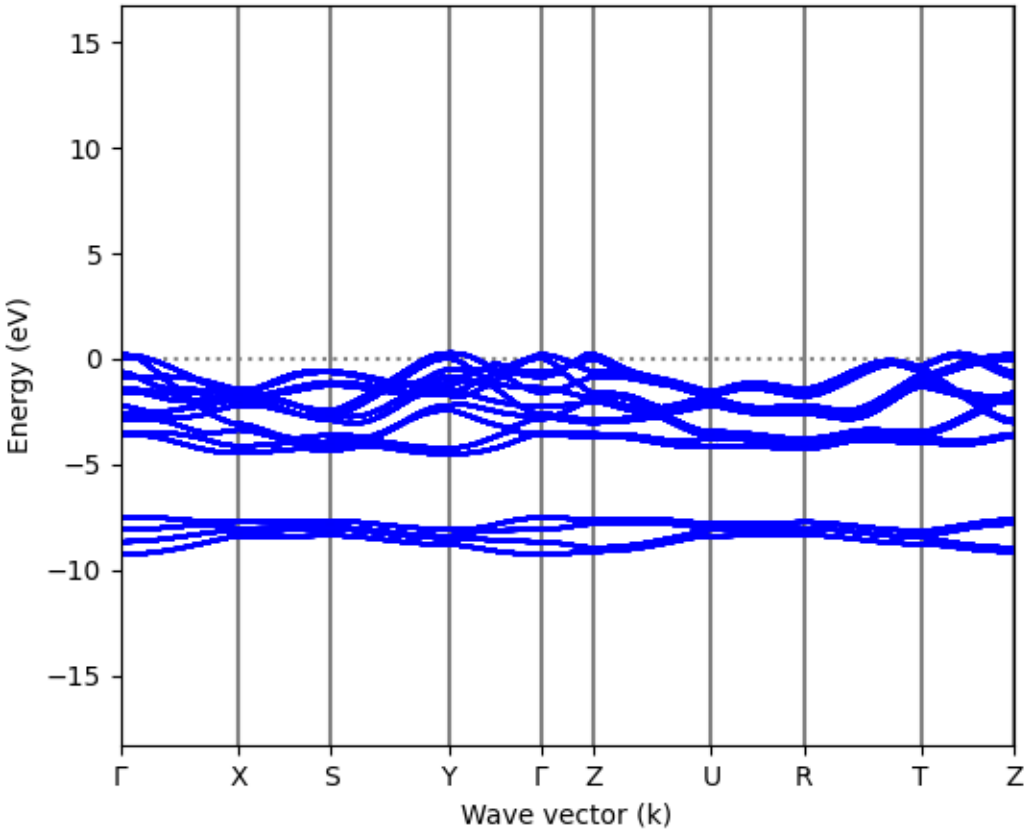
mp-1211307: https://next-gen.materialsproject.org/materials/mp-1211307?formula=Ge1Li3O4Zn0

CHAPTER

FIFTY

GE1LI2MG1O4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	Y	Γ	Z	Z1
6.0615	0.6559	0.5419	0.6559	0.2384	0.2384

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.870281837	0.000000000	0.000000000
0.000000000	1.103198899	0.000000000
0.000000000	0.000000000	1.945863239

ATOMIC_POSITIONS (crystal)

Li	0.2494143045	0.2508480085	0.0617315631
Li	0.2505856955	0.2508480085	0.5617315631
Li	0.7505856955	0.7508480085	0.9382684369
Li	0.7494143045	0.7508480085	0.4382684369
Li	0.2494370927	-0.001202185	0.3462775697
Li	0.7505629073	0.4987978149	0.6537224303
Li	0.7494370927	0.4987978149	0.1537224303
Li	0.2505629073	-0.001202185	0.8462775697
Mg	0.2504671659	0.5001488732	0.8315010844
Mg	0.7495328341	1.0001488732	0.1684989156
Mg	0.7504671659	1.0001488732	0.6684989156
Mg	0.2495328341	0.5001488732	0.3315010844
Ge	0.7504284237	0.2510203034	0.9035901283
Ge	0.7495715763	0.2510203034	0.4035901283
Ge	0.2495715763	0.7510203034	0.0964098717
Ge	0.2504284237	0.7510203034	0.5964098717

SYSTEM Parameters

System	
ntyp	4
nat	16
ibrav	0
celldm(1)	9.77822650
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

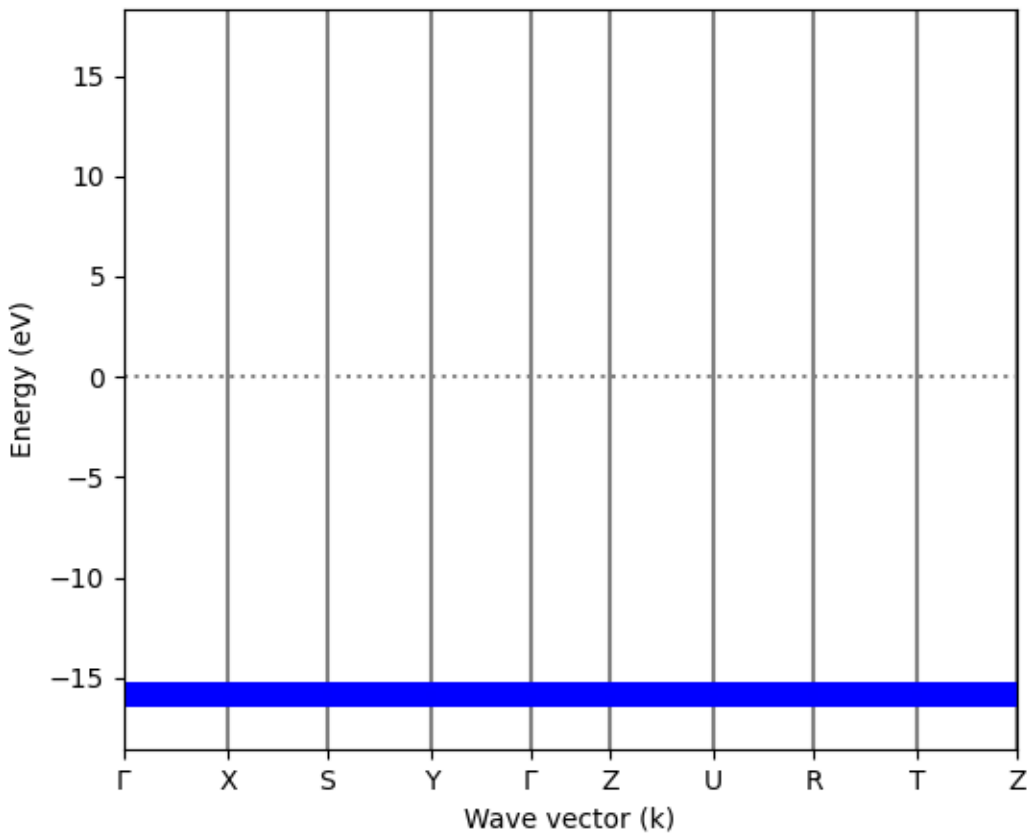
mp-1210797: https://next-gen.materialsproject.org/materials/mp-1210797?formula=Ge1Li2Mg1O4

CHAPTER

FIFTYONE

GA1LI1SE2

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
8.0472

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.930118649	0.000000000	0.000000000
0.000000000	0.968612032	0.000000000
0.000000000	0.000000000	1.210620887

ATOMIC_POSITIONS (crystal)

Li	0.1028372707	0.9302822638	0.6189740924
Li	0.6028372707	0.0697177362	0.3810259076
Li	0.6028372707	0.5697177362	0.1189740924
Li	0.1028372707	0.4302822638	0.8810259076
Ga	0.6314212456	0.5924648420	0.6233620911
Ga	0.1314212456	0.4075351580	0.3766379089
Ga	0.1314212456	0.9075351580	0.1233620911
Ga	0.6314212456	0.0924648420	0.8766379089
Se	0.4899162193	0.9341487693	0.6349256326
Se	-0.010083781	0.0658512307	0.3650743674
Se	-0.010083781	0.5658512307	0.1349256326
Se	0.4899162193	0.4341487693	0.8650743674
Se	1.0138242544	0.0713173210	0.8768095375
Se	0.5138242544	0.9286826790	0.1231904625
Se	0.5138242544	0.4286826790	0.3768095375
Se	1.0138242544	0.5713173210	0.6231904625

SYSTEM Parameters

System	
ntyp	3
nat	16
ibrav	0
celldm(1)	12.4524888
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

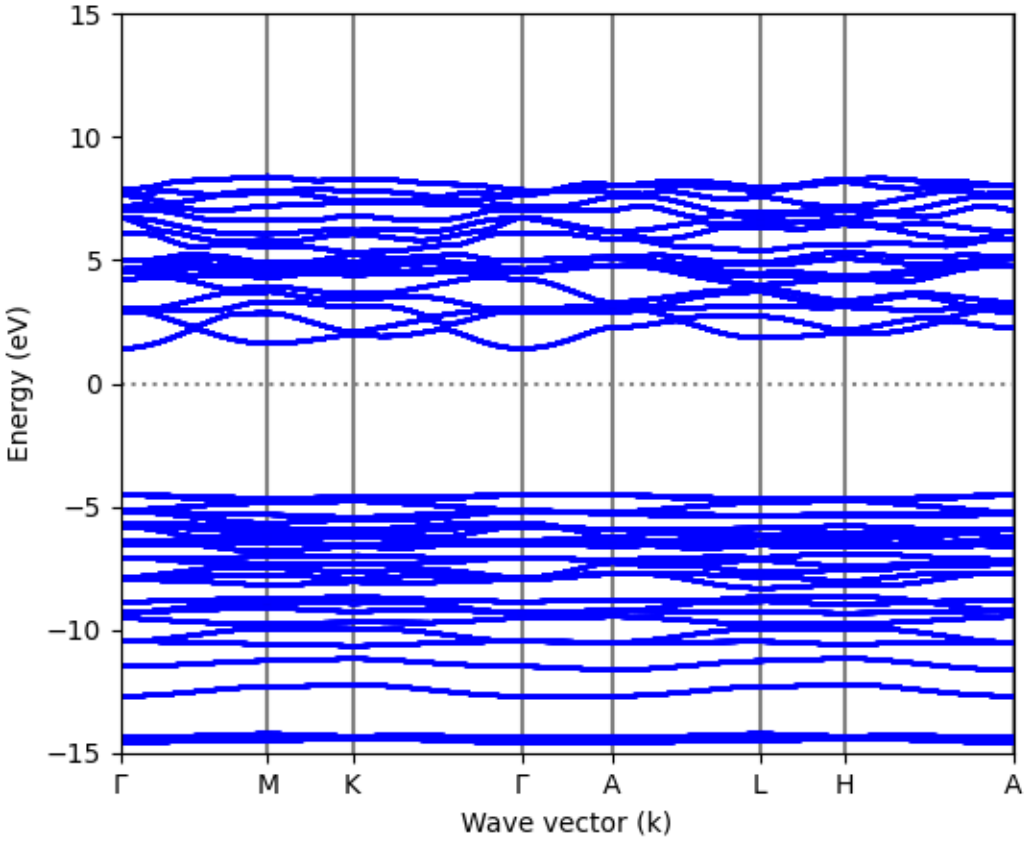
mp-11582: https://next-gen.materialsproject.org/materials/mp-11582?formula=Ga1Li1Se2

CHAPTER

FIFTYTWO

BA1LI2MG1O8P2

Band gap Diagram



Bravais lattice type

Trigonal

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	M	K	Γ	A	L	H	A2
10.7872	5.9345	6.2631	6.6029	5.9345	6.7928	6.5189	6.7543	6.7928

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.981643379	0.000000000	0.000000000
-0.490821690 | 0.850128104 | -0.0000000		
-0.000000000 | -0.000000000 | 1.3743967		

ATOMIC_POSITIONS (crystal)

Ba	0.000000000	0.000000000	0.500000000
Li	0.666666670	0.333333330	0.237987360
Li	0.333333330	0.666666670	0.762012640
Mg	0.000000000	0.000000000	-0.000000000
P	0.666666670	0.333333330	0.749389981
P	0.333333330	0.666666670	0.250610019
O	0.666666670	0.333333330	0.528293443
O	0.333333330	0.666666670	0.471706557
O	0.357024388	0.300182569	0.820554789
O	0.943158181	0.642975612	0.820554789
O	0.699817431	0.056841819	0.820554789
O	0.642975612	0.699817431	0.179445211
O	0.056841819	0.357024388	0.179445211
O	0.300182569	0.943158181	0.179445211

SYSTEM Parameters

System	
ntyp	5
nat	14
ibrav	0
celldm(1)	9.63906799
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

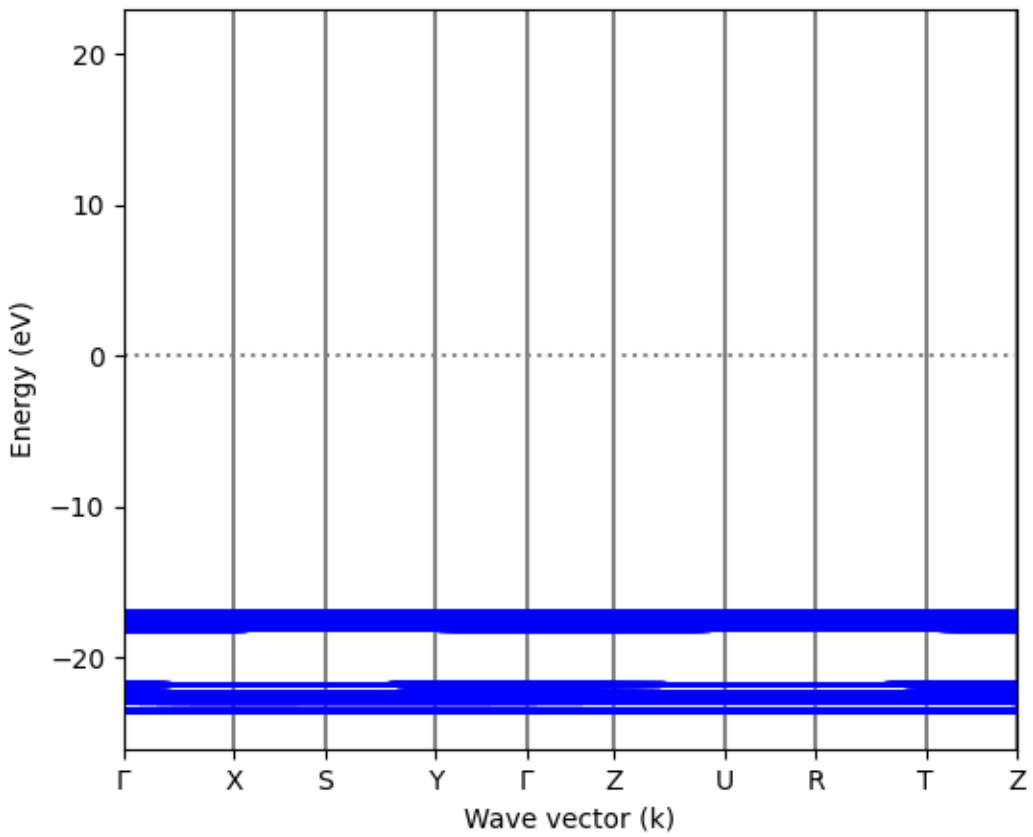
mp-1019547: https://next-gen.materialsproject.org/materials/mp-1019547?formula=Ba1Li2Mg1O8P2

CHAPTER

FIFTYTHREE

GA1LI3NA2O4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
6.2989

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.962851379	0.000000000	0.000000000
0.000000000	1.168786188	0.000000000
0.000000000	0.000000000	1.211393393

ATOMIC_POSITIONS (crystal)

Na	-0.000000000	0.255655022	0.749877452
Na	0.000000000	0.744344978	0.250122548
Na	0.500000000	0.755655022	0.750122548
Na	0.500000000	0.244344978	0.249877452
Na	0.741923126	0.500000000	-0.000000000
Na	0.758076874	0.000000000	0.500000000
Na	0.258076874	0.500000000	0.000000000
Na	0.241923126	-0.000000000	0.500000000
Li	0.758357264	0.486724128	0.350552504
Li	0.758357264	0.513275872	0.649447496
Li	0.741642736	0.986724128	0.149447496
Li	0.741642736	0.013275872	0.850552504
Li	0.241642736	0.513275872	0.649447496
Li	0.241642736	0.486724128	0.350552504
Li	0.258357264	0.013275872	0.850552504
Li	0.258357264	0.986724128	0.149447496
Li	0.000000000	0.261608614	0.414475054
Li	-0.000000000	0.738391386	0.585524946
Li	0.500000000	0.761608614	0.085524946
Li	0.500000000	0.238391386	0.914475054
Ga	-0.000000000	0.773577179	0.877847889
Ga	0.000000000	0.226422821	0.122152111
Ga	0.500000000	0.273577179	0.622152111
Ga	0.500000000	0.726422821	0.377847889
O	-0.000000000	0.554079149	0.779106985
O	-0.000000000	0.445920851	0.220893015
O	0.500000000	0.054079149	0.720893015
O	0.500000000	0.945920851	0.279106985

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Table 1 – continued from previous page

O	0.751885779	0.804213137	0.993061283
O	0.751885779	0.195786863	0.006938717
O	0.748114221	0.304213137	0.506938717
O	0.748114221	0.695786863	0.493061283
O	0.248114221	0.195786863	0.006938717
O	0.248114221	0.804213137	0.993061283
O	0.251885779	0.695786863	0.493061283
O	0.251885779	0.304213137	0.506938717
O	0.000000000	0.955523521	0.719029438
O	0.000000000	0.044476479	0.280970562
O	0.500000000	0.455523521	0.780970562
O	0.500000000	0.544476479	0.219029438

SYSTEM Parameters

System	
ntyp	4
nat	40
ibrav	0
celldm(1)	12.4573080
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

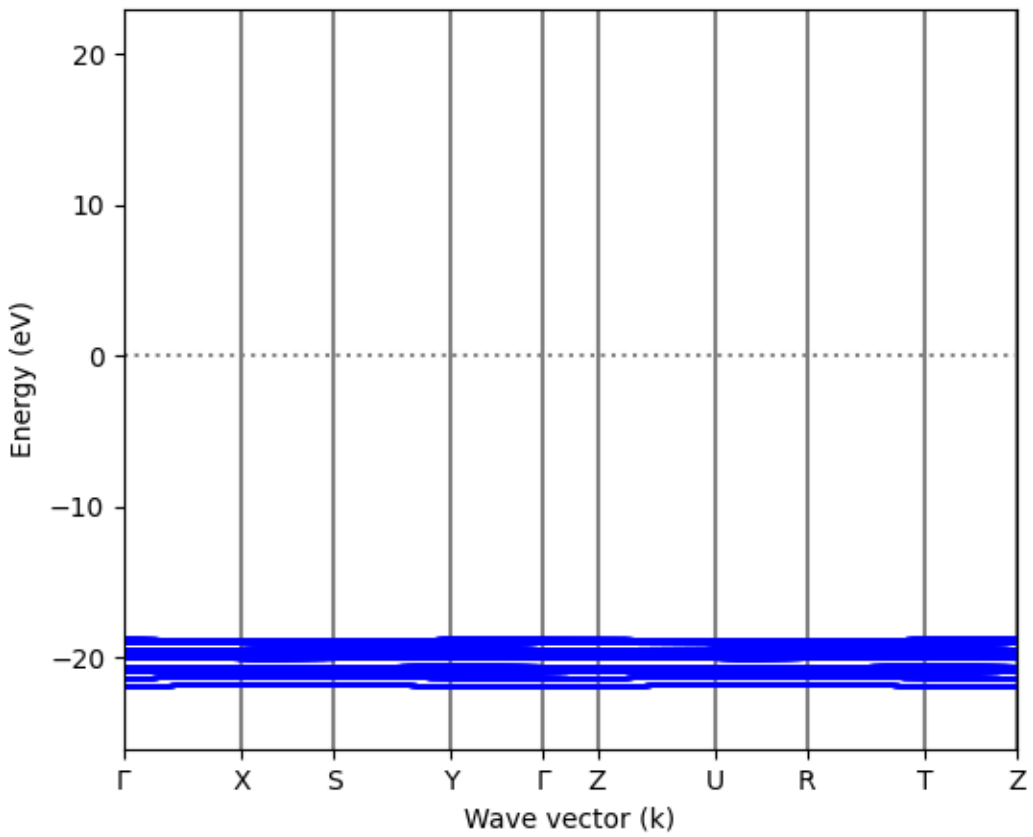
mp-540945: https://next-gen.materialsproject.org/materials/mp-540945?formula=Ga1Li3Na2O4

CHAPTER

FIFTYFOUR

LI2MG1O4SI1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
7.4437

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.964493003	0.000000000	0.000000000
0.000000000	1.222684692	0.000000000
0.000000000	0.000000000	2.086453241

ATOMIC_POSITIONS (crystal)

Li	0.189898114	0.002298689	0.338662657
Li	0.689898114	0.502298689	0.161337343
Li	0.810101886	0.502298689	0.661337343
Li	0.310101886	0.002298689	0.838662657
Li	0.291528858	0.253207043	0.077153884
Li	0.791528858	0.753207043	0.422846116
Li	0.708471142	0.753207043	0.922846116
Li	0.208471142	0.253207043	0.577153884
Mg	0.809597101	0.999834018	0.667492779
Mg	0.309597101	0.499834018	0.832507221
Mg	0.190402899	0.499834018	0.332507221
Mg	0.690402899	0.999834018	0.167492779
Si	0.183129629	0.751110201	0.084992147
Si	0.683129629	0.251110201	0.415007853
Si	0.816870371	0.251110201	0.915007853
Si	0.316870371	0.751110201	0.584992147
O	0.210815701	0.752196317	0.438616458
O	0.710815701	0.252196317	0.061383542
O	0.789184299	0.252196317	0.561383542
O	0.289184299	0.752196317	0.938616458
O	0.292762987	0.533176556	0.157860424
O	0.792762987	0.033176556	0.342139576
O	0.713195804	0.469016300	0.838853330
O	0.213195804	0.969016300	0.661146670
O	0.847234650	0.749819875	0.088525620
O	0.347234650	0.249819875	0.411474380
O	0.152765350	0.249819875	0.911474380
O	0.652765350	0.749819875	0.588525620

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Table 1 – continued from previous page

O	0.286804196	0.969016300	0.161146670
O	0.786804196	0.469016300	0.338853330
O	0.707237013	0.033176556	0.842139576
O	0.207237013	0.533176556	0.657860424

SYSTEM Parameters

System	
ntyp	4
nat	32
ibrav	0
celldm(1)	9.55948127
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

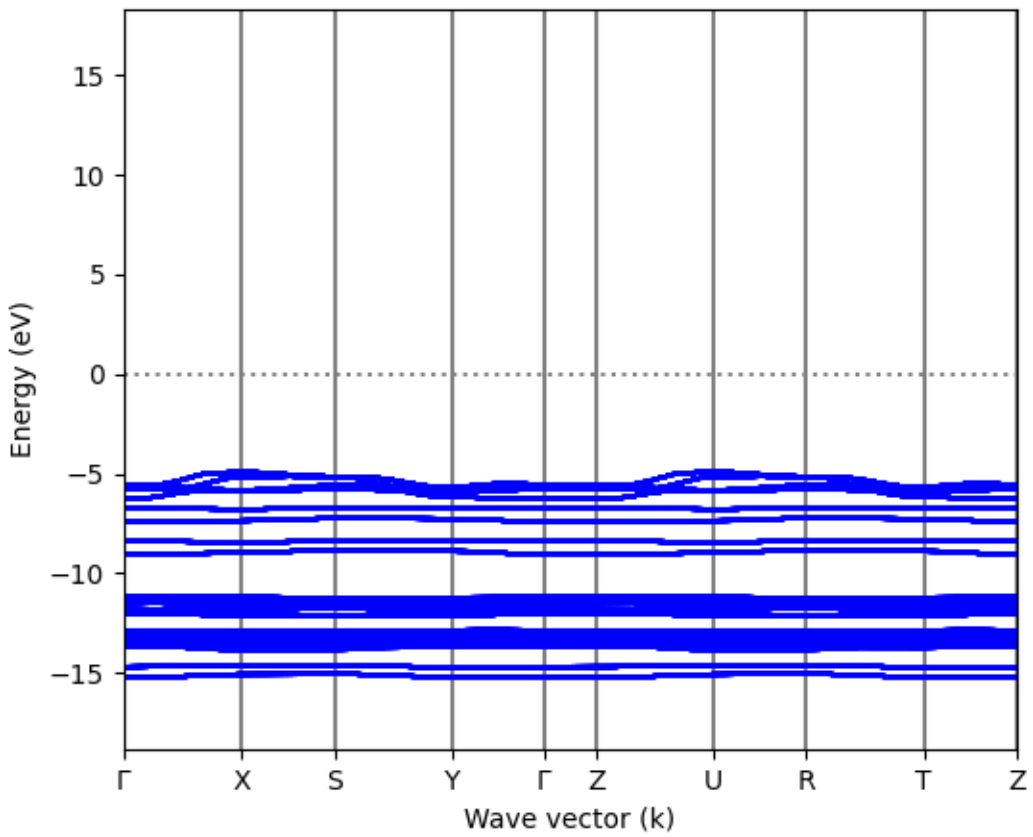
mp-1222742: https://next-gen.materialsproject.org/materials/mp-1222742?formula=Li2Mg1O4Si1

CHAPTER

FIFTYFIVE

GE1LI4S4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
4.6452

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.977684423	0.000000000	0.000000000
0.000000000	1.233872165	0.000000000
0.000000000	0.000000000	2.244875153

ATOMIC_POSITIONS (crystal)

Li	0.186251961	0.251824784	0.573547584
Li	0.313748039	0.748175216	0.073547584
Li	0.813748039	0.751824784	0.926452416
Li	0.686251961	0.248175216	0.426452416
Li	0.812065023	0.249873946	0.927024210
Li	0.687934977	0.750126054	0.427024210
Li	0.187934977	0.749873946	0.572975790
Li	0.312065023	0.250126054	0.072975790
Li	0.372796399	0.001502030	0.837989875
Li	0.127203601	0.998497970	0.337989875
Li	0.627203601	0.501502030	0.662010125
Li	0.872796399	0.498497970	0.162010125
Li	0.004653760	0.754336022	0.749687012
Li	0.495346240	0.245663978	0.249687012
Li	0.995346240	0.254336022	0.750312988
Li	0.504653760	0.745663978	0.250312988
Ge	0.644717123	0.000873449	0.660632520
Ge	0.855282877	0.999126551	0.160632520
Ge	0.355282877	0.500873449	0.839367480
Ge	0.144717123	0.499126551	0.339367480
S	0.770398575	0.001166156	0.814177363
S	0.729601425	0.998833844	0.314177363
S	0.229601425	0.501166156	0.685822637
S	0.270398575	0.498833844	0.185822637
S	0.276869726	1.000854575	0.666076639
S	0.223130274	-0.000854575	0.166076639
S	0.723130274	0.500854575	0.833923361
S	0.776869726	0.499145425	0.333923361

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Table 1 – continued from previous page

S	0.781041434	0.760712373	0.591859050
S	0.718958566	0.239287627	0.091859050
S	0.218958566	0.260712373	0.908140950
S	0.281041434	0.739287627	0.408140950
S	0.220780348	0.741512797	0.908397556
S	0.279219652	0.258487203	0.408397556
S	0.779219652	0.241512797	0.591602444
S	0.720780348	0.758487203	0.091602444

SYSTEM Parameters

System	
ntyp	3
nat	36
ibrav	0
celldm(1)	11.6477508
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

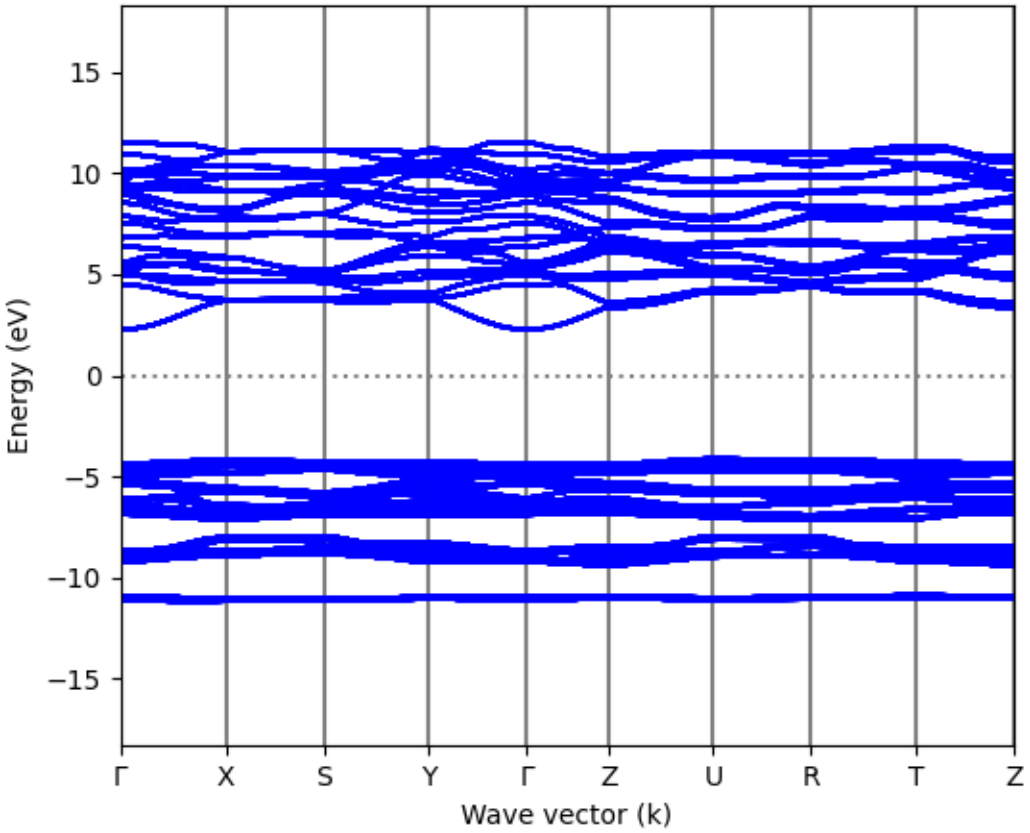
mp-1222529: https://next-gen.materialsproject.org/materials/mp-1222529?formula=Ge1Li4S4

CHAPTER

FIFTYSIX

LI3O4P1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
7.8172	6.6414	7.8732	7.9500	7.9920	6.6414	7.7448	8.2698	8.6023	8.4160	7.7448

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.980038739	0.000000000	0.000000000
0.000000000	1.053518933	0.000000000
0.000000000	0.000000000	1.267710588

ATOMIC_POSITIONS (crystal)

Li	0.519110656	0.822037825	-0.000000000
Li	0.019110656	0.177962175	0.500000000
Li	0.511404251	0.325735208	0.252014369
Li	0.011404251	0.674264792	0.247985631
Li	0.011404251	0.674264792	0.752014369
Li	0.511404251	0.325735208	0.747985631
P	0.499727750	0.828598960	0.500000000
P	-0.000272250	0.171401040	0.000000000
O	0.107928678	0.311481883	0.789895856
O	0.600313313	0.120357055	0.500000000
O	0.171085423	0.818253489	0.500000000
O	0.671085423	0.181746511	-0.000000000
O	0.607928678	0.688518117	0.289895856
O	0.107928678	0.311481883	0.210104144
O	0.100313313	0.879642945	-0.000000000
O	0.607928678	0.688518117	0.710104144

SYSTEM Parameters

System	
ntyp	3
nat	16
ibrav	0
celldm(1)	9.09602848
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

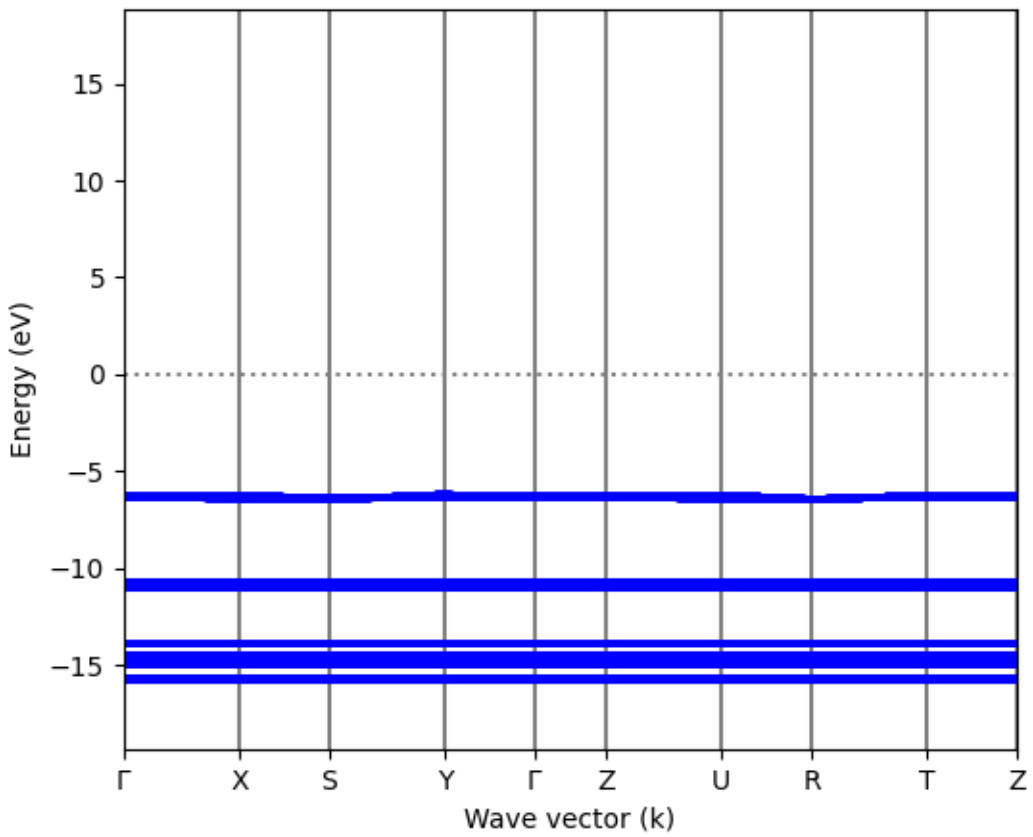
mp-13725: https://next-gen.materialsproject.org/materials/mp-13725?formula=Li3O4P1

CHAPTER

FIFTYSEVEN

K3LI1SI4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
4.5427

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.988773006	0.000000000	0.000000000
0.000000000	1.260850691	0.000000000
0.000000000	0.000000000	1.585457043

ATOMIC_POSITIONS (crystal)

K	0.655424617	0.250000000	0.399219426
K	0.155424617	0.250000000	0.100780574
K	0.344575383	0.750000000	0.600780574
K	0.844575383	0.750000000	0.899219426
K	0.654653945	0.445493844	0.693597536
K	0.154653945	0.054506156	0.806402464
K	0.345346055	0.945493844	0.306402464
K	0.845346055	0.554506156	0.193597536
K	0.345346055	0.554506156	0.306402464
K	0.845346055	0.945493844	0.193597536
K	0.654653945	0.054506156	0.693597536
K	0.154653945	0.445493844	0.806402464
Li	0.500000000	0.500000000	0.000000000
Li	-0.000000000	0.000000000	0.500000000
Li	0.500000000	-0.000000000	0.000000000
Li	-0.000000000	0.500000000	0.500000000
Si	0.361936727	0.750000000	0.898560583
Si	0.861936727	0.750000000	0.601439417
Si	0.638063273	0.250000000	0.101439417
Si	0.138063273	0.250000000	0.398560583
Si	0.714037631	0.627381346	0.451623803
Si	0.214037631	0.872618654	0.048376197
Si	0.285962369	0.127381346	0.548376197
Si	0.785962369	0.372618654	0.951623803
Si	0.010457766	0.250000000	0.583824301
Si	0.510457766	0.250000000	0.916175699
Si	0.989542234	0.750000000	0.416175699
Si	0.489542234	0.750000000	0.083824301

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Table 1 – continued from previous page

Si	0.214037631	0.627381346	0.048376197
Si	0.714037631	0.872618654	0.451623803
Si	0.785962369	0.127381346	0.951623803
Si	0.285962369	0.372618654	0.548376197

SYSTEM Parameters

System	
ntyp	3
nat	32
ibrav	0
celldm(1)	14.5014813
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

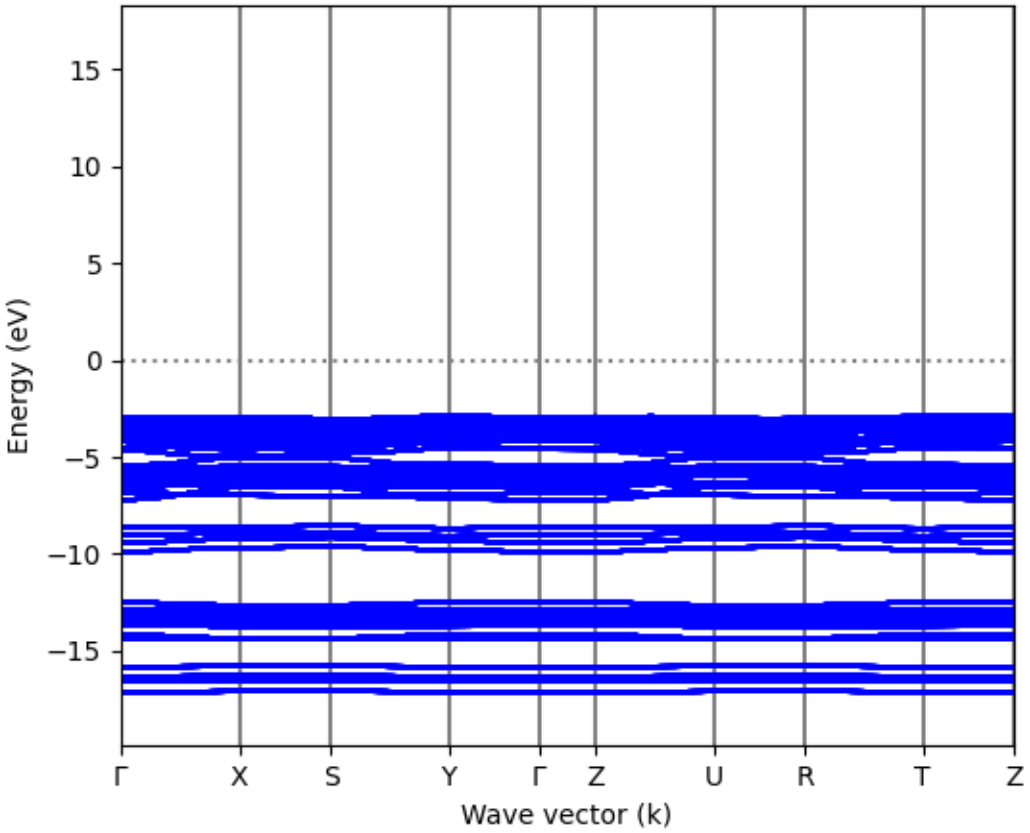
mp-28316: https://next-gen.materialsproject.org/materials/mp-28316?formula=K3Li1Si4

CHAPTER

FIFTYEIGHT

LI3P1S4

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F
4.5444

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.977286820	0.000000000	0.000000000
0.000000000	1.270126501	0.000000000
0.000000000	0.000000000	2.088730255

ATOMIC_POSITIONS (crystal)

Li	0.376791633	0.032793258	0.668660955
Li	0.623208367	0.967206742	0.331339045
Li	0.876791633	0.967206742	0.831339045
Li	0.123208367	0.032793258	0.168660955
Li	0.623208367	0.532793258	0.331339045
Li	0.376791633	0.467206742	0.668660955
Li	0.123208367	0.467206742	0.168660955
Li	0.876791633	0.532793258	0.831339045
Li	0.500000000	-0.000000000	-0.000000000
Li	-0.000000000	-0.000000000	0.500000000
Li	0.500000000	0.500000000	-0.000000000
Li	0.000000000	0.500000000	0.500000000
P	0.144981202	0.250000000	0.911023408
P	0.855018798	0.750000000	0.088976592
P	0.644981202	0.750000000	0.588976592
P	0.355018798	0.250000000	0.411023408
S	0.806862080	0.250000000	0.901607311
S	0.193137920	0.750000000	0.098392689
S	0.306862080	0.750000000	0.598392689
S	0.693137920	0.250000000	0.401607311
S	0.277858610	0.250000000	0.060648415
S	0.722141390	0.750000000	0.939351585
S	0.777858610	0.750000000	0.439351585
S	0.222141390	0.250000000	0.560648415
S	0.264578905	0.028797518	0.846220415
S	0.735421095	0.971202482	0.153779585
S	0.764578905	0.971202482	0.653779585
S	0.235421095	0.028797518	0.346220415

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Table 1 – continued from previous page

S	0.735421095	0.528797518	0.153779585
S	0.264578905	0.471202482	0.846220415
S	0.235421095	0.471202482	0.346220415
S	0.764578905	0.528797518	0.653779585

SYSTEM Parameters

System	
ntyp	3
nat	32
ibrav	0
celldm(1)	11.6629464
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

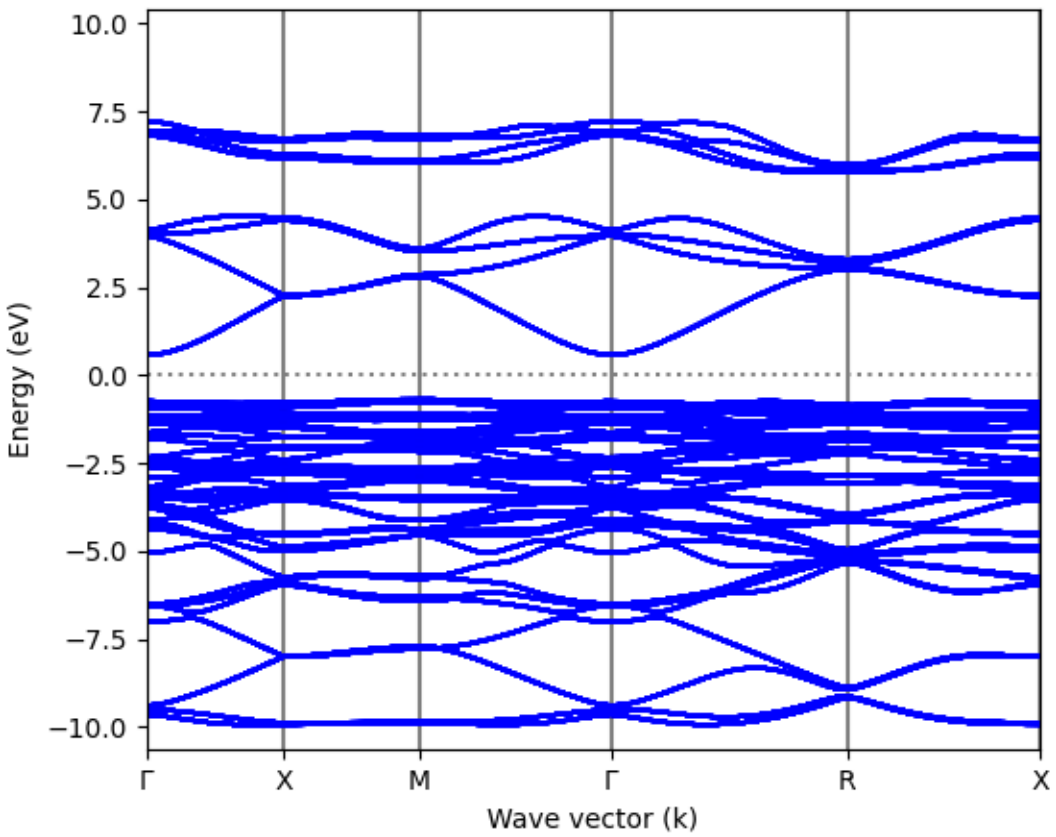
mp-1097036: https://next-gen.materialsproject.org/materials/mp-1097036?formula=Li3P1S4

CHAPTER

FIFTYNINE

LI1O3SB1

Band gap Diagram



Bravais lattice type

CubicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	R	X2
6.5186	1.3245	3.0316	3.4921	1.3245	3.8790	3.0316

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.988519343	0.000000000	0.000000000
0.000000000	0.988519343	0.000000000
0.000000000	0.000000000	0.988519343

ATOMIC_POSITIONS (crystal)

Li	0.872382241	0.872382241	0.872382241
Li	0.627617759	0.127617759	0.372382241
Li	0.372382241	0.627617759	0.127617759
Li	0.127617759	0.372382241	0.627617759
Sb	0.090635913	0.909364087	0.409364087
Sb	0.409364087	0.090635913	0.909364087
Sb	0.590635913	0.590635913	0.590635913
Sb	0.909364087	0.409364087	0.090635913
O	0.984912127	0.228656647	0.366826298
O	0.133173702	0.015087873	0.728656647
O	0.228656647	0.366826298	0.984912127
O	0.271343353	0.633173702	0.484912127
O	0.366826298	0.984912127	0.228656647
O	0.515087873	0.771343353	0.866826298
O	0.484912127	0.271343353	0.633173702
O	0.633173702	0.484912127	0.271343353
O	0.728656647	0.133173702	0.015087873
O	0.771343353	0.866826298	0.515087873
O	0.866826298	0.515087873	0.771343353
O	0.015087873	0.728656647	0.133173702

SYSTEM Parameters

System	
ntyp	3
nat	20
ibrav	0
celldm(1)	11.3774808
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

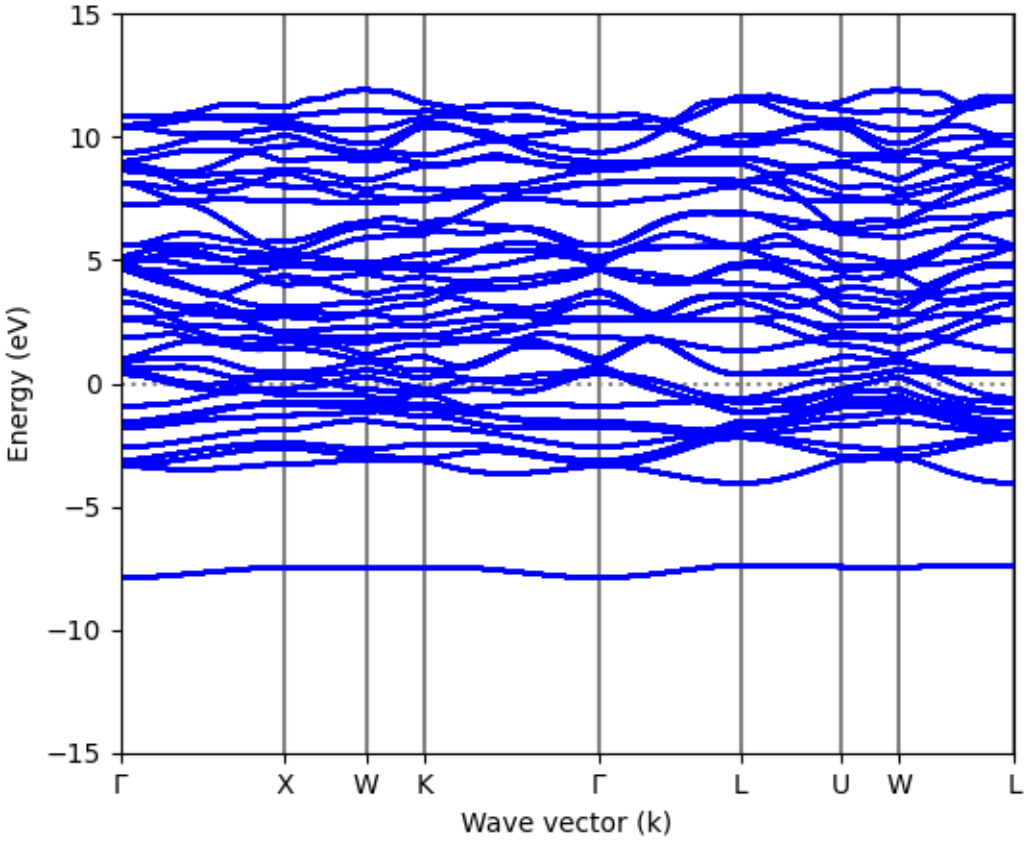
mp-752636: https://next-gen.materialsproject.org/materials/mp-752636?formula=Li1O3Sb1

CHAPTER

SIXTY

LI1SI1Y1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	W	K	Γ	L	U	W2	L2
10.9207	1.2783	0.4333	0.4614	0.4417	1.2783	0.9569	0.4223	0.4614	0.9569

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.62851015	-0.00456983	0.628510148
0.004569834	0.628510148	0.628510148
-0.62851015	0.628510148	-0.00456983

ATOMIC_POSITIONS (crystal)

Li	-0.032070904	0.102308528	0.102308528
Li	0.044880731	0.544880731	0.544880731
Li	0.602308528	0.102308528	0.467929096
Li	0.602308528	0.467929096	0.102308528
Y	0.235426175	0.225500139	0.225500139
Y	0.205849254	0.705849254	0.705849254
Y	0.725500139	0.225500139	0.735426175
Y	0.725500139	0.735426175	0.225500139
Si	0.390297410	-0.109702590	-0.109702590

SYSTEM Parameters

System	
ntyp	3
nat	9
ibrav	0
celldm(1)	12.5815159
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

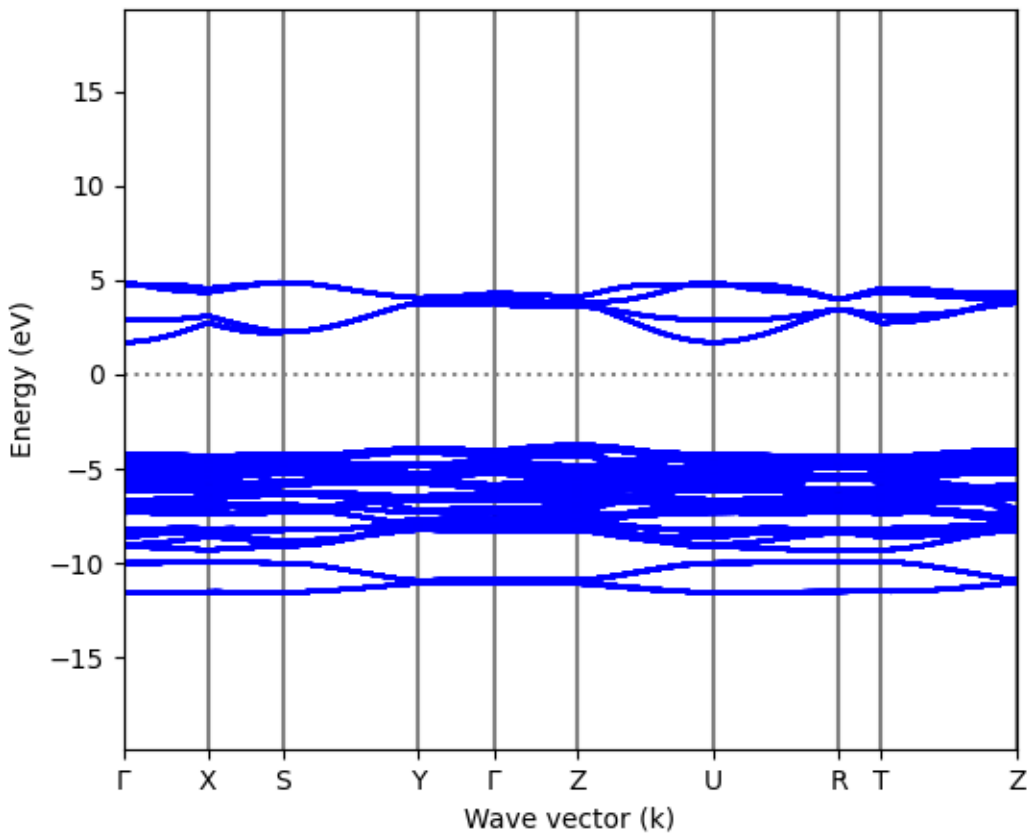
mp-1100428: https://next-gen.materialsproject.org/materials/mp-1100428?formula=Li1Si1Y

CHAPTER

SIXTYONE

LI2O3SI1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
8.2289	5.9604	7.0469	6.4958	7.7044	7.8495	7.3377	5.9604	7.7637	7.0469	7.8495

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

1.005645123	0.000000000	-0.000000005
0.000000000	1.746431068	0.000000000
-0.000000005	0.000000000	0.808044703

ATOMIC_POSITIONS (crystal)

Li	0.663418133	0.170915074	1.024104822
Li	0.836581878	0.329084905	0.524104750
Li	0.336581667	0.170915059	0.524104793
Li	0.163418307	0.329084965	1.024104727
Li	0.163418307	0.670915035	1.024104727
Li	0.336581677	0.829084931	0.524104793
Li	0.836581878	0.670915095	0.524104750
Li	0.663418123	0.829084916	1.024104822
Si	0.821082079	0.000000000	0.493143094
Si	0.178917718	0.000000000	0.993143087
Si	0.321082341	0.500000000	0.493143042
Si	0.678917953	0.500000000	0.993143037
O	0.876717493	0.000000000	0.110855138
O	0.184962497	0.352451113	0.571496103
O	0.684962302	0.147548905	0.571496144
O	0.123282264	0.000000000	0.610855158
O	0.815037682	0.352451077	0.071496109
O	0.315037473	0.147548912	0.071496162
O	0.376717748	0.500000000	0.110855109
O	0.684962302	0.852451095	0.571496144
O	0.184962497	0.647548887	0.571496103
O	0.623282508	0.500000000	0.610855073
O	0.315037473	0.852451088	0.071496162
O	0.815037682	0.647548923	0.071496109

SYSTEM Parameters

System	
ntyp	3
nat	24
ibrav	0
celldm(1)	10.1213095
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

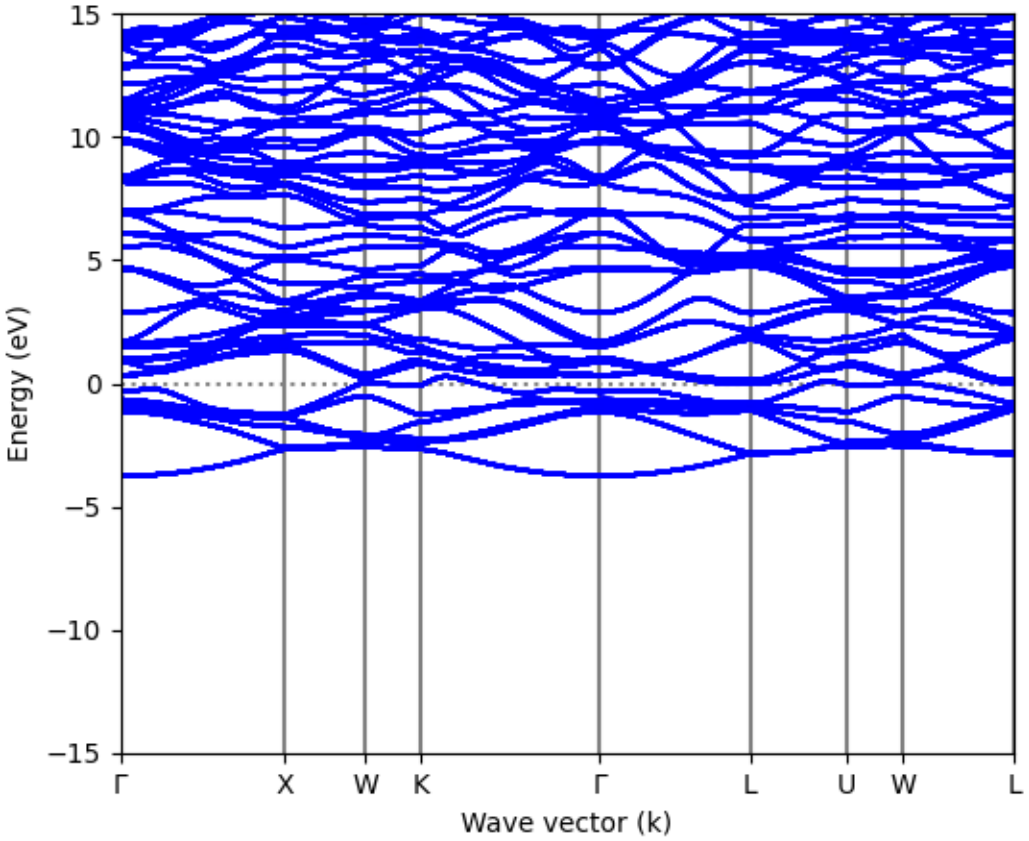
mp-5012: https://next-gen.materialsproject.org/materials/mp-5012?formula=Li2O3Si1

CHAPTER

SIXTYTWO

LI1S2TI1

Band gap Diagram



Bravais lattice type

CubicF

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	W	K	Γ	L	U	W2	L2
1.6382	0.6578	2.5530	0.5601	0.8936	0.6578	0.1617	0.8236	0.5601	0.1617

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

-0.45211224	-0.02954801	0.452112240
0.029548015	0.452112240	0.452112240
-0.45211224	0.452112240	-0.02954801

ATOMIC_POSITIONS (crystal)

Li	0.040959696	0.540959696	0.169623240
Li	0.080033157	0.920391909	0.250176629
Li	0.420391909	0.580033157	0.250176629
Li	0.458875494	0.958875494	0.329502989
Li	0.040959696	0.169623240	0.540959696
Li	0.080033157	0.250176629	0.920391909
Li	0.458875494	0.329502989	0.958875494
Li	0.420391909	0.250176629	0.580033157
Li	0.669623240	0.540959696	0.540959696
Li	0.829502989	0.958875494	0.958875494
Li	0.750176629	0.580033157	0.920391909
Li	0.750176629	0.920391909	0.580033157

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	19.0069395
occupations	'smearing'
smearing	'gauss'
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

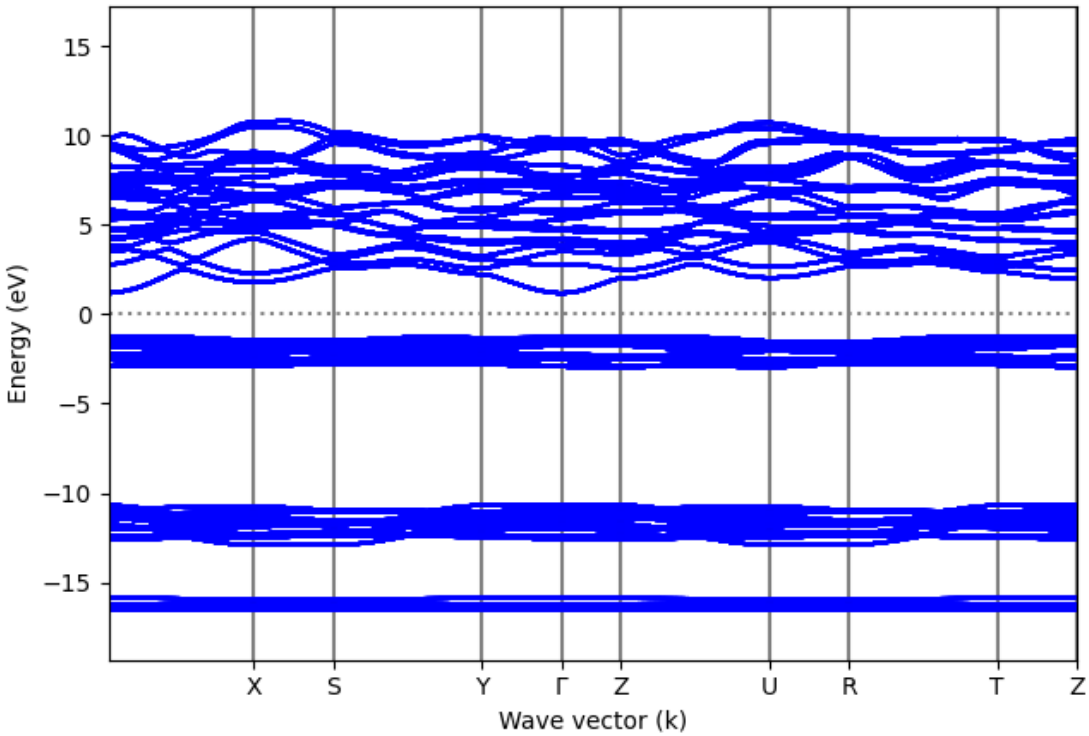
mp-755414: https://next-gen.materialsproject.org/materials/mp-755414?formula=Li1S2Ti1

CHAPTER

SIXTYTHREE

LI1O1RB1

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
6.7737	2.4777	3.2395	4.0345	3.5599	2.4777	3.2647	3.5656	4.1983	3.6421	3.2647

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.988901737	0.000000000	0.000000000
0.000000000	1.835986731	0.000000000
0.000000000	0.000000000	2.478334886

ATOMIC_POSITIONS (crystal)

Rb	0.250000000	0.999395630	0.819075495
Rb	0.750000000	0.000604370	0.180924505
Rb	0.250000000	0.499395630	0.680924505
Rb	0.750000000	0.500604370	0.319075495
Li	0.750000000	0.894100880	0.501350109
Li	0.250000000	0.605899120	1.001350109
Li	0.750000000	0.394100880	-0.001350109
Li	0.250000000	0.105899120	0.498649891
O	0.250000000	0.810706507	0.450484353
O	0.750000000	0.689293493	0.950484353
O	0.250000000	0.310706507	0.049515647
O	0.750000000	0.189293493	0.549515647

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	6.61878305
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

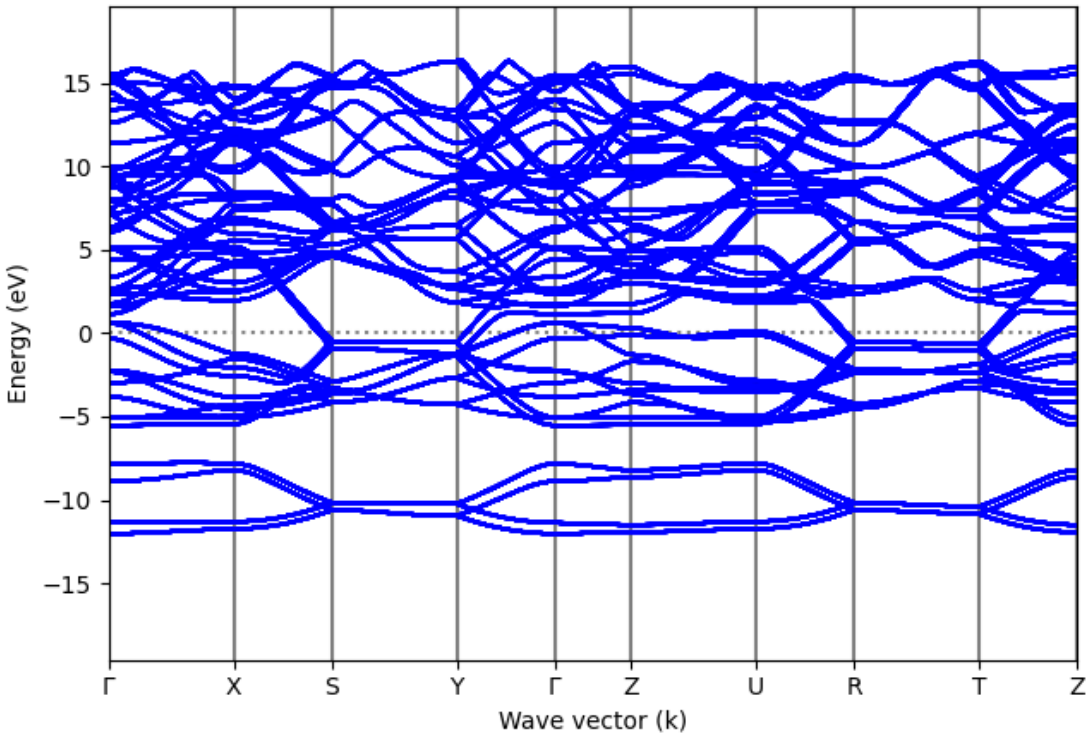
mp-28545: https://next-gen.materialsproject.org/materials/mp-28545?formula=Li1O1Rb1

CHAPTER

SIXTYFOUR

LI4P2S6

Band gap Diagram



Bravais lattice type

OrthorhombicP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	S	Y	Γ	Z	U	R	T	Z1
5.8963	0.9255	3.1947	5.0848	2.4066	0.9255	0.4281	0.2415	2.8953	2.6648	0.4281

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.686365828	0.000000000	0.000000000
0.000000000	0.874602801	0.000000000
0.000000000	0.000000000	1.123146991

ATOMIC_POSITIONS (crystal)

Li	0.750261043	0.510081125	0.077925597
Li	0.749738957	1.010081125	0.077925597
Li	0.250261043	-0.010081125	0.922074403
Li	0.249738957	0.489918875	0.577925597
Li	0.249738957	0.489918875	0.922074403
Li	0.250261043	-0.010081125	0.577925597
Li	0.749738957	1.010081125	0.422074403
Li	0.750261043	0.510081125	0.422074403
P	0.749915281	0.254665526	0.750000000
P	0.250084719	0.745334474	0.250000000
P	0.249915281	0.245334474	0.250000000
P	0.750084719	0.754665526	0.750000000

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	11.6476152
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

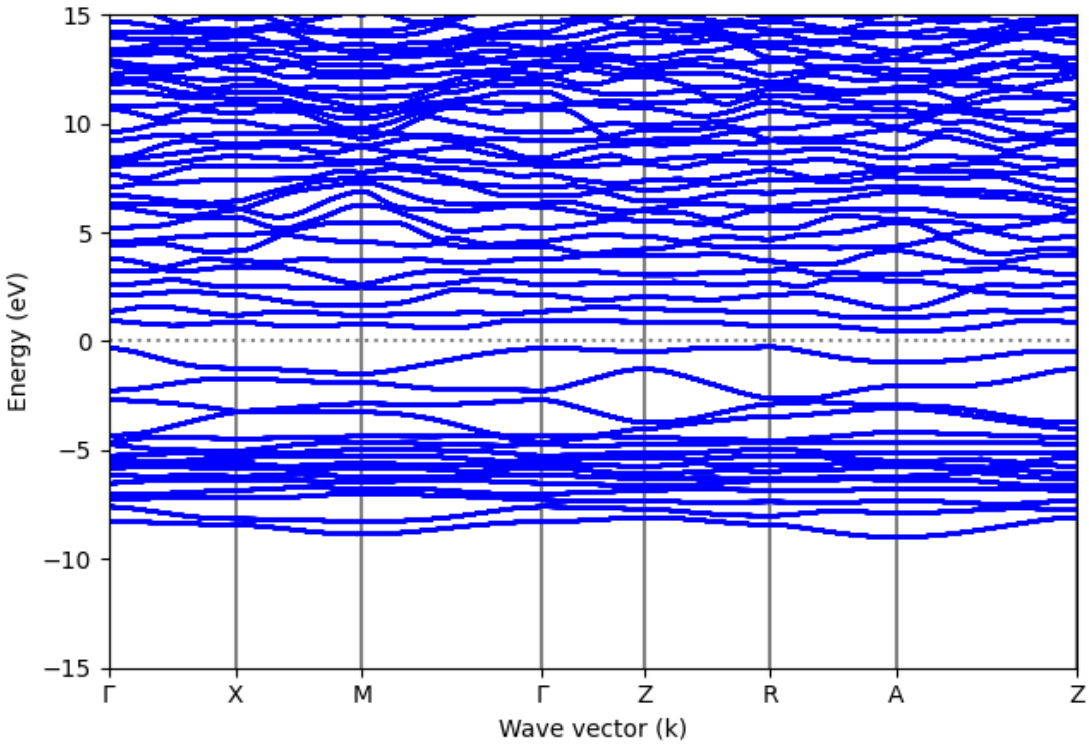
mp-38200: https://next-gen.materialsproject.org/materials/mp-38200?formula=Li4P2S6

CHAPTER

SIXTYFIVE

AL1LI1O2

Band gap Diagram



Bravais lattice type

TetragonalP

Band gap

Understand that fermi energy is denoted by :math: and all band gap values are in units of :math: (electron-Volt).

E_F	Γ	X	M	Γ	Z	R	A	Z2
6.0640	1.2697	2.1247	2.2771	1.2697	1.3903	0.9645	1.4319	1.3903

Optimized Structure

CELL_PARAMETERS (alat= 8.72223677)

0.963378522	-0.00922195	0.000734873
-0.00916869	0.965197334	0.014601177
0.000765357	0.017765158	1.198473674

ATOMIC_POSITIONS (crystal)

Li	0.700434731	0.267589163	0.254981609
Li	0.173327440	0.176611146	0.492599288
Li	0.295101639	0.696039254	0.741118081
Li	0.767335763	0.829269912	-0.006004204
Al	0.786083151	0.799508919	0.507680987
Al	0.667981723	0.346824765	0.755206818
Al	0.358269076	0.697678944	0.253351411
Al	0.224377204	0.179014603	-0.014207171
O	0.321464903	0.338268947	0.233098578
O	0.146274984	0.791243530	0.472817235
O	0.782504352	0.161214229	0.534303268
O	0.861289534	0.216736587	0.982441018

SYSTEM Parameters

System	
ntyp	3
nat	12
ibrav	0
celldm(1)	9.74043600
occupations	‘smearing’
smearing	‘gauss’
degauss	0.01
ecutwfc	40.0
ecutrho	400.0
edir	3
emaxpos	0
eopreg	0.05
eamp	0.00972345

Reference

mp-3427: https://next-gen.materialsproject.org/materials/mp-3427?formula=Al1Li1O2

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